

SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>110-20-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L018</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>MATSUYA, NORIHIRO</u>

1. SCOPE:

- 1.1 Title: Bumper Assembly; replace
- 1.2 Location of Work:
 - 1.2.1 Throughout the Craft (LARC-V)
- 1.3 Identification:
 - 1.3.1 Quantity (One AY), Bumper Assembly, Located in 1.2.1
 - 1.3.1.1 Quantity (2 EA), Re-Enforcement Bar of Lower Forward Bumper
 - 1.3.1.2 Quantity (12 EA), End Cap

2. REFERENCES:

- 2.1 Standard Items, (FY26)
- 2.2 1600916-10, Bumper System Arrangement and Details
- 2.3 1600901-1 Rev H, General Arrangement & Outboard Profile
- 2.4 1600910-2 Rev D, Bulwark Details
- 2.5 1600914-1 Rev D, Bow ASSY Hull
- 2.6 F8005-02-002, Aft ASSY Hull
- 2.7 F8005-02-008, Side ASSY Hull
- 2.8 0900-LP-060-4010 Chg-2, Fabrication, Welding and Inspection of Metal Boat and Craft Hulls
- 2.9 1600901-4 Rev F, Paint Schedule and Markings

3. REQUIREMENTS:

(V) "REPORT"

- 3.1 Submit one legible copy of Marine Chemist certificate to SUPERVISOR including certified location.
- 3.2 Remove each existing, template, cut and install each new bumper assembly listed in 1.3.1 with each new fastener, re-enforcement bar and end-cap where missing or damaged, using 2.2 through 2.7 for guidance.
 - 3.2.1 Template from existing and fabricate each new re-enforcement bar listed in 1.3.1.1 out of each new 3/16 inch thick aluminum sheet metal (Alloy 5086).
 - 3.2.2 Template from existing and fabricate each new end cap listed in 1.3.1.2 out of each new 3/16 inch thick aluminum sheet metal (Alloy 5086) and aluminum beam (T-Shape, 2-1/2 inch by 2-1/2 inch, 3/8 inch thick, Alloy 5083).
 - 3.2.3 Clean each exposed part/fastener/thread free of foreign matter, leaving no residue or injurious effects.

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- 3.2.4 Chase and tap each exposed threaded surface of equipment.
- 3.2.5 Straighten and align each existing flange of bumper channel.
- 3.2.6 Drill each fastening hole to each new bumper assembly.
- 3.2.7 Weld each end cap for the fabrication and installation.

(I) "VISUAL"

3.2.8 Accomplish the requirements of 2.8 including Sections 6 through 8 for welding and visual (VT) inspection using 5X magnifying glass.

3.3 Accomplish the requirements of 009-032 of 2.1 for cleaning and preservation of each new and disturbed surface to match surrounding areas, using 2.9 and manufacturer's instruction for guidance.

(V) (G) "VERIFICATION"

3.4 Inspect and verify completion of this work. -S130

3.5 Transport each material listed in 5.1 as excess to YOKOSE via SHIPSUPT.

4. NOTES:

4.1 Initial Condition - ALL BUMPERS ON THE LARC NEED REPLACEMENT DUE TO CRACKING AND CHIPPING.

4.2 SRF Tool room will issue torque wrenches when required.

4.3 GFM listed in 5.1 have been ordered under ICN 38NBA11007-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	3 EA	BUMPER, V06-001-100, BOW BUMPER, SUPERSEDED F/N.	1	2.2	9B814000 0699027	3.2
2.	3 EA	BUMPER, V06-001-106, BOW BUMPER, (ALT) SUPERSEDED	1	2.2	9B814000 9245530	3.2
3.	2 EA	BUMPER, V06-001-101, CENTER BUMPER, SUPERSEDED F/N	3	2.2	9B534001 6324514	3.2
4.	4 EA	BUMPER, V06-001-107, CENTER BUMPER, (ALT) SUPERSED	3	2.2	9B814000 9245316	3.2
5.	4 EA	BUMPER, V06-001-103, SIDE AFT BUMPER RUBBER, SUPER	4	2.2	9B534001 6324771	3.2
6.	2 EA	BUMPER, V06-001-	5	2.2	9B534001	3.2

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		104, CORNER UPPER BUMPER, SUPERSED			6324643	
7.	1 EA	BUMPER, V06-001- 105, STERN BUMPER, SUPERSEDED F/N.	6	2.2	9B534001 6324521	3.2
8.	4 EA	BUMPER, V06-001- 108, STERN BUMPER, (ALT) SUPERSEDE	6	2.2	9B814000 9244645	3.2
9.	2 EA	BUMPER, V06-001- 102, CORNER LOWER BUMPER, SUPERSED	7	2.2	9B534001 6324639	3.2
10.	10 PG	SCREW,CAP,HEXAGO N H, HEXAGON HEAD SCREW, 3/8- 16UNC	8	2.2	9B530501 5172222	3.2
11.	50 EA	SCREW,CAP,HEXAGO N H, HEXAGON HEAD SCREW, 3/8- 16UNC	9	2.2	9B530501 5263037	3.2
12.	2 HD	WASHER,FLAT, WASHER, FLAT FOR 3/8", P/N. 92141A031	10	2.2	9B531001 4759305	3.2
13.	8 PG	NUT,SELF- LOCKING,HE, SELF-LKG NUT, 3/8-16UNC-3B, P	11	2.2	9B531001 4759244	3.2
14.	1 SH	SHEET,METAL, SHEET METAL, AL 5086, 0.1875"THK X 5	-		9B953501 0122427	
15.	4 SH	SHEET,METAL, SHEET METAL, AL 5086 H-32, 0.1875"THK	13	2.2	9B953500 8660294	3.2
16.	3 CN	ROD,WELD, ROD, WELDING, AL 5356, 1/16"DIA X 36"LG,	-		9B343901 2435257	3.2
17.	2 CO	ROD,WELD, ROD, WELDING, AL 5356, 3/32"DIA X 36"LG,	-		9B343901 0123285	3.2
18.	2 CO	ROD,WELD, ROD, WELDING, AL 5356, 1/8"DIA X 36"LG,	-		9B343900 0635200	3.2
19.	1 EA	BEAM, T-SHAPR, AL, BEAM, T- SHAPE, 2-2/12" X	14	2.2	9B9540SA 2600626	3.2

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	2-1/2"				
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5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

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BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>OOKUBO, TAKASHI</u> <u>KAWASAKI, YUZO</u> <u>NAGAMINE, YUUICHI</u> <u>TERASHITA, SHUUJI</u>

1. SCOPE:

1.1 Title: ENG System Component; replace and maintain

1.2 Location of Work:

1.2.1 Operator's Cabin

1.2.2 Cargo Deck

1.2.3 Midships Compartment

1.2.4 Engine Compartment

1.2.5 AFT Deck

1.3 Identification:

1.3.1 Quantity (One EA), Diesel Engine located in 1.2.4, MFR John Deere Inc, Model 6081AFM75 Marine

1.3.1.1 Quantity (One EA), Alternator of 1.3.1, Parts No. RE65414 (Option Code 3111), Figure 1G9-6 of 2.6

1.3.1.2 Quantity (One EA), Starter of 1.3.1, Parts No. RE533571 (Option Code 3022), Refer to MFR's web site

1.3.1.3 Quantity (One EA), Element of Fuel Water Separator of 1.3.1, Parts No. RE503676 (Option Code 3545), Figure 1G16-42 of 2.6

1.3.1.4 Quantity (One EA), Element of Fuel Filter of 1.3.1, Parts No. RE506428 (Option Code 3545), Figure 1G16-23 of 2.6

1.3.1.5 Quantity (One EA), Drive Belt of 1.3.1, Parts No. R133694 (Option Code 2452), Figure 1E12-1 of 2.6

1.3.1.6 Quantity (One EA), Air Filter of 1.3.1, Parts No. RD-1460, Find No.23 of 2.7

1.3.2 Quantity (One EA), Air Intake Canister and Air Inlet Hood Assembly (ASSY) located in 1.2.2, Parts No. PDI-10012, Find No.01 of 2.7 and 2.8, Existing in Missing

1.3.2.1 Quantity (One EA), Element of 1.3.2, Parts No. AF1743M, Find No.02 of 2.7

1.3.3 Quantity (One EA), Engine Instrument and Gauge Panel with Diagnostic Display Unit, Tachometer, Alarm, Oil Pressure Gauge, Coolant Temperature Gauge, Voltmeter Gauge and Percent Load Gauge (except Ignition Panel) located in 1.2.1, P/N RE555597, MFR John Deere Inc, See ENCLOSURE (A)

1.3.4 Quantity (One EA), Ignition Panel with Ignition Switch, Engine Start Push Button and Dimmer Switch located in 1.2.1

1.3.4.1 Quantity (One EA), Ignition Key-On Switch, P/N 30G264, P/N 3SU1002-2BF10-0AA0, Key-On Holder P/N 3SU15000AA100AA0, Key-On Module P/N 3SU14001AA101FA0, Find No. 140 through 143 of 2.10, MFR - Siemens Inc

1.3.4.2 Quantity (One EA), Engine Start Push Button, P/N 3916302, Find No. 113 of 2.9, MFR- Cummins Inc

1.3.5 Quantity (One EA), 4 Inch OD and 5 Inch OD Exhaust Piping with Silencer and Bellows located in 1.2.5, Each Mechanical Joint, CRES, See ENCLOSURE (B)

1.3.6 Quantity (One AY), Engine Coolant System located in 1.2.3 and 1.2.4, Medium Antifreeze Mixture, A-A-52624A, Total Capacity Approximately 27 Gallon, Contained in each engine, radiator, heat exchanger, and hose

1.3.7 Quantity (2 EA), Port / STBD Fuel Tank located in 1.2.3, Capacity 72 Gallon

1.3.7.1 Quantity (2 EA), Tank Level Sending Unit of 1.3.7, Parts No. CGF-36, Find No.86 of 2.10

2. REFERENCES:

2.1 Standard Items, (FY26)

2.2 F9YC0-AB-TMM-010 Rev 5, Change U, Systems Technical Manual for LARC-V A1/A2, Organizational and Intermediate Maintenance

2.3 F9YC0-AL-IPB-010, Systems Technical Manual for LARC-V A1/A2, Illustrated Parts Breakdown

2.4 F9YC0-AC-TMM-010, John Deere Marine Engine KEC6081AFM75MPK24, Commercial Off the Shelf Operators Manual

2.5 F9YC0-AC-TMM-020, John Deere Marine Engine KEC6081AFM75MPK24, Commercial Off the Shelf Engine Service Manual

2.6 F9YC0-AC-TMM-030, John Deere Marine Engine KEC6081AFM75MPK24, Commercial Off the Shelf Parts Manual

2.7 1600912-2 Rev E, Engine Intake System

2.8 PDI-10012 Rev B, Air Intake Canister

2.9 1600909-1 Rev AB, Operator's Console & Controls

2.10 1600907-1 Rev AA, Electrical System Installation & Details

2.11 1600907-3 Rev M, Electronic Transmission Control Schematic Diagram

2.12 1600907-4, LARC V Vehicle Electrical Junction Box

2.13 1600912-4 Rev A, Engine Exhaust System LARC-V

2.14 1600906-1 Rev M, Cooling System INST & DETS

2.15 1600912-7, Fuel Tank Assembly & Details

2.16 1600901-4 Rev F, Paint Schedule and Markings

2.17 Test Procedure Test No.58331-4-100_26126, ENGINE SYSTEM COMPONENT

3. REQUIREMENTS:

3.1 Remove each existing and install new engine component listed in 1.3.1.1 through 1.3.1.6 installing each new gasket and packing, using Paragraphs 6.2.14(alternator and drive belt), 6.2.15(starter), 6.2.16(fuel water separator element), 6.2.17(fuel filter element) of 2.2, and 2.4 through 2.7 for guidance.

3.1.1 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.

3.1.2 Inspect each part not specified for renewal for wear and defect.

3.1.3 Remove each existing and install new component in place of those identified to be missing or defective.

3.1.4 Remove each high spot, burr, abrasion, nick, corrosion, gasket material, and foreign matter from each flange and mating surface exposed by disassembly and removals.

3.1.5 Remove each burr and high spot from each sliding surface, screw thread, key, and keyway exposed by disassembly and removals.

3.1.6 Dress and true each metal-to-metal mating surface and gasket mating surface.

3.1.7 Tension the drive belt listed in 1.3.1.5 in accordance with Page 35-5 Step 4 of 2.4.

3.2 Install new air intake canister and air inlet hood ASSY listed in 1.3.2 installing each new element listed in 1.3.2.1, gasket, and packing, using Paragraph 6.2.12 of 2.2 and 2.7 for guidance.

3.2.1 Fabricate each new air intake canister from 16 GA (0.0625 inches) CRES 304 sheeting and assemble fabricated air intake canister and new air inlet hood, using 2.8 for guidance.

3.3 Remove each existing and install new component listed in 1.3.3, 1.3.4.1, and 1.3.4.2, using Paragraphs 6.2.7(engine panel gauge), 6.2.8(engine ignition switch), and 6.2.9(engine start pushbutton) of 2.2, 2.4 through 2.6, 2.9 through 2.12, and ENCLOSURE (A) for guidance.

3.3.1 Matchmark, identify and retain disconnected hardware.

3.3.2 Install retained hardware of 3.3.1.

3.4 Remove each existing and install new exhaust piping listed in 1.3.5, using 2.2, 2.3, and 2.13 for guidance.

3.4.1 Accomplish the requirements of 009-011 of 2.1 for reusable pad on exhaust piping.

3.5 Remove each existing and fill new coolant of engine coolant system listed in 1.3.6 in accordance with Paragraph 6.2.10 of 2.2, using Paragraph 3.7.4 of 2.2, 2.4, 2.5, and 2.14 for guidance.

(V) (G) "VISUAL INSPECTION"

3.5.1 Prior to filling, inspect and verify no residue of the existing coolant in the engine coolant system. -S130

3.6 Remove each existing and fill new fuel oil of each fuel tank listed in 1.3.7 in accordance with Paragraph 6.2.18 of 2.2, using Figure 3-3 of

2.2 and 2.15 for guidance.

(V) (G) "VISUAL INSPECTION"

3.6.1 Prior to filling, inspect and verify no condensation, sediment, or contaminated fuel oil in each fuel tank. -S130

(V) (G) "CALIBRATION"

3.6.2 Calibrate each tank level sending unit listed in 1.3.7.1 in accordance with Paragraph 6.2.3 of 2.2. -S130

3.7 Accomplish the requirements of 009-032 of 2.1 for each new and disturbed surface, using 2.16 for guidance.

(V) (G) "OPERATIONAL TEST"

3.8 Accomplish the requirements of 2.17 (TP) for an operational test of each component listed in 1.3.1 through 1.3.7.1. -S246

4. NOTES:

4.1 Initial Condition - S/F DISCOVERED ENGINE SYSTEM COMPONENT DAMAGE ON LARC 936 DUE TO CORROSION TO ENGINE COMPONENTS TO INCLUDE: FUEL TANK SENDING UNIT CALIBRATION, ENGINE PANEL GAUGE, ENGINE IGNITION SWITCH, ENGINE START PUSHBUTTON, COOLANT DRAIN/FILL, EXHAUST PIPING, ENGINE ALTERNATOR, ENGINE STARTER, ENGINE FUEL WATER SEPARATOR, FUEL FILTER, FUEL TANK STRIPPING, ENGINE AIR FILTER, ENGINE DRIVE BELT.

4.2 SRF Tool Room will issue each torque wrench when required.

4.3 Ship's Force will provide fluids for operation such as coolant, hydraulic oil, and fuel oil when refilling.

4.4 Fabrication and installation of air intake canister listed in 1.3.2 were directed by S245 under ESR MISC-167 response.

4.5 The alternator listed in 1.3.1.1 is 7H material advice code 5S. The carcass item must be turned in the warehouse building No.487 when remove the existing one.

4.6 GFM listed in 5.1 have been ordered under ICN 38NBA23303-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	1 EA	PULLEY,GROOVE, R114844	1G9-5	2.6	9B302001 5456374	3.1
2.	1 EA	GENERATOR,ALTERN ATI, RE65414	1G9-6	2.6	7H611501 5453694	3.1
3.	1 EA	GASKET, R53108	1G3-1	2.6	9B533001 3258987	3.1
4.	1 EA	STARTER,ENGINE,E LEC, RE43300 (RE38632)	1G3- 10 (APL)	2.6	9B292001 2975199	3.1
5.	1 EA	STARTER MOTOR, P/N; RE533571 OR	MFR's web site	2.6	9B2920SA 2301387	3.1

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		RE501150 (OF CODE 3				
6.	2 EA	PACKING, PREFORMED, 1G16-18&22	1G16-18&22	2.6	9B533001 5682337	3.1
7.	2 EA	PACKING, PREFORMED, P/N; R501632, FIG.19 OF PAGE 1G	1G16-19	2.6	9B5330SA 2300184	3.1
8.	1 EA	FILTER, FLUID, RE506428	1G16-23	2.6	9B433001 5464645	3.1
9.	1 EA	FILTER, FLUID, RE503676	1G16-42	2.6	9B433001 5464648	3.1
10.	1 EA	BELT, V, R133694	1E12-1	2.6	9B303001 5457090	3.1
11.	1 EA	FILTER ELEMENT, INTA, RD-1460	23	2.7	9B294001 6193831	3.1
12.	1 EA	FILTER ELEMENT, INTA, AF1743M	2	2.7	9B294001 5454214	3.2
13.	2 EA	MOUNTING BAND, P/N; 3918233S, ITEM NO.3 OF DWG 16	3	2.7	9B5310SA 2300185	3.2
14.	1 KT	FASTENER KIT, P/N; Q01368, ITEM NO.13 OF DWG 16009	13	2.7	9B5310SA 2300186	3.2
15.	1 EA	GASKET, P/N; 8525T33, NEOPRENE, 12"X12"X1/8" THK	16	2.7	9B5310SA 2300187	3.2
16.	8 EA	BOLT, MACHINE, 1/4" -20 UNC X 1-1/2" LG SST	17	2.7	9B530601 4759900	3.2
17.	8 EA	NUT, SELF-LOCKING, HE, 1/4" -20 UNC, SST	18	2.7	9B531000 2416604	3.2
18.	1 EA	WIRING HARNESS, BRAN, P/N RE5225692D9 KEY 26	-	2.6	9B615001 5461120	3.3
19.	1 EA	ENGINE CONTROLLER, P/N RE5225152D9 KEY 31	-	2.6	3B292001 5453685	3.3
20.	1 EA	SWITCH, REED, P/N 3916302 ITEM NO.113 STARTER ENGINE	113	2.9	9B593001 4856573	3.3
21.	1 EA	KEY ON SWITCH, P/N 3SU1002-2BF10-0AA0 ITEM NO.110	141	2.1	9B5930SA 2300189	3.3

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22.	1 EA	RETAINER, ELECTRICAL, P/N 3SU15000AA100AA0 ITEM NO.1	142	2.1	9B593001 7053344	3.3
23.	1 EA	CONTACT, KEY-ON MODULE ELECTRIC CONTACT P/N 3SU140	143	2.1	9B5930SA 2300191	3.3
24.	2 OZ	CHEMICAL CORROSION, P/N TEF-GEL	132	2.1	9B685001 4504009	3.3
25.	1 PG	SCREW, CAP, HEXAGON H, P/N 93190A5421/4 - 20, 1" L, C	-		9B530501 5286392	3.3
26.	1 PG	SCREW, CAP, HEXAGON H, P/N 93190A5461/4-20, 1.5" L,	-		9B530501 5294413	3.3
27.	10 EA	LOCK NUT, P/N 90715A1241/4"- 20UNC 316 SST	-		9B531001 5045702	3.3
28.	1 TU	INSULATING COMPOUND, 3140 RTVITEM 133	133	2.1	9B597001 4871943	3.3
29.	1 SL	WIRE, P/N 184003, 14 AWG BLACK 18FT/ SPOOL	39	2.1	9B6145SA 2300192	3.3
30.	1 SL	WIRE, P/N 117010, 2/0 AWG BLACK 100FT/ SPOOL	36	2.1	9B6125SA 2300193	3.3
31.	1 EA	DISPLAY, MICRO DISPLAY AND MOUNTING NUT KIT P/N DZ	EN (A)		9B5925SA 2300769	3.3
32.	1 EA	HARNESS, CONVERSION HARNESS P/N DZ101719	EN (A)		9B5925SA 2300771	3.3
33.	1 EA	NUT, INSTALLATION NUT FOR GAGE DZ119769P/N DZ10265	EN (A)		9B5925SA 2300770	3.3
34.	1 EA	GAUGE, PRESSURE GAUGE P/N RE521036	EN (A)		9B5925SA 2300772	3.3
35.	1 EA	GAUGE, TEMPERATURE GAUGE P/N RE521037	EN (A)		9B5925SA 2300773	3.3
36.	1 EA	METER, VOLT	EN (A)		9B5925SA	3.3

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		METERP/N RE521038			2300774	
37.	1 EA	METER, LOAD METERP/N RE521039	EN (A)		9B5925SA 2300775	3.3
38.	1 EA	METER, TACHOMETERP/N RE521040	EN (A)		9B5925SA 2300776	3.3
39.	1 EA	ALARM, AUDIBLE WARNING ALARMP/N RE521041	EN (A)		9B5925SA 2300777	3.3
40.	1 EA	SWITCH, IGNITION SWITCH P/N RE521819ITEM NO13 OF P	EN (A)		9B5925SA 2501313	3.3
41.	1 EA	SWITCH, PUSH BUTTON START SWITCHP/N RE507239ITEM N	EN (A)		9B5925SA 2501314	3.3
42.	1 EA	BOOT, INSTRUMENT PANEL, BLACK DUST BOOTP/N R504168	EN (A)		9B5925SA 2501315	3.3
43.	1 EA	SWITCH, DIMMER POTENTIOMETER SWITCHP/N RE521820ITE	EN (A)		9B5925SA 2501316	3.3
44.	1 EA	SENSOR, POTENTIOMETER ENGINE SPEED SENSORP/N AT312	EN (A)		9B5925SA 2501317	3.3
45.	1 EA	DIMMER SWITCH, P/N RE162449FIND NO 114 OF DWG 1600	114	2.9	9B5925SA 2600806	3.3
46.	1 EA	KEY-ON SWITCH, P/N 30G264FIND NO. 109 OF DWG 16009	109	2.9	9B5925SA 2600807	3.3
47.	1 EA	POTENTIOMETER, SINGLE TURN /LINER , 1K OHMS, CARBO	108	2.9	9B5925SA 2501867	3.3
48.	1 EA	KNOB, RIBBED WITH SKIRT INDICATOR LINE 0.80 INCH D	107	2.9	9B5925SA 2600808	3.3
49.	1 EA	SILENCER, SILENCER, 6", 300 SERIES SST, FIG&IDX# 1	1-5-1	2.3	9C4730SA PTE0070	3.4
50.	1 EA	EXHAUST BELLOWS ASSY, 5"BELLOWS PORTION OF DWG:	1-5-2	2.3	9C4730SA PTE0072	3.4

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51.	1 EA	EXHAUST ASSY, ELBOW PORTION OF 28638, 90 DEG ELBOW	1-5-2	2.3	9C4730SA PTE0073	3.4
52.	2 EA	CLAMP, 5" EASY SEAL STAINLESS STEEL BAND CLAMP TEC	1-5-3	2.3	9C4730SA PTE0074	3.4
53.	2 EA	GASKET, GASKET, JOHN DEERE P/N: T151881	1-5-4	2.3	9B533001 5461431	3.4
54.	2 EA	GASKET, 5" CUMMINS FLANGE NUT BILT GASKET KIT, P/N	1-5-5	2.3	9B5330SA PTE0076	3.4
55.	2 EA	GASKET, 5" CUMMINS FLANGE GASKET, P/N: 48CUM500	1-5-5	2.3	9B5330SA PTE0077	3.4
56.	1 EA	LAGGING, BLANKET FOR GA304717 AND GA304718 ASSEMBL	1-5-10	2.3	9Q5640SA PTE0078	3.4
57.	20 EA	BOLT,MACHINE, BOLT, HEX, 3/4"- 10UNC-2A X 1- 1/2"LG	1-5-8	2.3	9B530600 2270939	3.4
58.	20 EA	NUT,SELF, NUT, SELF LOCKING, 3/4"-10UNC, CRES 316	1-5-9	2.3	9B531001 5265282	3.4
59.	20 EA	WASHER,FLAT, WASHER, FLAT, 3/4", CRES316	1-5-15	2.3	9B531001 5400898	3.4
60.	4 PG	SCREW,CAP,HEXAGO N H, BOLT, 3/8"- 16UNC X 1- 1/2"LG,	1-5-6	2.3	9B530501 5260747	3.4
61.	20 EA	NUT,SELF- LOCKING,HE, NUT, SELF LOCKING, 3/8-16UNC.	1-5-7	2.3	9B531001 3863517	3.4
62.	1 EA	CLAMP,LOOP, JOHN DEERE CLAMP, P/N: RE47537	-		9B534001 5460668	3.4
63.	4 EA	PLUG, JOHN DEERE PIPE PLUG, P/N: L2646N	-		9B4730SA PTE0083	3.4
64.	1 EA	ELBOW, JOHN DEERE ELBOW FITTING, P/N: R110108	-		9B4730SA PTE0084	3.4
65.	1 RO	INSULATI, FELT,1" X 5' X	-		9B5640SA 9607458	3.4

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		45', L/TEMP, MIL-I-16411				
66.	1 EA	WIRE FAB, WIRE, MESH, 0.008" WIRE, 30 INCH WIDE X	-		9B533500 5989404	3.4
67.	1 RO	CLOTH, PL, CLOTH, PLAIN WEAVE, GLASS CLOTH, 5 FT W X	-		9B830500 1487364	3.4
68.	1 EA	AIR INLET HOOD, AIR INLET HOOD, STACK CAP, 7"PLAST	-	2.8	9B4730SA PTE0088	3.2
69.	4 EA	COCK, DRAIN, DRAIN COCK, 1/8 PIPE MALE, BRASS	-	2.8	9B482000 6840880	3.2
70.	4 EA	MOUNTING BAND, MOUNTING BANDS, 13" STAINLESS STEEL	-	2.7	9B4730SA PTE0089	3.2
71.	20 EA	FASTENER KIT, FASTENER KIT, 5/16"X18 CAP NUT	-	2.7	9B4730SA PTE0090	3.2
72.	10 EA	HOSE, PREFORMED, RADIATOR HOSE, P/N: R502404	-		9B472001 5458551	
73.	10 EA	CLAMP, HOSE, HOSE CLAMP FOR RADIOATOR HOSE, P/N: AR	-		9B473000 2782523	

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

ESR RESPONSE ON ESR MISC-289

This email serves as S245 official email response to ESR MISC-289.

BLUF: Procure each component part of RE555597 per Attachments (1) and (3), and install per references (d) and (e).

ESR Title: Engine Gauge Panel for LARC 936; procure

REF:

- (a) ESR MISC-289, Engine Gauge Panel for LARC 936; procure
- (b) DWG 1600907-1 Rev W, LARC V Vehicle Electrical System Installation & Details
- (c) DWG 1600909-1 Rev R, LARC V Vehicle Operator's Console & Controls
- (d) DWG 1600907-1 Rev AA, LARC V Vehicle Electrical System Installation & Details
- (e) DWG 1600909-1 Rev V, LARC V Vehicle Operator's Console & Controls

Attachment:

- (1) 97HA Field Installed Instrumentation Panel Kit – ST724771
- (2) Email correspondence - NAVFAC EXWC Code EX262 and SRF-JRMC Det Sasebo Design Branch Code S245
- (3) LAR No. LV23001 (SRF# REPAIR/LARCV/343), Replacement Part on Engine Gauge Panel

Background:

1. Reference (a) requested SRF-JRMC Det Sasebo Design Branch Code S245 for the replacement material information for the discontinued engine gauge panel, P/N RE521033 (John Deere P/N 32-35-0001) which is listed as Find No. 12 in reference (b), installed on LARC 936.
2. Design Branch Code S245 sent a LAR to NAVFAC EXWC to request the replacement material information for the discontinued RE521033.
3. Per information from John Deere's Japanese distributor, RE521033 has been replaced by RE555597. RE555597 has been discontinued as an instrument panel assembly, but each component of RE555597 listed in Attachment (1) is available to order.
4. Design Branch Code S245 sent the information that RE555597 can be used as replacement of RE521033 to NAVFAC EXWC as additional information of LAR. NAVFAC EXWC responded that RE555597 is not a direct replacement for the existing RE521033 and will require craft modification to outfit the new parts. See Attachment (2) for details.
5. NAVFAC EXWC revised references (b) and (c) via Attachment (3) to install RE555597 based on the correspondence of Attachment (2). Revised drawings are references (d) and (e) and have been uploaded to EDCTM.

Designer's Comments:

1. Procure each component part of RE555597 per Attachments (1) and (3), and install per references (d) and (e). Refer to Attachment (1) for individual components of RE555597, such as oil pressure gauge, coolant temperature gauge, voltmeter, percent load gauge, tachometer, and audible alarm.

ENCLOSURE (A) 1/4

SHIP:

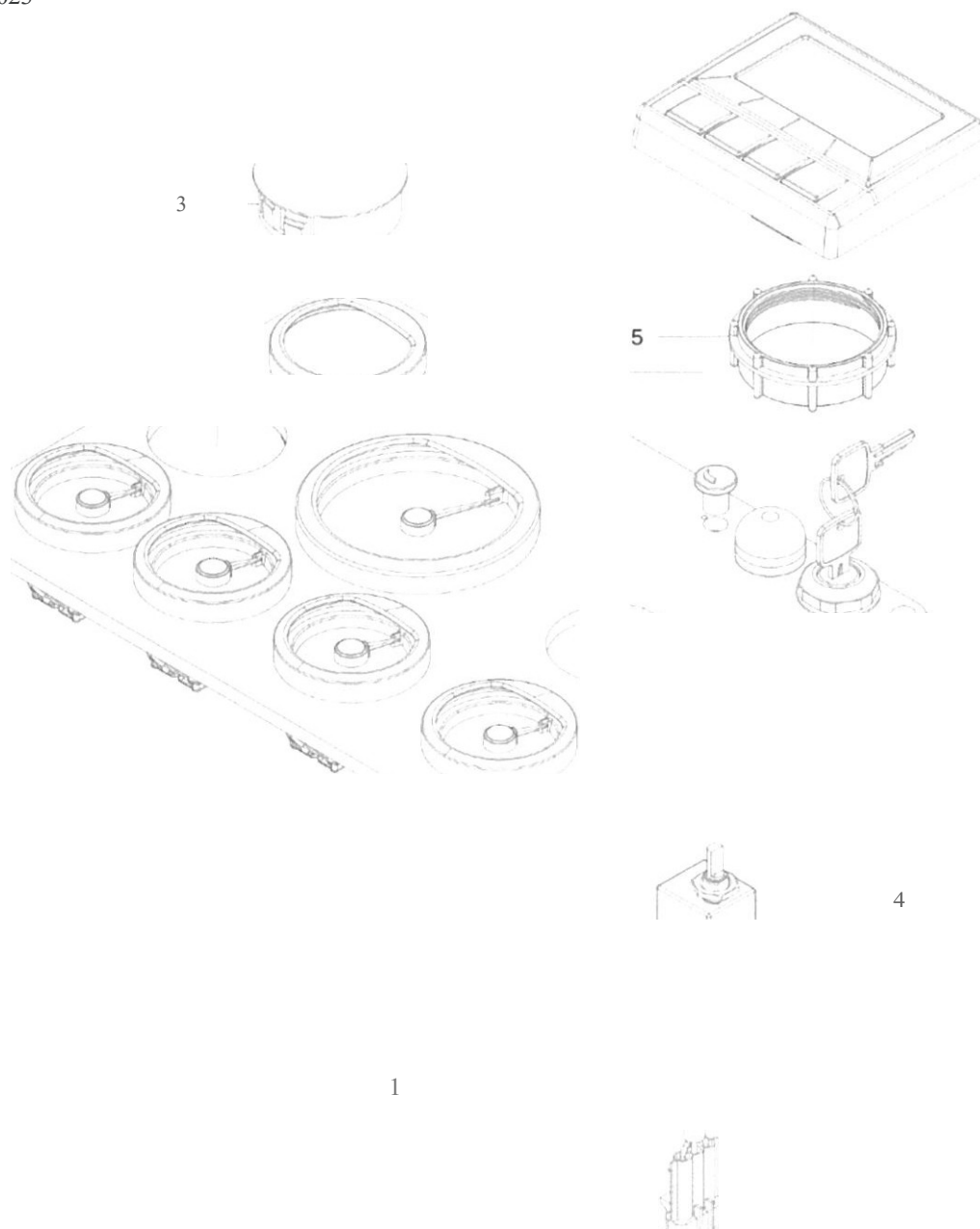
NBU-7 (RMC 57078)

12.5 L - ENGINE, POWERTECH

PC9769

97HA Field Installed Instrumentation Panel Kit - ST724771

Apr 11 2023



(ATTACHMENT(1))

ENCLOSURE (A) 2/4

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SHIP:

NBU-7 (RMC 57078)**12.5 L - ENGINE, POWERTECH**

PC9769

97HA Field Installed Instrumentation Panel Kit - S1724771

Apr 11 2023

KEY	PART NO.	PART NAME	QTY	REMARKS
1	RE555597	Instrument Panel	1	12 V / 24 V, Wheel House INCLUDES RE555553, R130486, AT340550, AT312606, R518539, RE521036, RE521037, RE521038, RE521039, RE521040, RE521041, RE521819, RE507239, R504168 AND RE521820, SUB FOR RES46692 OR RE521033
2	DZ119769	Gauge	1	Premium Diagnostic Module No Logo SUB FOR RE555553
3	R130486	Plug	1	Special
4	AT340550	Potentiometer	1	Engine Speed Gold
5	DZ102651	Nut	1	
6	R518539	Faceplate	1	Not Illustrated Wheel House
7	RE521036	Pressure Gauge	1	Not Illustrated Oil Pressure Gauge
8	RE521037	Temperature Gauge	1	Not Illustrated Coolant Temperature Gauge
9	RE521038	Voltmeter	1	Not Illustrated
10	RE521039	Meter	1	Not Illustrated Load Meter
11	RE521040	Tachometer	1	Not Illustrated
12	RE521041	Audible Warning Alarm	1	Not Illustrated Siren Alarm
13	RE521819	Switch	1	Not Illustrated Ignition
14	RE507239	Switch	1	Not Illustrated Push Button Start
15	R504168	Boot	1	Not Illustrated Black Dust

(ATTACHMENT(1))

ENCLOSURE (A) 3/4

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KEY	PART NO.	PART NAME	QTY	REMARKS
16	RES21820	Switch	1	Not Illustrated Dimmer, Potentiometer
17	AT312606	Sensor	1	Not Illustrated Potentiometer Engine Speed Sensor

(ATTACHMENT(1))

ENCLOSURE (A) 4/4

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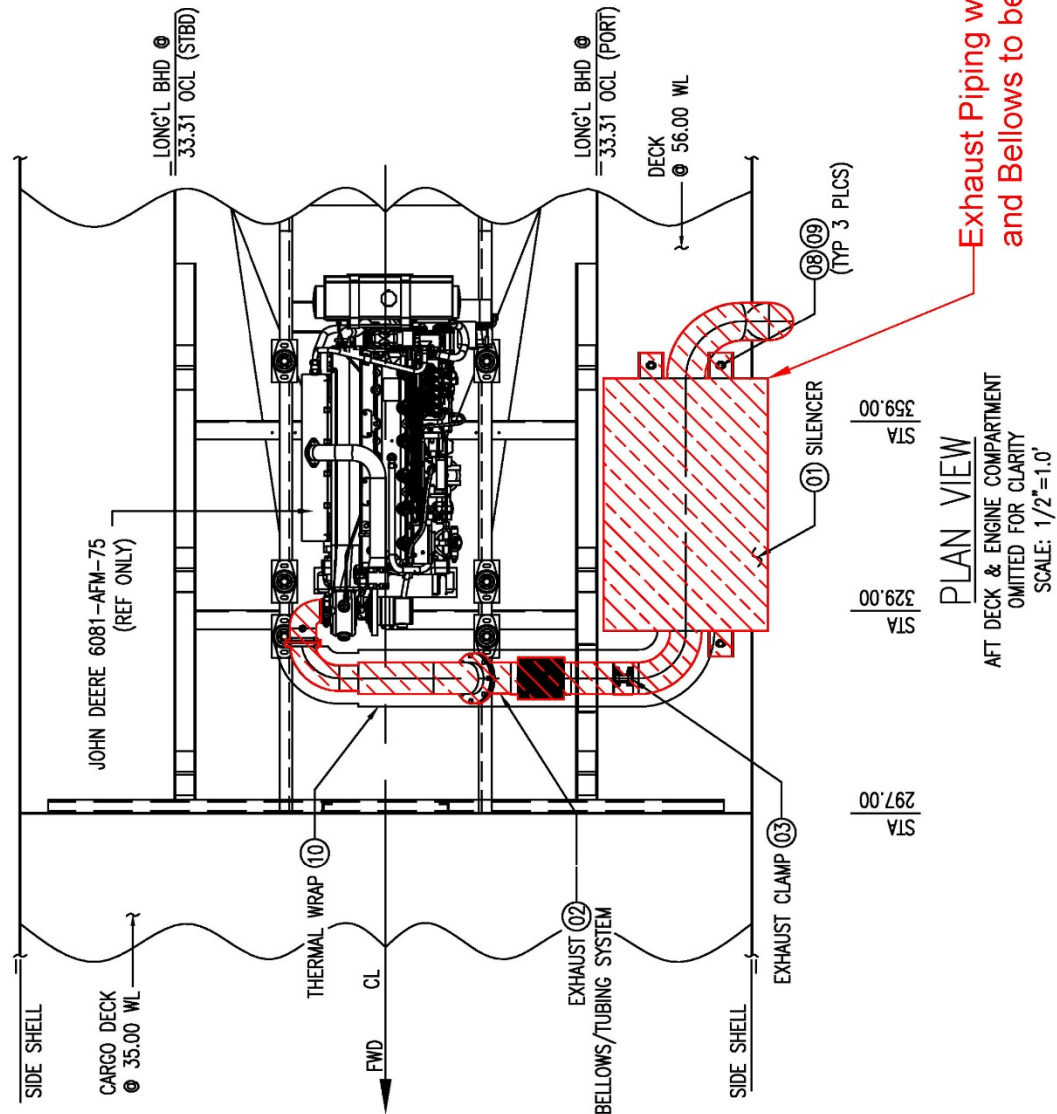
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ENCLOSURE (B) 1/1

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SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>243-11-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L012</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>OOKUBO, TAKASHI</u>

1. SCOPE:

1.1 Title: Propulsion Shafting and Rudder Component; replace

1.2 Location of Work:

1.2.1 Midships Compartment

1.2.2 Under Hull

1.2.3 Engine Compartment

1.2.4 Throughout the Craft

1.3 Identification:

1.3.1 Quantity (One EA), Universal Joint located in 1.2.1

Parts No. - 5-7202X

Figure - 30 of 2.4

1.3.2 Quantity (One EA), Propeller Shaft located in 1.2.2

Parts No. - 1600905-3

Material - AQUAMET 17 (Precipitation-Hardening, Martensitic
Stainless Steel)

Figure - 16 of 2.5

1.3.3 Quantity (One EA), Propeller located in 1.2.2

Parts No. - PDI-10040

Material - Ni-Br-Al Alloy

Figure - 20 of 2.5

1.3.4 Quantity (One EA), Rudder Tiller Arm located in 1.2.3

Parts No. - 1600904-9/2 V08-008-006

Figure - 1-26-2 / 1-27-3 of 2.3, 02 of 2.6

1.3.5 Quantity (One EA), Rudder located in 1.2.2

Parts No. - EL08-008-67

Figure - 1-27-15 of 2.3

1.3.6 Quantity (2 EA), Rudder Cable located in 1.2.4

Parts No. - 005-8722-282

Figure - 1-26-1 of 2.3, 01 of 2.6

2. REFERENCES:

2.1 Standard Items, (FY26)

2.2 F9YC0-AB-TMM-010 Rev 5, Change U, Systems Technical Manual for
LARC-V A1/A2, Organizational and Intermediate Maintenance

- 2.3 F9YC0-AL-IPB-010, Systems Technical Manual for LARC-V A1/A2, Illustrated Parts Breakdown
- 2.4 1600905-1 Rev G, Sea Drive Motor Installation and Foundation
- 2.5 1600905-2, Propeller and Propeller Shaft Assembly
- 2.6 1600904-9 Rev F, Rudder Steering System
- 2.7 Maintenance Requirement Card, MIP Control Number: 59A32/B02, MRC: 45 2UG6 N, Lubricate LARC V (Enclosed)
- 2.8 Test Procedure Test No.58331-4-100_26125, Propulsion Shafting & Rudder Component

3. REQUIREMENTS:

3.1 Remove each existing and install each new propeller component listed in 1.3.1 through 1.3.3 installing each new seal, bearing, gasket, and packing, using Paragraphs 6.6.1 and 6.6.2 of 2.2, 2.3 through 2.5 for guidance.

3.1.1 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.

3.1.2 Inspect each part not specified for renewal for wear and defect.

3.1.3 Remove each existing and install new component in place of those identified to be missing or defective.

3.1.4 Remove each high spot, burr, abrasion, nick, corrosion, gasket material, and foreign matter from each flange and mating surface exposed by disassembly and removals.

3.1.5 Remove each burr and high spot from each sliding surface, screw thread, key, and keyway exposed by disassembly and removals.

3.1.6 Handwork and skim cut each machined, sealing, aligning, mating, and gasket surface, taking precautions to ensure minimum metal removal.

3.1.7 Dress and true each metal-to-metal mating surface and gasket mating surface.

3.1.8 Measure and record each final size and clearance of each wearing part and fit area for propeller shaft listed in 1.3.2.

(V) "REPORT"

3.1.8.1 Submit one legible copy of a report listing results of the requirements of 3.1.8 to SUPERVISOR.

3.1.9 Apply new sealant to each threaded hardware in accordance with Note One of 2.4 and 2.5.

3.1.10 Apply new sealant to each gasket (No.09 and 29 of 2.5) in accordance with Note 3 of 2.5.

3.1.11 Apply new grease conforming to A-A-50433A to each propeller shaft lubrication point, propeller bearing housing, and shaft universal joint, using 2.7 for guidance.

3.1.12 Tighten each fastener at each torque value stated in

Paragraphs 6.6.1 and 6.6.2 of 2.2.

3.2 Remove each existing and install new rudder component listed in 1.3.4 through 1.3.6 installing each new seal, bearing, gasket, and packing, using Paragraphs 6.6.3 through 6.6.5 of 2.2, 2.3 and 2.6 for guidance.

3.2.1 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.

3.2.2 Inspect each part not specified for renewal for wear and defect.

3.2.3 Remove each existing and install new component in place of those identified to be missing or defective.

3.2.4 Remove each high spot, burr, abrasion, nick, corrosion, gasket material, and foreign matter from each flange and mating surface exposed by disassembly and removals.

3.2.5 Remove each burr and high spot from each sliding surface, screw thread, key, and keyway exposed by disassembly and removals.

3.2.6 Handwork and skim cut each machined, sealing, aligning, mating, and gasket surface, taking precautions to ensure minimum metal removal.

3.2.7 Measure and record each final size and clearance of each wearing part and fit area for rudder listed in 1.3.5.

(V) "REPORT"

3.2.7.1 Submit one legible copy of a report listing results of the requirements of 3.2.7 to SUPERVISOR.

3.2.8 Apply new seal compound to each threaded hardware in accordance with Note One of 2.6.

3.2.9 Apply new grease conforming to A-A-50433A to each rudder post and rudder cable lubrication point, using 2.7 for guidance.

3.3 Accomplish the requirements of 009-032 of 2.1 for cleaning and preservation of each new and disturbed surface to match each surrounding area.

(V) (G) "OPERATIONAL TEST"

3.4 Accomplish the requirements of 2.8 (TP) for an operational test of each component listed in 1.3.1 through 1.3.6. -S246

4. NOTES:

4.1 Initial Condition - S/F discovered shafting, rudder and propeller damage on LARC 936 due to corrosion to primary shafting, rudder, and propeller systems to include: propeller shafting u-joints, propeller shafting and propeller, rudder linkage (rudder tiller), rudder, rudder cables.

4.2 SRF Tool Room will issue each torque wrench and dial indicator when required.

4.3 Ship's Force will provide fluids for operation such as coolant,

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hydraulic oil, and fuel oil when refilling.

4.4 GFM listed in 5.1 have been ordered under ICN 38NBA24302-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	1 EA	SCREW, CAP, HEXAGON HE, CAP SCREW, HEX HEAD, P/N: 2-	27	2.4	9B530500 8548322	3.1
2.	1 EA	LOCK PLATE, RETAINER, 3551J	28	2.4	9B252000 5301357	3.1
3.	1 EA	BOLT, MACHINE, 3562J	29	2.4	9B530600 4250003	3.1
4.	1 EA	PARTS KIT, UNIVERSAL, 5-7202X	30	2.4	9B252000 9662070	3.1
5.	1 EA	SPIDER AND BEARING, 114-5164	30FmAPL	2.4	9B201000 9302463	3.1
6.	1 EA	YOKE, SEA DRIVE, SEA DRIVE YOKE, P/N: 1600905-4, M	02	2.4	9B5310SA 2300231	3.1
7.	1 EA	SEAL, PLAIN ENCASED, 31250	4	2.5	9B533001 1131316	3.1
8.	1 EA	BEARING, ROLLER, SELF, 22310CCW33	6	2.5	9B311000 0878224	3.1
9.	1 EA	HOUSING, BEARING UNI, V09-006-010	7	2.5	9B313000 9227065	3.1
10.	1 PG	GREASE FITTING, 1/8"NPTF THREAD, P/N: 1095K15, 10 E	8	2.5	9B5310SA 2300225	3.1
11.	1 EA	SPACER, SLEEVE, V09-006-013 (906013)	10	2.5	9B536500 0812659	3.1
12.	1 EA	SEAL, PLAIN ENCASED, 00044	12	2.5	9B533000 0011995	3.1
13.	1 EA	SEAL, PLAIN ENCASED, 262-3623-24	13	2.5	9B533000 0166911	3.1
14.	1 PG	WASHER, FLAT, FLAT, 1/2", STEEL	14	2.5	9B531000 8093079	3.1
15.	4 EA	NUT, SELF-LOCKING, HE, 1/2"-13UNC, STEEL	15	2.5	9B531000 2256993	3.1
16.	1 EA	PROPELLER SHAFT, VEH,	16	2.5	9B252001 6432450	3.1

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		1600905-3				
17.	1 EA	MACHINE KEY, P/N; 98500A206, BRASS, 5/8"X5/8"X12"L	17	2.5	9B5310SA 2300226	3.1
18.	1 PG	SET SCREW, P/N; 91375A906, 3/8"- 16UNC- 5/8"LG,STEEL	18	2.5	9B5310SA 2300227	3.1
19.	1 EA	PROPELLER,MARINE , PDI-10040	20	2.5	9B201001 6516321	3.1
20.	1 EA	WASHER,KEY, V09- 006-022	21	2.5	9B531000 8086572	3.1
21.	1 EA	NUT,PLAIN,HEXAGO N, V09-006-023	22	2.5	9B531000 0614962	3.1
22.	1 EA	SPACER,SLEEVE, V09-006-025	23	2.5	9B536500 0812660	3.1
23.	1 EA	SEAL,PLAIN ENCASED, 17387	24	2.5	9B533001 0347868	3.1
24.	1 EA	SEAL,PLAIN ENCASED, 50352S20	25	2.5	9B533000 0815411	3.1
25.	1 EA	BEARING,BALL,ANN ULA, 1207	27	2.5	9B311000 5555432	3.1
26.	1 EA	RETAINING RING, P/N;98410A646, FOR 3-1/16" OD,0.09	28	2.5	9B5310SA 2300228	3.1
27.	1 EA	RETAINING RING, P/N; 98410A380, FOR 4- 3/4"ODX0.109	5	2.5	9B5310SA 2300229	3.1
28.	1 EA	GASKET, GASKET, BEARING HOUSING, P/N; V09-006- 029,	29	2.5	9B533000 0879423	3.1
29.	1 PG	HEAD CAP SCREW, P/N;90128A637, SIZE; 3/8"-16UNC X	34	2.5	9B5305SA 2400150	3.1
30.	1 EA	CAP,FAIRWATER, V09-006-034	35	2.5	9B201000 0618220	3.1
31.	1 HD	WASHER,LOCK, 1/2"	36	2.5	9B531000 5845272	3.1
32.	1 PG	SOCKET CAP SCREW, P/N;91274A456, 1/2"-13UNCX1- 3/4"	37	2.5	9B5310SA 2300230	3.1
33.	1 EA	YOKE,PROPELLER DRIVE, PART NO.1600905- 5(116-15970J	3	2.5	9B5310SA 2401099	3.1
34.	4 EA	STUD,PLAIN, V09-	11	2.5	9B530700	3.1

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		006-016			1809832	
35.	1 EA	CAP, FAIRWATER, V09-006-019	19	2.5	9B201000 0618221	3.1
36.	1 EA	GASKET, SOLID TYPE	9	2.5	9B533000 0879424	3.1
37.	2 EA	CABLE ASSEMBLY, CONT, 005-8722-282	1-26-1	2.3	9B259001 5680542	3.2
38.	1 EA	TILLER ARM, TILLER ARM FOR RUDDER, P/N; 1600904-9/2	1-27-3	2.3	9B5310SA 2300232	3.2
39.	1 EA	CLEVIS PIN, 316 STAINLESS STEEL CLEVIS PINS WITH S	1-26-3	2.3	9B5310SA 2300330	3.2
40.	1 PG	PIN, COTTER, 98355A180, 25EA/P G	1-26-7	2.3	9B531501 5520725	3.2
41.	4 EA	CLEVIS ASSY, P/N; 89003-1, FIG 1-26-8 OF T/M F9YC0-	1-26-8	2.3	9B5310SA 2300194	3.2
42.	1 EA	NUT, SELF- LOCKING, HE, 79NTE126	1-27-1	2.3	9B531001 1277600	3.2
43.	1 EA	WASHER, FLAT, MS15795-823	1-27-2	2.3	9B531001 4282151	3.2
44.	2 EA	BEARING, SLEEVE, 08DU08	1-27-4	2.3	9B312000 0380796	3.2
45.	1 EA	KEY, MACHINE, MS20066-251	1-27-5	2.3	9B531500 8099641	3.2
46.	1 EA	RING, RETAINING, MS9013-26	1-27-6	2.3	9B532500 6336585	3.2
47.	1 EA	WASHER, KEYWAY, V08-002-004	1-27-7	2.3	9B531000 9730253	3.2
48.	1 HD	PIN, SPRING, MS171536	1-27-8	2.3	9B531500 0589756	3.2
49.	1 EA	SEAL, PLAIN ENCASED, 21158- 0904	1-27-9	2.3	9B533000 5995247	3.2
50.	2 EA	BEARING, SLEEVE, MS35772-134	1-27-10	2.3	9B312000 8099640	3.2
51.	1 EA	RUDDER AND SHAFT AS, EL08-008-67	1-27-15	2.3	9B204000 0806312	3.2
52.	1 EA	RING, RETAINING, MS16624-4100	1-27-17	2.3	9B532500 5981785	3.2
53.	1 CN	GREASE, UTILITY, A-A- 50433, 35LB/CN	-	2.7	9B915001 3069167	3.1
54.	1 EA	CORROSION PREVENTIV, TEF- GEL, 16 OZ/TU	2.5	2.6	9B803001 5767112	3.1
55.	1 PM	PLATE, METAL,	For 1-27-	2.3	9B951501	

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		CRES AMS 5516(302),0.25"X 36"X10'	15		3887430	
56.	1 CN	PAINT,ACRYLIC,DT M,2, SHERWIN- WILLIAMS SHER- CRYL HI	-		9B8010LH A035587	3.3
57.	1 KT	EPOXY COATING KIT, GREEN, MIL- DTL-24441/20, 150 F	-		9B801001 3504742	3.3

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

SSP: SASN-003-26

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ITEM NO: 243-11-001

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SHIP: NBU-7 (RMC 57078)

DISTRIBUTION STATEMENT A:

Approved for public release: distribution unlimited.

Date: April 2025 **MIP** 59A3 **MRC:** 45 2UG6 N **Periodicity:** M-4R
Series:

Location:

Ship System: Special Purpose Systems 59000
System: Logistics Over The Shore (LOTS) Support 59A00
SubSystem: Logistics Over The Shore (LOTS), Personnel Transport 59A30
Equipment: Lighter, Amphibious, Resupply, Cargo V (LARC V) 59A32

Rates	Man-Hours	Rates	Man-Hours	Rates	Man-Hours
CM3	1.5	CMCN	0.7		
Total Man-Hours:	2.2	Elapsed Time:	1.5		

MAINTENANCE REQUIREMENT DESCRIPTION

1. [Lubricate LARC V.](#)

SAFETY PRECAUTIONS

1. Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.
2. To prevent personnel injury, Tag-Out procedures shall be in accordance with the Tag-out Users Manual (TUM), and/or local tag-out instruction.
3. Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.

TOOLS, PARTS, MATERIALS, TEST EQUIPMENT

MATERIALS

1. [\[000073\]](#)Grease, Termalene 2, water-wash resistant, CID A-A-50433
Hazardous Material User's Guide (HMUG) Group 13
2. [\[001144\]](#)Tag, safety
3. [\[018450\]](#)Grease, white lithium, (As required in Step [1.c.\(3\).](#)) (Contingency)
Hazardous Material User's Guide (HMUG) Group 13

TOOLS

1. [\[000771\]](#)Lubricating gun, hand, Lever-operated, 16 OZ 12" rub hose ext & adpt kit incl
2. [\[018451\]](#)Creeper, mechanics, W/C provide, (If desired) (Contingency)
3. [\[020301\]](#)Tool kit, CESE maintenance, basic

MISCELLANEOUS

1. [\[010949\]](#)Spectacles, industrial with side cover
2. [\[017934\]](#)Gloves, 4-mil Nitrile, Light use only
3. [\[001365\]](#)Technical manuals/drawings, W/C provide (F9YC0-AM-TMM-030 LARC V; Technical Manual Supplement for Lighter Amphibious Resupply Cargo, Five Ton (LARC-V A1/A2))
4. [\[011922\]](#)S9086-WK-STM-020/NSTM 670 Vol II, Hazardous Material Users Guide (HMUG)
5. [\[015244\]](#)S0400-AD-URM-010/TUM; TAG OUT USER'S MANUAL

NOTE: Numbers in brackets can be referenced to Standard PMS Materials Identification Guide (SPMIG) for identification.

HAZARDOUS MATERIALS CONTROL STATEMENT(U)

SSP: SASN-003-26

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ITEM NO: 243-11-001

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For all Navy commands, units, and activities regardless of warfare community, OPNAV M-5100.23 Navy Safety and Occupational Health Manual provides guidance for Hazardous Material Control and Management (HMC&M) and directs Safety Data Sheets (SDS) be maintained and accessible to review, for all HM at the command. For forces afloat, the Hazardous Material Users Guide (HMUG), S9086-WK-STM-020, NSTM Chapter 670 Volume II supplements the technical data found in SDSs by providing additional control measures, precautions, personal protective equipment (PPE), and spill controls for the hazardous material(s) identified in the Tools, Parts, Materials, Test, Equipment block. Maintenance personnel shall determine if additional

PPE is necessary to accomplish the MRC and take appropriate action to obtain and wear such PPE to ensure the safety of maintenance personnel. Final disposition of all HAZMAT, both shipboard and ashore, shall be IAW local HAZMINCEN/CHRIMP guidance as required by OPNAV-M 5090.1 series Table 35-3.

PROCEDURE

SCHEDULING CRITERIA

Accomplish this maintenance requirement when any of the following periodicities or situations occur.

- a. Monthly.
- b. After each days use.

WARNING: Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.

Preliminary

- a. Ensure that the LARC is in a maintenance-ready condition.

NOTE 1: The steering ball valve handle is accessible in the operator's cabin under the port passenger seat. During normal operation the ball valve handle should be in the OPEN position.

- b. Ensure the emergency steering ball valve is in the OPEN position, if applicable.

WARNING: To prevent personnel injury, Tag-Out procedures shall be in accordance with the Tag-out Users Manual (TUM), and/or local tag-out instruction.

- c. Attach safety tag in accordance with the Tag-out Users Manual (TUM) and/or local tag-out instruction.

WARNING: Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.

- d. Put on eye protection and gloves.

1. Lubricate LARC V.

- a. Lubricate Propeller drive shaft lubrication points inside center bay (#1 on Figure 1 and Figure 2):
 - (1) Inspect propeller drive shaft bearing lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.a.(2) through Step 1.a.(3) and proceed to Step 1.a.(4). If fresh or excessive grease is not present proceed to Step 1.a.(2).
 - (2) Clean propeller drive shaft bearing lubrication fittings.
 - (3) Using grease gun, apply grease three to four pumps to propeller drive shaft grease fitting. Do not over-grease to avoid damaging craft component seals.
 - (4) Remove excess grease from bearing.
 - (5) Repeat Step 1.a.(1) through Step 1.a.(4) for remaining propeller drive shaft grease fittings.
- b. Lubricate Rudder post. Inside Aft Stbd engine bay near hatch with Lubrication Access improvements (#2 on Figure 1 and Figure 4), without Lubrication Access improvements (#2 on Figure 6):
 - (1) For craft without the Lubrication Access improvements proceed to Step 1.b.(2). Inspect rudder post remote lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.b.(3) through Step 1.b.(4) and proceed to Step 1.b.(5). If fresh or excessive grease is not present proceed to Step 1.b.(3).

- (2) Inspect rudder post cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.b.(3) through Step 1.b.(4) and proceed to Step 1.b.(5). If fresh or excessive grease is not present proceed to Step 1.b.(3).
- (3) Clean rudder post remote lubrication fitting.
- (4) Using grease gun, apply grease to rudder post remote lubrication fitting. Do not over-grease to avoid damaging craft component seals.
- (5) Remove excess grease from remote lubrication fitting.
- c. Lubricate Aft Port/Stbd rudder cable lube points (Located either side of engine) with Lubrication Access improvements (#3 on Figure 1 and Figure 4), without Lubrication Access improvements (#3 on Figure 6):
 - (1) Inspect Aft Port/Stbd rudder cables lube points for fresh or excessive lubricant. If fresh or excessive lubricant is present omit Step 1.c.(2) through Step 1.c.(3) and proceed to Step 1.c.(4). If fresh or excessive lubricant is not present proceed to Step 1.c.(2).
 - (2) Clean Aft Port/Stbd rudder cables and linkages.
 - (3) Apply lithium grease to Aft Port/Stbd rudder cables and linkages.
 - (4) Remove excess grease from Aft Port/Stbd rudder cables and linkages.
 - (5) For craft without the Lubrication Access improvements proceed to Step 1.c.(6). Inspect remote Port rudder cable lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.c.(7) through Step 1.c.(8) and proceed to Step 1.c.(9). If fresh or excessive grease is not present proceed to Step 1.c.(7).
 - (6) Inspect Aft Port/Stbd rudder cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.d.(8) through Step 1.d.(9) and proceed to Step 1.d.(10). If fresh or excessive grease is not present proceed to Step 1.d.(8).
 - (7) Clean rudder cable lubrication fitting.
 - (8) Using grease gun, apply grease to remote Port rudder cable lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (9) Remove excess grease from remote Aft Port rudder cable lubrication fitting.
 - (10) Repeat Step 1.c.(7) through Step 1.c.(9) for remote Stbd rudder cable lubrication fitting.
- d. Lubricate Fwd Port/Stbd steering cable lube points (Located inside hull) with Lubrication Access improvements (#4 on Figure 1 and Figure 4), without Lubrication Access improvements (#4 on Figure 5):
 - (1) Inspect Fwd Port/Stbd steering cables lube points for fresh or excessive lubricant. If fresh or excessive lubricant is present omit Step 1.d.(2) through Step 1.d.(3) and proceed to Step 1.d.(4).
 - (2) Clean Fwd Port/Stbd steering cables and linkages.
 - (3) Apply white lithium grease to Fwd Port/Stbd steering cables and linkages.
 - (4) Remove excess lubricant from Fwd Port/Stbd steering cables and linkages.
 - (5) For craft without the Lubrication Access improvements proceed to Step 1.d.(6). Inspect remote Fwd Port/Stbd steering cables lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.d.(7) through Step 1.d.(8) and proceed to Step 1.d.(9).
 - (6) Inspect Fwd Port/Stbd steering cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.d.(7) through Step 1.d.(8) and proceed to Step 1.d.(9). If fresh or excessive grease is not present proceed to Step 1.d.(7).
 - (7) Clean steering cable lubrication fitting.
 - (8) Using grease gun, apply grease to remote Port steering cable lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (9) Remove excess grease from remote Fwd Port steering cable lubrication fitting.
 - (10) Repeat Step 1.d.(7) through Step 1.d.(9) for remote Stbd steering cable lubrication fitting.
- e. Lubricate hatch dogging mechanism (6 ea) with Lubrication Access improvements (#5 on Figure 6), without Lubrication Access improvements (#5 on Figure 2):
 - (1) Inspect hatch hinges and dogging mechanisms for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.e.(2) through Step 1.e.(3) and proceed to Step 1.e.(4). If fresh or excessive grease is not present proceed to Step 1.e.(2).
 - (2) Clean hatch dogging mechanism.
 - (3) Using grease gun, apply grease to hatch hinge and dogging mechanism lubrication fitting until grease extrudes from hinge and pivot point.
 - (4) Remove excess grease.
 - (5) Repeat Step 1.e.(1) through Step 1.e.(4) for remaining hatch hinge and dogging mechanisms.

NOTE 2: For craft with Steering System Modification completed proceed to Step 1.e.(6). For craft without the Steering System Modification completed omit Step 1.e.(6) through Step 1.e.(9). and proceed to Step 1.f.

NOTE 3: For craft with Lubrication Access Improvements complete proceed to Step 1.f.

- (6) Inspect cabin floor bulkhead fittings for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.e.(7) through Step 1.e.(8) and proceed to Step 1.e.(9). If fresh or excessive grease is not present proceed to Step 1.e.(7).
- (7) Clean cabin floor bulkhead fitting between center seat and port seat.
- (8) Using grease gun, apply grease to cabin floor bulkhead fitting until grease extrudes from bulkhead fitting if required.
- (9) Remove excess grease.
- f. Lubricate Engine/bow/top hatch (16 points) (#6 on Figure 2):
 - (1) Inspect Engine/bow/top hatch for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.f.(2) through Step 1.f.(3) and proceed to Step 1.f.(4). If fresh or excessive grease is not present proceed to Step 1.f.(2).
 - (2) Clean hatch hinge and lubrication fitting.
 - (3) Using grease gun, apply grease to hatch hinge and lubrication fitting until grease extrudes from hinge and pivot point.
 - (4) Remove excess grease from lubrication fitting.
 - (5) Repeat Step 1.f.(1) through Step 1.f.(4) for remaining hatch hinge lubrication fittings.
- g. Lubricate Steering post inside Fwd bilge area with Lubrication Access improvements (#7 on Figure 1), without Lubrication Access improvements (#7 on Figure 6):
 - (1) For craft without the Lubrication Access improvements proceed to Step 1.g.(2). Inspect remote steering post inside Port Fwd bilge area for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.g.(3) through Step 1.g.(4) and proceed to Step 1.g.(5). If fresh or excessive grease is not present proceed to Step 1.g.(3).
 - (2) Inspect steering post inside Port Fwd bilge area for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.g.(3) through Step 1.g.(4) and proceed to Step 1.g.(5). If fresh or excessive grease is not present proceed to Step 1.g.(3).
 - (3) Clean steering post and remote lubrication fitting, if applicable.
 - (4) Using grease gun, apply grease to remote steering post lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (5) Remove excess grease from steering post and lubrication fitting.
- h. Lubricate Tie rod ends underneath Fwd hull between front wheels (4 points) (#8 on Figure 3):
 - (1) Inspect tie rod ends underneath Fwd hull between front wheels for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.h.(2) through Step 1.h.(3) and proceed to Step 1.h.(4). If fresh or excessive grease is not present proceed to Step 1.h.(2).
 - (2) Clean tie rod ends.
 - (3) Using grease gun, apply grease to tie rod ends until grease extrudes from ends.
 - (4) Remove excess grease from tie rod ends.
 - (5) Repeat Step 1.h.(2) through Step 1.h.(4) for remaining tie rod ends.
- i. Lubricate Aft Propeller bearing bushing (2 points each) (#9 on Figure 3):
 - (1) Inspect Aft Port/Stbd Propeller bearing bushing for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.i.(2) through Step 1.i.(3) and proceed to Step 1.i.(4). If fresh or excessive grease is not present proceed to Step 1.i.(2).
 - (2) Clean Aft Port/Stbd Propeller bearing bushing.

CAUTION: Do not over-grease to avoid damaging craft component seals.

- (3) Using grease gun, apply two to three pumps of grease to Aft Port/Stbd Propeller bearing bushing.
- (4) Remove excess grease from bearing bushing.
- (5) Repeat Step 1.i.(1) through Step 1.i.(4) for remaining propeller bearing bushings.
- j. Wheel motor steering bearings (4 points) (#11 on Figure 1):

NOTE 4: There are two fittings on each steering bearing cover plate. Craft have one

lubrication fitting and a relief valve.

CAUTION:

Do not grease wheel motor lubrication fittings.

- (1) Inspect wheel motor steering bearing cover plate grease fittings for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.j.(2) through Step 1.j.(3) and proceed to Step 1.j.(4). If fresh or excessive grease is not present proceed to Step 1.j.(2).
- (2) Clean lubrication fitting and relief valve; remove relief valve.
- (3) Using grease gun, apply grease to the steering bearing cover plate until grease exits the open hole.
- (4) Reinstall relief valve.
- (5) Using grease gun, apply 3 more pumps of grease.
- (6) Remove excess grease from lubrication fitting.
- (7) Repeats Step 1.j.(2) through Step 1.j.(6) for remaining three steering bearing cover plates.
- k. Lubricate Fwd propeller bearing bushing and shaft universal joint (2 points) (#10 on Figure 2):
 - (1) Inspect Fwd propeller bearing bushing and shaft universal joint for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.k.(2) through Step 1.k.(3) and proceed to Step 1.k.(4). If fresh or excessive grease is not present proceed to Step 1.k.(2).
 - (2) Clean Fwd propeller bearing bushing and shaft universal joint lubrication fitting.
 - (3) Using grease gun, apply grease to Fwd propeller drive shaft bearing bushing and shaft universal joint lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (4) Remove excess grease from Fwd bearing bushing and universal joint lubrication fitting.

NOTE 5: For craft with Lubrication Access Improvements and Hydraulic Cover Modification proceed to Step 1.i. For craft without Lubrication Access Improvements and Hydraulic Cover Modification omit Step 1.i through Step 1.o.(4) proceed to Step 1.p.

- l. Lubricate hydraulic fill cover hinge and slide-bolts (10 points) (#12 on Figure 2):
 - (1) Inspect Port hydraulic fill cover hinge and slide-bolts for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.l.(2) through Step 1.l.(3) and proceed to Step 1.l.(4). If fresh or excessive grease is not present proceed to Step 1.l.(2).
 - (2) Clean cover hinge and slide-bolts.
 - (3) Using hand, apply grease to cover hinge and slide-bolts. Remove excessive grease.
 - (4) Repeat Step 1.m.(1) through Step 1.l.(4) for Starboard cover hinge and slide-bolts.
- m. Lubricate steering cylinder rod pin (1 point) (#13 on Figure 1):
 - (1) Inspect steering cylinder rod pin for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.m.(2) through Step 1.m.(3) and proceed to Step 1.m.(4). If fresh or excessive grease is not present proceed to Step 1.m.(2).
 - (2) Clean steering cylinder rod pin and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote steering cylinder rod pin lubrication fitting. Do not over-grease to avoid craft damage.
 - (4) Remove excessive grease from steering cylinder rod pin and remote lubrication fitting.
- n. Lubricate linear alignment coupling (1 point) (#14 on Figure 1):
 - (1) Inspect linear alignment coupling for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.n.(2) through Step 1.n.(3) and proceed to Step 1.n.(4). If fresh or excessive grease is not present proceed to Step 1.n.(2).
 - (2) Clean linear alignment coupling and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote linear alignment coupling lubrication fitting. Do not over-grease to avoid craft damage.
 - (4) Remove excessive grease from linear alignment coupling and remote lubrication fitting.
- o. Lubricate land steering shaft (1 point) (#15 on Figure 1):
 - (1) Inspect steering shaft for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.o.(2) through Step 1.o.(3) and proceed to Step 1.o.(4). If fresh or excessive grease is not present proceed to Step 1.o.(2).
 - (2) Clean land steering shaft and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote steering shaft lubrication fitting. Do not over-grease to avoid craft damage.

SHIP:

NBU-7 (RMC 57078)

- (4) Remove excessive grease from steering shaft and remote lubrication fitting.
- p. Remove safety tag.
- q. Take off gloves and then eye protection.
- r. Return LARC to readiness condition.

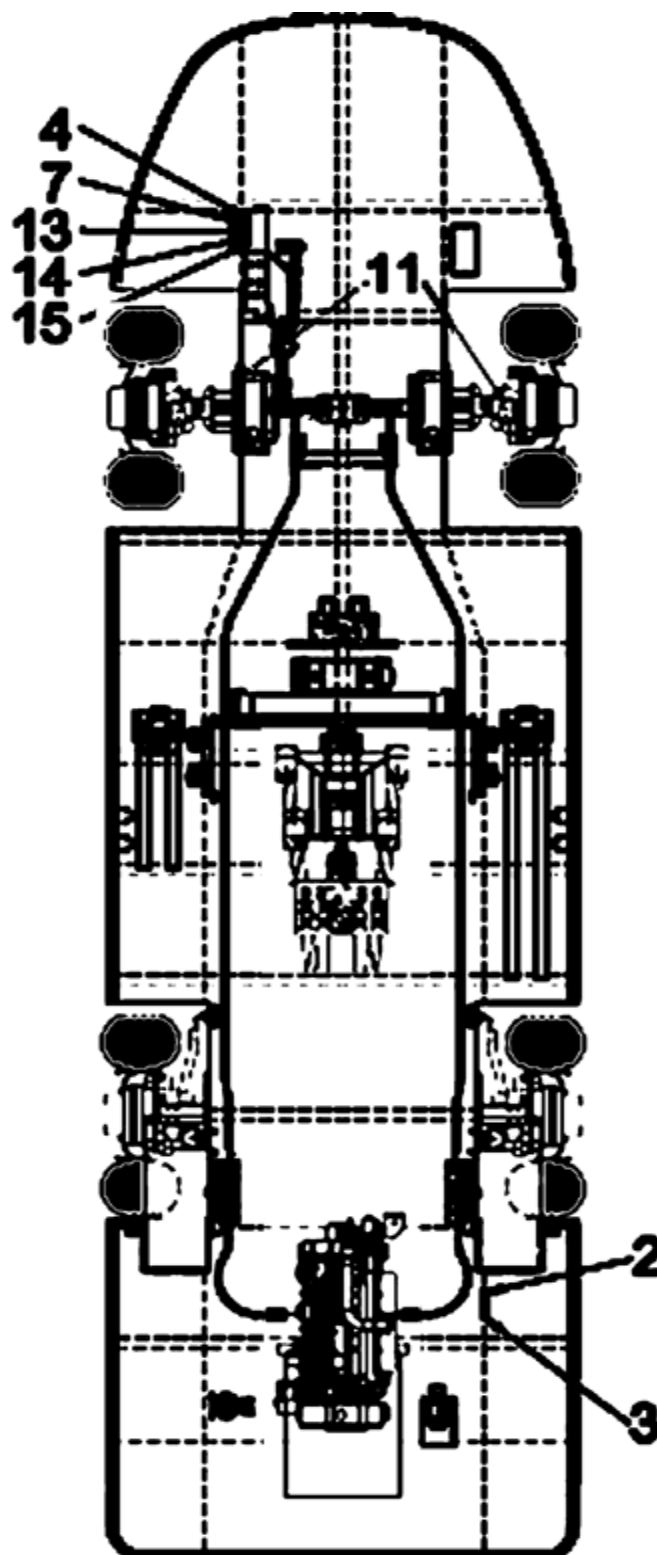
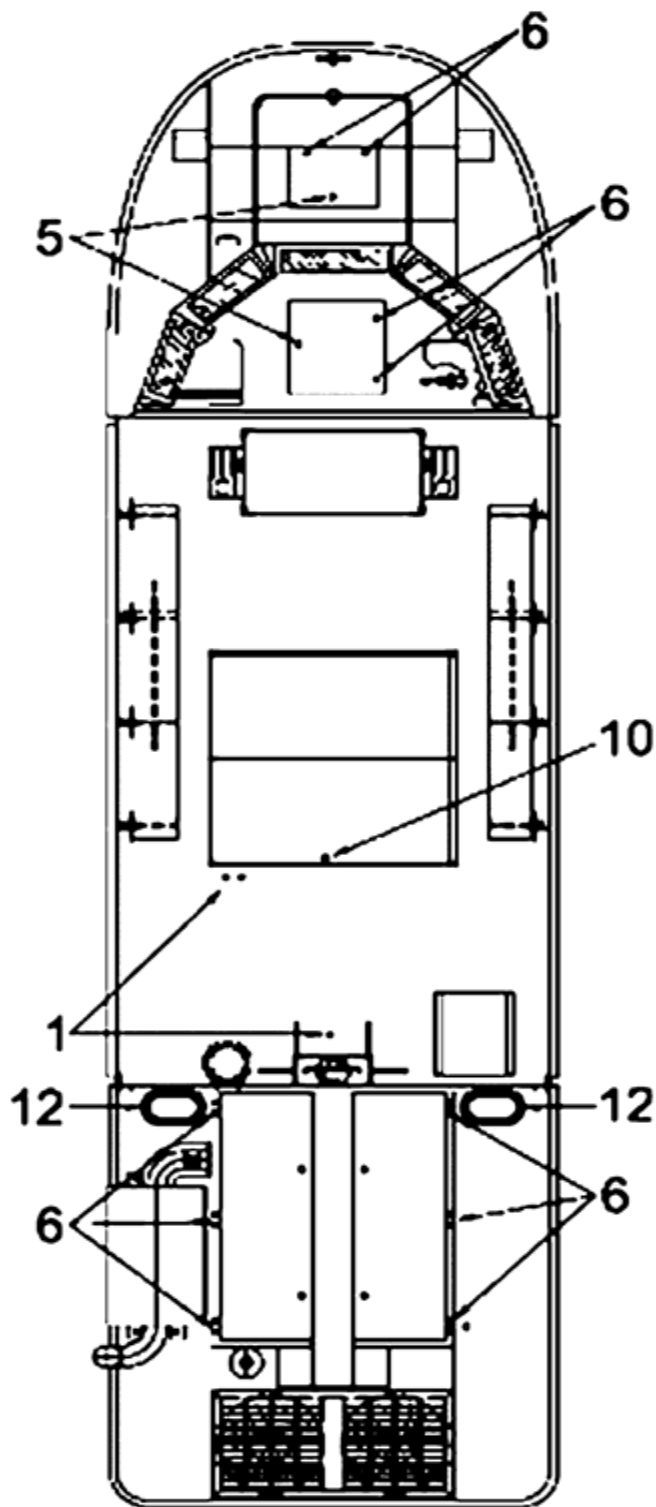


Figure 1 Lube Points

**Figure 2 Lube Points**

SHIP:

NBU-7 (RMC 57078)

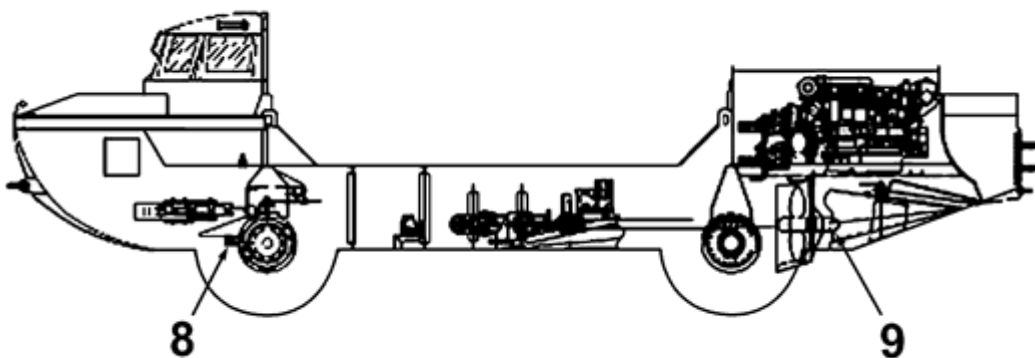


Figure 3 Lube Points

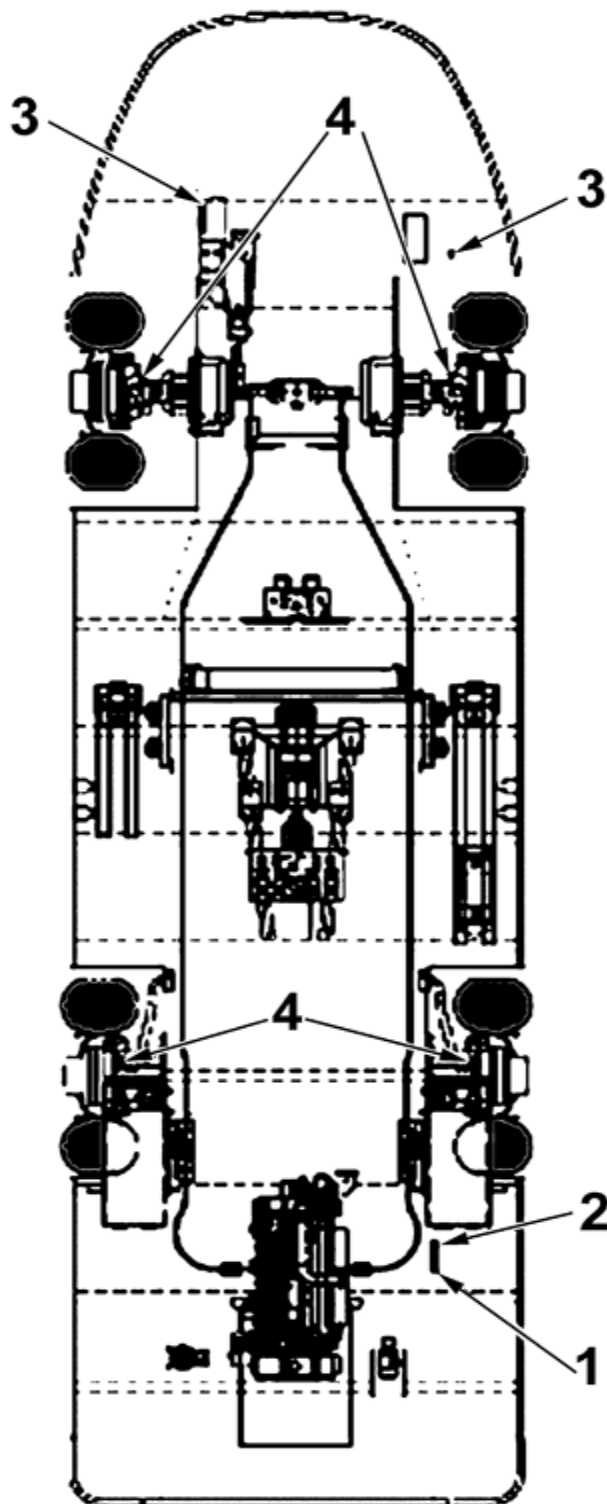


Figure 4 Lube Points

SHIP:

NBU-7 (RMC 57078)

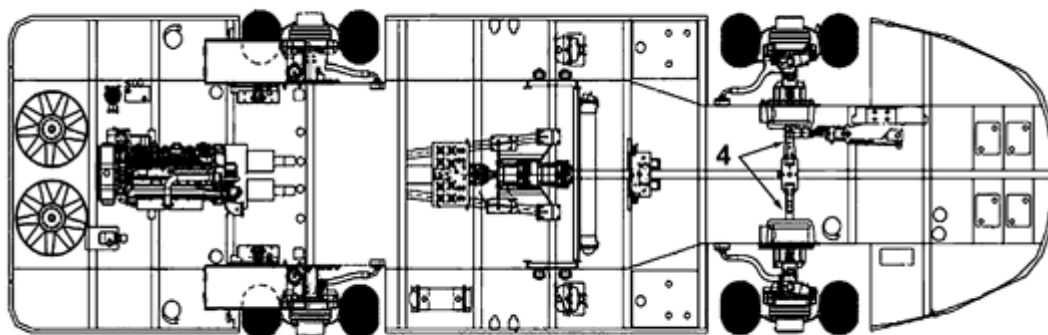


Figure 5 Lube Points

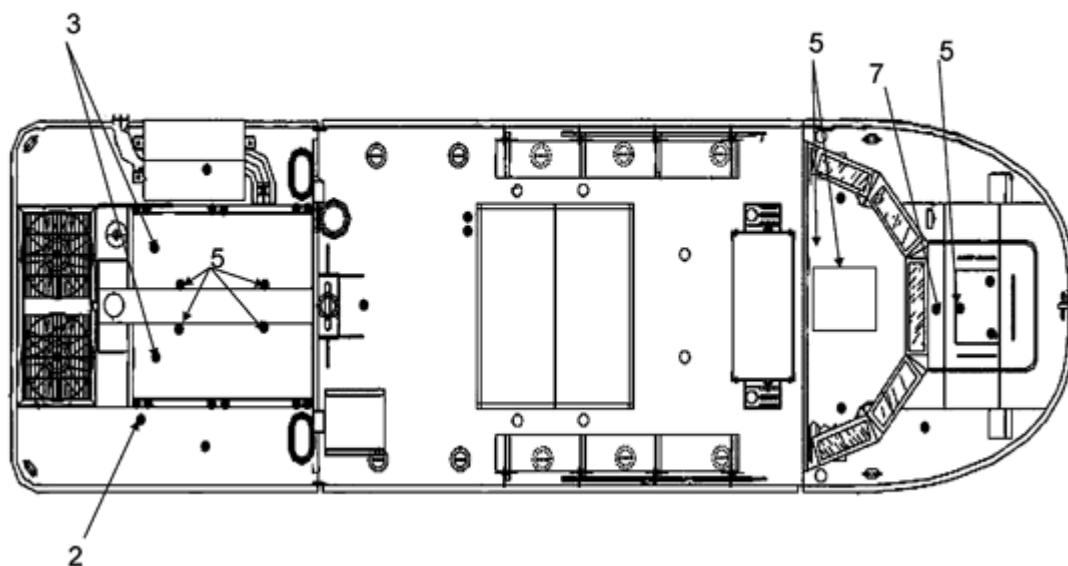


Figure 6 Lube Points

SHIP:

NBU-7 (RMC 57078)

Item Number	Nomenclature
1	Propeller drive shaft lubrication points inside center bay
2	Rudder post (1 Aft remote lubrication fitting)
3	Aft Port/Stbd rudder cable lube points (2 Aft remote lubrication fittings)
4	Forward Port/Stbd steering cable lube points (2 Fwd remote lubrication fittings)
5	Hatch dogging handles
6	Engine/bow/top hatch (16 points)
7	Steering post inside forward bilge area centerline (1 Fwd remote lubrication fitting)
8	Tie rod ends underneath forward hull (between front wheels) (4 points)
9	Aft propeller bearing (2 points)
10	Forward propeller bearing and shaft universal joint (2 points)
11	Wheel motor steering bearing cover plate (4 points)
12	Hydraulic fill cover hinge and slide-bolts (10 points)

Table 1 LARC Lubrication Point Identification

SHIP: NBU-7 (RMC 57078) ITEM NO: 422-11-001
 COAR: 18 PCN: RS01-L016
 BWST FILE NO: 000-001 CMP: NONE
 REVISED: 05 APR 2019 PLANNER: NAGAMINE, YUUICHI

1. SCOPE:

- 1.1 Title: Navigation System Component; replace
- 1.2 Location of Work:
 - 1.2.1 LARC 936
- 1.3 Identification:
 - 1.3.1 Quantity (One EA), Electric Horn located in 1.2.1, P/N 10002, Twin Horn 12VDC, NSN 6350-01-526-0746, MFR - Ongaro Marine Inc

2. REFERENCES:

- 2.1 Standard Items, (FY26)
- 2.2 F9YC0-AB-TMM-010 Rev 5, Systems Technical Manual for LARC-V A1/A2
- 2.3 1600901-1 Rev H, General Arrangement & Outboard Profile
- 2.4 1600907-1 Rev AA, Electrical System Installation & Detail

3. REQUIREMENTS:

- 3.1 Remove each existing equipment listed in 1.3.1, using paragraph 6.11.3.3 of 2.2, 2.3 and 2.4 for guidance.
 - 3.1.1 Matchmark, identify and retain disconnected hardware.
- 3.2 Install each new equipment listed in 1.3.1 and retained hardware of 3.1.1, using paragraph 6.11.3.4 of 2.2, 2.3 and 2.4 for guidance.
- 3.3 Accomplish the requirements of 009-032 of 2.1 for cleaning and preservation of new and disturbed surfaces to match surrounding areas.

(V) (G) "OPERATIONAL TEST"

- 3.4 Accomplish an operational test of each new electric horn listed in 1.3.1 in accordance with paragraph 6.11.3.4 of 2.2. -S130

4. NOTES:

- 4.1 Initial Condition - S/F discovered that navigation equipment damage on LARC 936 due to corrosion to navigation equipment to include electric horn.
- 4.2 GFM listed in 5.1 have been ordered under ICN 38NBA42201-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	1 EA	HORN, SIGNAL	91	2.4	63500152 60746	3.2
2.	1 PG	SCREW, CAP, HEXAGO N H	92	2.4	53050152 86392	3.2

SHIP:

NBU-7 (RMC 57078)

3.	1 HD	NUT, PLAIN, HEXAGON	93	2.4	53100152 86210	3.2
4.	1 PG	WASHER, FLAT	94	2.4	53100152 86210	3.2
5.	2 EA	SILICONE COMPOUND	97	2.4	68500152 94861	3.2
6.	1 PG	TERMINAL SPADE, FEMALE	72625K77		5940SA23 00331	3.2
7.	2 EA	CONDUIT, 3/8" LENGTH 5FT /EA	7840K82		5975SA23 00332	3.2
8.	2 EA	FUSE, ENCLOSED LINK	7460K45		59200161 57500	3.2
9.	15 EA	STRAP, TIE DOWN, ELEC	7130K42		59750157 23833	3.2
10.	1 PG	TERMINAL, RING	9983K18		5940SA23 00179	3.2
11.	1 PG	TERMINAL SPADE, MALE	72625K76		5940SA23 00180	3.2
12.	1 SL	WIRE BLACK, 14AWG, 18FT/SPOOL	184003		6145SA23 00192	3.2
13.	1 OZ	CHEMICAL CORROSION, P/N TEF-GEL	132	2.4	68500145 04009	3.2

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

SHIP: NBU-7 (RMC 57078) ITEM NO: 491-11-001
COAR: 18 PCN: RS01-L010
BWST FILE NO: 000-001 CMP: NONE
REVISED: 05 APR 2019 PLANNER: NAGAMINE, YUUICHI

1. SCOPE:

1.1 Title: Electrical System Component; replace

1.2 Location of Work:

1.2.1 LARC 936

1.3 Identification:

1.3.1 Quantity (2 EA), Power Converter located in 1.2.1, P/N DCVR-20A, NSN 6130-98-206-5635, MFR - BEP Marine Inc

1.3.2 Quantity (One EA), Dead Man Foot Pedal located in 1.2.1, Item No 121 of 2.3, FD-5-20-T-F(AL), 20A, 2NO/NC, 3/4 Inch Conduit, MFR - Suns International Inc

1.3.3 Quantity (One EA), Hand Throttle located in 1.2.1, P/N M115SL15F59, P-Q Controls Inc

1.3.4 Quantity (One EA), Propeller Speed Sensor located in 1.2.1, P/N 0009730004, MFR - Linde Inc

1.3.5 Quantity (3 SE), Wind Shield Wiper Motor located in 1.2.1, Motor P/N : WWF24C191020, MFR United Technologies Inc, NSN 2540-01-424-0868, Adjustable Arm ASSY P/N: 4-730603, MFR - Cleveland Ignition Inc, Blade P/N: 13610M, MFR - Wexco Industries Inc

1.3.6 Quantity (3 EA), Blue Sea Rocker Switch Assembly located in 1.2.1, 8 Position Breaker Panel Assembly : P/N 8371, NSN 5925-01-545-5972 (One EA) and P/N 8287, NSN 5930-01-595-4870 (One EA): 3 Position Breaker Panel Assembly: P/N 8374, NSN 5925-01-545-5960 (One EA): 4 Position Breaker Panel Assembly: P/N 8372, NSN 5925-01-545-3083 (One EA), MFR - Blue Sea System Inc

1.3.7 Quantity (3 EA), Bilge Pump Rocker Switch Assembly located in 1.2.1, P/N 8263 (Control Panel Switch Assembly), NSN 5930-01-549-4180, MFR - Blue Sea System Inc

1.3.8 Quantity (2 EA), Hydraulic Oil Temperature Gauge (FWD and AFT) located in 1.2.1, P/N 310-916, NSN 6685-01-474-4923, MFR - VDO

1.3.9 Quantity (One AY), Electrical Junction Box Assembly with All Inside Parts located in 1.2.1, Parts Number and MFR on 2.8

2. REFERENCES:

2.1 Standard Items, (FY26)

2.2 F9YC0-AB-TMM-010 Rev 5, Systems Technical Manual for LARC-V A1/A2

2.3 1600911-2 Rev W, LARC V Hydraulic System Schematic

2.4 1600909-1 Rev AB, Operator's Console & Controls

2.5 1600907-1 Rev AA, Electrical System Installation & Details

2.6 1600907-2 Rev E, LARC V Vehicle Electronic Control System Layout & Details

- 2.7 1600907-3 Rev M, Electronic Transmission Control Schematic Diagram
- 2.8 1600907-4, LARC V Vehicle Electrical Junction Box
- 2.9 1600907-5 Rev A, Electrical Wire Lists
- 2.10 Test Procedure, Test# 58331-4-100_26135, Title: Electrical System Component

3. REQUIREMENTS:

3.1 Remove each existing equipment listed in 1.3.1 through 1.3.9, using paragraph 6.4.1.3, 6.4.4.3, 6.4.5.3, 6.4.6.3, 6.4.7.3, 6.4.8.3, 6.4.9.3, 6.4.10.3, 6.4.11.3, 6.4.12.3, 6.4.13.3, 6.4.14.3, 6.4.15.3, 6.4.16.3 and 6.4.17.3 of 2.2 and 2.3 through 2.9 for guidance.

3.1.1 Matchmark, identify and retain disconnected hardware.

3.2 Install each new equipment listed in 1.3.1 through 1.3.9 and retained hardware of 3.1.1, using paragraph 6.4.1.4, 6.4.4.4, 6.4.5.4, 6.4.6.4, 6.4.7.4, 6.4.8.4, 6.4.9.4, 6.4.10.4, 6.4.11.4, 6.4.12.4, 6.4.13.4, 6.4.14.4, 6.4.15.4, 6.4.16.4 and 6.4.17.4 of 2.2 and 2.3 through 2.9 for guidance.

3.2.1 Assemble new electrical junction box assembly listed in 1.3.9 in accordance with 2.8.

3.2.2 Seal each thread fastener, electrical end connection, and electrical box, using note 4, 6 and 7 of 2.5 for guidance.

3.3 Accomplish the requirements of 009-032 of 2.1 for cleaning and preservation of new and disturbed surfaces to match surrounding areas.

(V) (G) "OPERATIONAL TEST"

3.4 Accomplish an operational test of electrical system in accordance with 2.10. -S246

4. NOTES:

4.1 Initial Condition - S/F discovered electrical system issues on LARC 936 due to corrosion to primary electrical components to include : power converter, dead man foot pedal, hand throttle, propeller speed sensor, windshield wiper motor, bluesea rocker switch, bilge pump rocker switch, hydraulic oil temperature gauge, electrical junction box receptacle pin, electrical junction box receptacle socket, junction box plug, junction box HD20 connector receptacle, junction box diodes, junction box resistors, junction box wires.

4.2 GFM listed in 5.1 have been ordered under ICN 38NBA49102-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
2.	2 EA	CONVERTER, DIRECT CU, P/N DCVR-20A	41	2.4	9B613098 2065635	3.2
3.	1 EA	SWITCH, FOOT, DEAD MAN FOOT	60	2.4	9B593001 6916287	3.2

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		DEDAL P/N 88SN2-05 (OLD				
4.	1 EA	PROPELLER SPEED SENS, PROPELLER SPPED SENSOR P/N 0009730004 ITEM NO. 158 OF DWG 1600911-2	158	2.3	9B 6680 12390745 0	3.2
5.	1 TU	INSULATING COMPOUND, P/N 3140 RTV ITEM 133 OF DWG	133	2.5	9B597001 4871943	3.2
6.	3 EA	ARM ASSEMBLY, ADJUSTABLE ARM ASSEMBLY OF WIND SHIE	104	2.5	9B2540SA 2300221	3.2
7.	3 EA	MOTOR, WINDSHIELD WI, WIPER MOTOR P/N WWF24C191020	103	2.5	9B254001 4240868	3.2
8.	3 EA	BLADE, WIND SHIELD WIPER BLADE STAINLESS STEEL, 10	105	2.5	9B2540SA 2300196	3.2
9.	1 EA	CIRCUIT BREAKER BOX, P/N 8371	3	2.4	9B592501 5455972	3.2
10.	1 EA	SWITCH, TOGGLE, P/N 8287	59	2.4	9B593001 5954870	3.2
11.	1 EA	CIRCUIT BREAKER BOX, P/N 8374	4	2.4	9B592501 5455960	3.2
12.	1 EA	CIRCUIT BREAKER BOX, P/N 8372	5	2.4	9B592501 5453083	3.2
13.	3 EA	SWITCH ASSEMBLY, P/N 8263	11	2.4	9B593001 5494180	3.2
14.	2 EA	INDICATOR, TEMPER ATU, P/N 310-916	50	2.4	9B668501 4744923	3.2
15.	2 EA	HEADER, SHORTING, P/N 280-402	11	2.8	9B599901 4160314	3.2
16.	4 EA	TERMINAL JUNCTION B, END STOP ITEM NO 13	13	2.8	9B594001 5478605	3.2
17.	28 EA	FUSE, ENCLOSED LINK, 5 AMP ITEM NO 15	15	2.8	9B592001 1235212	3.2
18.	7 EA	CONNECTOR, PLUG, E LEC, P/N HDP24-24-31-PE	16	2.8	9B593501 5722349	3.2
19.	7 EA	CONNECTOR, PLUG, E LEC, P/N HDP26-24-31-SE	17	2.8	9B593501 5722319	3.2
20.	11 EA	NUT, PLAIN, ROUND, P/N 2411-001-2405	18	2.8	9B531001 5190209	3.2
21.	11 EA	WASHER, LOCK, P/N 2414-001-2486	19	2.8	9B531001 5625734	3.2

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22.	8 EA	CONTACT,ELECTRIC AL, P/N 0460- 204-12141	20	2.8	9B599901 2865843	3.2
23.	8 EA	CONTACT,ELECTRIC AL, P/N 0462- 203-12141	21	2.8	9B599901 2801438	3.2
24.	154 EA	CONTACT,ELECTRIC AL, P/N 0460- 202-16141	22	2.8	9B599901 2163648	3.2
25.	154 EA	CONTACT,ELECTRIC AL, P/N 0462- 201-16141	23	2.8	9B599901 2036687	3.2
26.	88 EA	CONTACT,ELECTRIC AL, P/N 0460- 010-20141	24	2.8	9B599901 6010977	3.2
27.	168 EA	CONTACT,ELECTRIC AL, P/N 0462- 005-20141	25	2.8	9B599901 5884916	3.2
28.	4 EA	CONNECTOR,PLUG,E LEC, P/N HDP24- 24-47-PE	26	2.8	9B593501 6519821	3.2
29.	4 EA	CONNECTOR,PLUG,E LEC, P/N HDP26- 24-47-SE	27	2.8	9B593501 6519823	3.2
30.	1 EA	CONNECTOR,RECEPT ACL, P/N HDP24- 18-8-PE	28	2.8	9B593501 5722856	3.2
31.	1 EA	CONNECTOR BODY,PLUG, P/N HDP26-18-8-SE	29	2.8	9B593501 6299315	3.2
32.	1 EA	NUT,TUBE COUPLING, P/N 2411-002-1805	30	2.8	9B473001 5720515	3.2
33.	1 EA	WASHER,LOCK, P/N 2414-002-1886	31	2.8	9B531001 5798214	3.2
34.	4 EA	PLUG,END SEAL,ELECT, P/N 114017	32	2.8	9B593501 1741235	3.2
35.	31 EA	PLUG,END SEAL,ELECT, P/N 0413-204-2005	33	2.8	9B593501 4791823	3.2
36.	1 SL	WIRE,ELECTRICAL, P/N 1859/19 BK005	35	2.8	9B614501 6522304	3.2
37.	1 SL	WIRE,ELECTRICAL, P/N1859/19 RD005	36	2.8	9B614501 6522302	3.2
38.	2 EA	RESISTOR,FIXED,C OMP, P/N 153JE	40	2.8	9B590501 6516660	3.2
39.	2 EA	RESISTOR,FIXED,W IRE, P/N OF472JE	41	2.8	9B590501 6518493	3.2
40.	4 EA	SEMICONDUCTOR DEVIC, P/N 1N5408	42	2.8	9B596100 2520995	3.2
41.	3 HD	STRAP,TIE	43	2.8	9B597500	3.2

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		DOWN,ELEC, 4 INCH			7275153	
42.	11 EA	GASKET, P/N 16- 04477	48	2.8	9B533001 5391502	3.2
43.	1 EA	GASKET, P/N 16- 04978	49	2.8	9B533001 5393236	3.2
44.	1 EA	HAND THROTTLE, P/N M115SL15F59JOYST ICK	6	2.4	9B3040SA 2300197	3.2
45.	20 EA	CONTACT,ELECTRIC AL, P/N 0462- 005-20141	25	2.8	9B599901 5884916	3.2
46.	1 EA	BOX ENCLOSURE, P/N BN4161406SSITEM NO.1	1	2.8	9B5975SA 2300219	3.2
47.	1 EA	BACK PANEL, P/N P1614ITEM NO.2	2	2.8	9B5975SA 2300220	3.2
48.	1 EA	DIN RAIL, P/N 210-162STEEL RAIL DIN 35 X H 7.5 MM1	3	2.8	9B5999SA 2300233	3.2
49.	28 EA	FUSED CONNECTOR, P/N 282-698/281- 413FUSED TERMINAL	4	2.8	9B5999SA 2300234	3.2
50.	12 EA	GROUND CONNECTOR, P/N 280-517DIN RAIL MOUNT TERMIN	5	2.8	9B5999SA 2300235	3.2
51.	43 EA	2 LAYER CONNECTOR, P/N 280-519DIN RAIL MOUNT TERMI	6	2.8	9B5999SA 2300236	3.2
52.	7 EA	2 LAYER CONNECTOR, P/N 280-520DIN RAIL MOUNT TERMI	7	2.8	9B5999SA 2300237	3.2
53.	2 EA	END PLATE, P/N 280-340CONNECTOR ACCESSORY, SEPARAT	8	2.8	9B5999SA 2300238	3.2
54.	1 EA	MARKER CARD 1- 50, ITEM NO.9	9	2.8	9B534001 5496485	3.2
55.	1 EA	MARKER, 51-100, P/N 209-507ITEM NO.10	10	2.8	9B5999SA 2300239	3.2
56.	13 EA	JUMPER WIRE, P/N 249-127PUSH IN TYPE WIRE JUMPER,	12	2.8	9B5999SA 2300240	3.2
57.	1 EA	SLEEVEING,TEXTILE ,EL, P/N PSBXP-	14	2.8	9B597001 6521409	3.2

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		114-187				
58.	1 SL	WIRE, P/N 1857/19 WH001WIRE 18 AWG, MIL- 16878E/1 (	34	2.8	9B6145SA 2300198	3.2
59.	2 PG	SHRINK TUBE, P/N MW-3/16-6"- BLACKHEAT SHRINK TUBIN	37	2.8	9B5970SA 2300199	3.2
60.	1 SL	SOLDER, P/N 13223NO CLEAN FLUX WIRESN63/PB37 ALLOY	38	2.8	9B5970SA 2300244	3.2
61.	1 EA	TAPE, P/N 784301PRIMARY WIRE & BATTERY TERMINAL SE	39	2.8	9B5970SA 2300300	3.2
62.	7 EA	SCREW,MACHINE, P/N 92314A831	44	2.8	9B530501 4867793	3.2
63.	7 EA	NUT,SELF- LOCKING,HE, P/N 91831A411	45	2.8	9B531001 5091545	3.2
64.	3 EA	TERMINAL,LUG, P/N 7113K24	46	2.8	9B594001 5131400	3.2
65.	1 BX	WASHER,FLAT, P/N 90107A011	47	2.8	9B531001 4969571	3.2
66.	1 EA	MARKERS 101-150, P/N 209- 508TERMINAL BLOCK MARKER1	50	2.8	9B5999SA 2300241	3.2
67.	1 EA	MARKERS, 151- 200, P/N 209- 509TERMINAL BLOCK MARKER	51	2.8	9B5999SA 2300242	3.2
68.	5 EA	FUSE,ENCLOSED LINK, 20A P/N 7460K47	116	2.5	9B592001 0850825	3.2
69.	13 EA	FUSE,ENCLOSED LINK, 15A P/N 7460K46	115	2.5	9B592001 6115772	3.2
70.	2 EA	FUSE,ENCLOSED LINK, 10A P/N 7460K45	101	2.5	9B592001 6157500	3.2
71.	5 PG	FUSE,ENCLOSED LINK, 5A P/N 7460K43	114	2.5	9B592001 6625196	3.2
72.	1 EA	FUSE,CARTRIDGE, 3A P/N 7460K41	113	2.5	9B592001 5986391	3.2
73.	1 OZ	CHEMICAL CORROSION, P/N TEF-GEL	132	2.5	9B685001 4504009	3.2
74.	1 BX	INHIBITOR,CORROS	134	2.5	9B685001	3.2

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SHIP:

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		ION, P/N VCI-101			3381392	
75.	1 CN	DESICCANT,ACTIVA TED, P/N 8790670-8	135	2.5	9B685000 2646564	3.2
76.	1 CN	CONFORMAL COATING, P/N CTAR-12ITEM NO 136 OF DWG N	136	2.5	9B597001 4174158	3.2
77.	1 PG	SCREW,CAP,HEXAGO N H, P/N 93190A542	71	2.4	9B530501 5286392	3.2
78.	4 EA	LOCK NUT, P/N 90715A125	72	2.4	9B531001 5045702	3.2
79.	8 EA	WASHER,F, P/N 90107A029	73	2.4	9B531001 3824068	3.2
80.	8 EA	NUT,SELF- LOCKING,HE, P/N 91831A030	23	2.6	9B531001 4993569	3.2
81.	1 HD	WASHER,LOCK, P/N 91475A030	38	2.6	9B531000 9746623	3.2
82.	4 EA	MOUNT,RESILIENT, UTI, P/N 9213K29	54	2.6	9B534001 6486797	3.2
83.	2 EA	LAMP,INCANDESCEN T, P/N 18522- H5001	102	2.5	9B624001 5560452	3.2
84.	1 EA	RETAINER,LENS, P/N M5939712	110	2.5	9B622000 7526020	3.2
85.	1 EA	STOP LIGHT- TAILLIGH, P/N 8378785	110	2.5	9B622000 6695623	3.2
86.	1 EA	LIGHT,NAVIGATION AL,, P/N 0170BSN24L	33	2.5	9B622001 5975592	3.2
87.	2 EA	LIGHT,NAVIGATION AL,, P/N 3839-1	87&88	2.5	9B622001 6290190	3.2
88.	1 EA	CAP,NAVIGATION LIGH, P/N 0964DP0STS	80	2.5	9B622001 6416992	3.2
89.	1 EA	FIXTURE,LIGHTING , P/N 1044DP2	9	2.5	9B621001 5457049	3.2
90.	1 EA	CONTROL,ELECTRIC LI, P/N 7501	14	2.5	9B622001 6189864	3.2
91.	3 EA	ARM ASSEMBLY, P/N 201211DS11- 14INCH ADJUSTABLE RAD	-		9B2540SA 2300222	3.2
92.	10 FT	CABLE ASSEMBLY, P/N 8219K6218 AWG, MIL-W- 16878/4,	62	2.4	9B6145SA 2400064	3.2
93.	1 PG	CONNECTOR, FEMALE SPADE CONNECTORINSULAT ED BARREL	63	2.4	9B5970SA 2400065	3.2

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94.	1 PG	CONNECTOR, INSULATED BARREL RING CONNECTORHEAT SHR	64	2.4	9B5970SA 2400066	3.2
95.	1 EA	CONNECTOR, PLUG, E LEC, P/N DT06-3S	65	2.4	9B593501 5238855	3.2
96.	1 EA	CONNECTOR, PLUG, E LEC, P/N DT04-3P	66	2.4	9B593501 5150573	3.2
97.	3 EA	CONTACT, ELECTRIC AL, P/N 0460- 202-16141	67	2.4	9B599901 2163648	3.2
98.	3 EA	CONTACT, ELECTRIC AL, P/N 0462- 201-16141	68	2.4	9B599901 2036687	3.2
99.	1 EA	SLEEVE, LOCKING, E LEC, P/N W3P	69	2.4	9B593501 4708337	3.2
100.	1 EA	POLARIZING KEY, ELEC, P/N W3S	70	2.4	9B593501 4708342	3.2
101.	1 EA	FOOT PEDAL, DEAD MAN FOOT PEDAL P/N FS-5-20-T- A20A	121	2.4	9B5930SA 2400067	3.2
102.	1 HD	SCREW, MACHINE, P/N 91770A542	122	2.4	9B530501 2857712	3.2
103.	1 EA	NUT, SELF- LOCKING, HE, P/N 94238A101	123	2.4	9B531001 6747438	3.2
104.	1 PG	FLAT WASHER, P/N 98019A2551/4 INCH SCREW SIZE, SST,	124	2.4	9B5310SA 2400069	3.2
105.	1 PG	CORD CONNECTOR, STRAIN RELIEF CORD CONNECTOR FOOT	125	2.4	9B5935SA 2400142	3.2
106.	1 PG	CORD CONNECTOR, STRAIN RELIEF CORD CONNECTOR BULKH	126	2.4	9B5935SA 2400143	3.2
107.	1 EA	FUSE HOLDER, P/N: 704- ANL, HEAVY DUTY FUSE HOLDER 5	129	2.5	9B 5920 98208079 1	3.2
108.	1 EA	POTENTIOMETER, P/N: 5AAA-B28- A10L, 1K OHM 1 GANG L	-		9B5930SA 2501870	3.2
109.	1 EA	RESISTOR, VARIABLE, N, P/N: RV4NAYSD102A / 501-0005	-		9B590500 5525478	3.2
110.	1 EA	BUS COVER, P/N:	127	2.5	9B5930SA	3.2

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SHIP:

NBU-7 (RMC 57078)

		703-500A/B, HEAVY DUTY BUS BAR, 3 X			2501868	
111.	6 TU	INSULATING COMPOUND, P/N 3140 RTV ITEM 133 OF DWG	133	2.4	9B597001 4871943	3.2

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>511-11-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L015</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>OOKUBO, TAKASHI</u> <u>NAGAMINE, YUUICHI</u> <u>TERASHITA, SHUUJI</u>

1. SCOPE:

1.1 Title: Cabin Heater Component; replace

1.2 Location of Work:

1.2.1 Forward Compartment

1.2.2 Engine Compartment

1.3 Identification:

1.3.1 Quantity (One EA), Heater Blower Assembly (ASSY) located in 1.2.1, Parts No. A-A-52545-1, Item No. 02 of 2.3 / Item No.35 of 2.4

1.3.2 Quantity (One EA), Heater (Hot Coolant Circulating) Pump and Motor ASSY located in 1.2.2, Parts No. T41E AB/T410E Wet-End (Pump)/CDP3410-V24 (Motor), Type MCC, Item No. 01 of 2.3 / Item No.62 of 2.4

2. REFERENCES:

2.1 Standard Items, (FY26)

2.2 F9YC0-AB-TMM-010 Rev 5, Change U, Systems Technical Manual for LARC-V A1/A2

2.3 1600909-2 Rev F, CAB Heater System Installation

2.4 1600907-1 Rev AA, Electrical System Installation & Details

3. REQUIREMENTS:

3.1 Remove each existing and install each new equipment listed in 1.3.1 and 1.3.2 installing each new gasket and packing, using Paragraph 6.10 of 2.2, 2.3 and 2.4 for guidance.

3.1.1 Matchmark, identify and retain disconnected hardware.

3.1.2 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.

3.1.3 Inspect each part not specified for renewal for wear and defect.

3.1.4 Remove each existing and install new component in place of those identified to be missing or defective.

3.1.5 Remove defective wiring of the equipment listed in 1.3.1 and install each new wiring listed in material list of Paragraph 5.

3.1.6 Install retained hardware of 3.1.1.

3.1.7 Apply new sealant to each threaded fastener in accordance with Notes One of 2.3.

3.2 Accomplish the requirements of 009-032 of 2.1 for each new and disturbed surface.

(V) (G) "OPERATIONAL TEST"

3.3 Accomplish an operational test of each equipment listed in 1.3.1 and 1.3.2 to verify correct operation, using Paragraph 6.10 of 2.2 for guidance. -S130

3.3.1 Verify each disturbed joint free of leaks.

4. NOTES:

4.1 Initial Condition - S/F discovered cabin heater equipment inoperable on LARC 936 due to corrosion to primary cabin heater components to include; heater blower assembly, heater coolant circulating pumps.

4.2 SRF Tool Room will issue each torque wrench when required.

4.3 Ship's Force will provide fluids for operation such as coolant, hydraulic oil, and fuel oil when refilling.

4.4 GFM listed in 5.1 have been ordered under ICN 38NBA51101-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	1 EA	PUMP&MOTOR ASSY, PUMP&MOTOR ASSEMBLY, P/N;0C-T410E	01 (31&62 of Ref.2. 4)	2.3	9B5310SA 2300301	3.1
2.	1 EA	HEATER, VEHICULAR , CO, A-A-52545- 1 (HW30)	2	2.3	9B254000 0208591	3.1
3.	3 EA	CLAMP, HOSE, 5011T14, SST 316	8	2.3	9B473001 6106830	3.1
4.	1 PG	SCREW, CAP, HEXAGO N H, 93190A712	25	2.3	9B530501 5867692	3.1
5.	4 EA	NUT, SELF- LOCKING, SL, 90715A165	26	2.3	9B531001 4664115	3.1
6.	1 OZ	CHEMICAL CORROSION, TEF- GEL, TG-01	27	2.3	9B685001 4504009	3.1
7.	1 OZ	CHEMICAL CORROSION, TEF- GEL, TG-01	27	2.3	9B685001 4504009	3.1
8.	1 EA	DC MOTOR, P/N CDP3410- V240.25HP, 18000 RPM, DC, 56	62	2.4	9B6105SA 2300302	3.1
9.	1 EA	MOTOR, DC CURRENT BLOWER FAN MOTORP/N 74022092BCON	-		9B610501 6083974	3.1
10.	1 EA	RELAY- SOLENOID, ENGI,	34	2.4	9B292001 2619814	3.1

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		P/N AT71088PUMP RELAY				
11.	1 EA	SWITCH,TOGGLE, P/N 8287	29	2.4	9B593001 5954870	3.1
12.	5 EA	CONDUIT, NONMETAL LIC, 3/8" CONDUIT P/N 7840K82	95	2.4	9B597501 5749289	3.1
13.	100 EA	STRAP, TIE DOWN, ELEC, P/N 7130K42	96	2.4	9B597501 5723833	3.1
14.	4 SL	WIRE, P/N 18400314 AWG BLACK 18FT/ SPOOL	39	2.4	9B6145SA 2300192	3.1
15.	1 EA	RESISTOR, P/N HS150-3.0F3.0 OHM RESISTOR, 150 WAT,	82	2.4	9B5905SA 2300245	3.1
16.	1 EA	RESISTOR, P/N HS150-3.0F3.0 OHM RESISTOR, 150 WAT,	82	2.4	9B5905SA 2300245	3.1
17.	1 EA	FUSE KIT, P/N 4603315A FUSE KIT (HOLDER)	69	2.4	9B5920SA 2300246	3.1
18.	2 OZ	CHEMICAL CORROSION, TEF- GEL, TG-01	27	2.3	9B685001 4504009	3.1
19.	1 TU	INSULATING COMPOUND, 3140 RTV ITEM NO.133	133	2.4	9B597001 4871943	3.1
20.	1 PG	SCREW, CAP, HEXAGO N H, P/N 93190A6223/8" - 16 UNC CRE	-		9B530501 5290058	3.1
21.	4 EA	SCREW, CAP, HEXAGO N H, P/N 93190A6243/8" - 16 UNC CRE	-		9B530501 5090161	3.1
22.	1 HD	WASHER, LOCK, P/N 98437A1153/8"-16 CRESI ITEM NO 89	89	2.4	9B531001 5616188	3.1
23.	1 PG	NUT, SELF- LOCKING, HE, P/N 91831A127NUT3/8" -16 CRESI	90	2.4	9B531001 4759244	3.1
24.	1 EA	SWITCH, TOGGLE, P/N 8232 ITEM 15 OF 1600909-1	-	2.4	9B593001 6054007	3.1
25.	2 EA	FUSE, ENCLOSED LINK, P/N 7460K4615A	-	2.4	9B592001 6115772	3.1
26.	4 EA	ADAPTER, STRAIGHT	15	2.3	9B473000	3.1

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SHIP:

NBU-7 (RMC 57078)

		,PI, FITTING, PARKER, 1", CRES 316			8983758	
27.	20 FT	HOSE, HOSE, PUSH LOCK, PARKER #801-12-BLUE	07	2.3	9C4720SA PTE0079	3.1
28.	20 EA	CLAMP,HOSE, 5011T14, SST 316	8	2.3	9B473001 6106830	3.1
29.	10 EA	FITTING, FITTING, PARKER, 3/4"BARBED END X 3/4"PLA	06	2.3	9B4730SA PTE0086	3.1
30.	4 FT	DUCT, FLEXIBLE DUCT, 8"ID, 0.013"THK, W/UNFINISHED	-		9B2040SA PTE0080	3.1
31.	10 FT	HOSE, HOSE, PUSH LOCK, BLUE, PARKER #801-10- BLU	16	2.3	9C4720SA PTE0081	3.1
32.	10 EA	FITTING, FITTING, PARKER, CRIMP HOSE FITTING, HOSE	18	2.3	9B4730SA PTE0085	3.1
33.	4 EA	FITTING, FITTING, PARKER, #1 X 1/2 PTR-SS	19	2.3	9B4730SA PTE0087	3.1
34.	10 EA	CONNECTOR,90 DEGREE, FITTING, PARKER, CONNECTOR 90	20	2.3	9B473001 4957859	3.1

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

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SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>514-11-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L011</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>AKAGI, TAKUYA</u> <u>NAGAMINE, YUUICHI</u> <u>SAITO, KAZUYUKI</u> <u>TERASHITA, SHUUJI</u>

1. SCOPE:

- 1.1 Title: Cooling System Component; replace
- 1.2 Location of Work:
 - 1.2.1 Radiator Enclosure
 - 1.2.2 Engine Compartment
 - 1.2.3 Midships Compartment
- 1.3 Identification:
 - 1.3.1 Quantity (2 EA), Fan Hydraulic Motor located in 1.2.1
 - Part No. - 3799855
 - Figure - No.55 of 2.4
 - 1.3.2 Quantity (2 EA), Fan Blade Assembly located in 1.2.1
 - Part No. - 700/9-9/32.5/PAG-4ZL/40mmX8mm/AHC SSHDW
 - Figure - Fig.1-15-3 / 1-16-3 of 2.3, No.3 of 2.6 and 2.7
 - 1.3.3 Quantity (2 EA), Fan Motor Solenoid located in 1.2.2
 - Part No. - DS163BD024W-A16T
 - Figure - No.65 of 2.5
 - 1.3.4 Quantity (One EA), Fan Motor Relief Valve located in 1.2.2
 - Part No. - RAH161T30
 - Figure - Fig.1-10-19 of 2.3, No.27 of 2.4
 - Set Press. - 2,300 PSI
 - 1.3.5 Quantity (One EA), Engine Coolant Radiator located in 1.2.1
 - Part No. - 00660001
 - Figure - Fig.1-15-1 of 2.3, No.1 of 2.5
 - 1.3.6 Quantity (One EA), Hydraulic Oil Radiator located in 1.2.1
 - Part No. - 00660002
 - Figure - Fig.1-16-1 of 2.3, No.2 of 2.5
 - 1.3.7 Quantity (One EA), Coolant Heat Exchanger located in 1.2.3
 - Part No. - AB1602D6FP-CNT-B-2Z
 - Figure - Fig.1-14-3 of 2.3, No.3 of 2.5
 - 1.3.8 Quantity (One EA), Hydraulic Oil Heat Exchanger located in 1.2.3

Part No. - AB1206D4FP-CNT-B-3Z
Figure - Fig.1-14-4 of 2.3, No.4 of 2.5
1.3.9 Quantity (2 EA), Sea Water Pump Solenoid located in 1.2.2
Part No. - DS163BD024W-A16T
Figure - Fig.1-14-53 of 2.3, No.65 of 2.5
1.3.10 Quantity (2 EA), Sea Water Pump located in 1.2.3
Part No. - SE11/2ELHYC
Figure - Fig.1-1-23 of 2.3, No.17 of 2.5
1.3.11 Quantity (4 EA), Seawater Strainer Assembly located in 1.2.3
Part No. - 18029
Figure - Fig.1-14-17 of 2.3, No.18 of 2.5
1.3.11.1 Quantity (4 EA), Strainer Basket of 1.3.11
Part No. - 14380
Figure - SH 14380 of ENCLOSURE (B)

2. REFERENCES:

- 2.1 Standard Items, (FY26)
- 2.2 F9YC0-AB-TMM-010 Rev 5, Change U, Systems Technical Manual for LARC-V A1/A2, Organizational and Intermediate Maintenance
- 2.3 F9YC0-AL-IPB-010, Systems Technical Manual for LARC-V A1/A2, Illustrated Parts Breakdown
- 2.4 1600911-2 Rev W, Hydraulic System Schematic
- 2.5 1600906-1 Rev M, Cooling System INST & DETS
- 2.6 1600906-2 Rev H, Engine Radiator Assembly
- 2.7 1600906-3 Rev H, Oil Cooler Assembly
- 2.8 1600907-2 Rev E, LARC V Vehicle Electronic Control System Layout & Detail
- 2.9 1600907-3 Rev M, LARC V Electronic Transmission Control Schematic Diagram
- 2.10 Test Procedure, Test No.58331-4-100_26127, Cooling System Component

3. REQUIREMENTS:

(V) (G) "SHOP TEST"

- 3.1 Accomplish the shop test and set each new relief valve listed in 1.3.4, using Table 3-26 of 2.2 and ENCLOSURE (A) for guidance. -S130
 - 3.1.1 Ensure the test medium is clean hydraulic oil conforming to MIL-PRF-17672.
 - 3.1.2 Ensure the lifting pressure is 2,300 (+50/-50) PSIG.
 - 3.1.3 Attach a metal tag to each relief valve, stamped with the following information:

- 3.1.3.1 Ship name and hull number.
- 3.1.3.2 Valve Number or identification.
- 3.1.3.3 Valve lifting pressure.
- 3.1.3.4 Date Valve tested and set.
- 3.1.3.5 Name of repair facility.

3.2 Remove existing and install each new cooling system component listed in 1.3.1 through 1.3.10 and 1.3.11.1 installing each new seal, gasket, and packing, using Paragraph 6.5 of 2.2, 2.3 through 2.9, ENCLOSURE (A) and ENCLOSURE (B) for guidance.

- 3.2.1 Matchmark, identify and retain disconnected hardware.
- 3.2.2 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.
- 3.2.3 Inspect each part not specified for renewal for wear and defect.
- 3.2.4 Remove each high spot, burr, abrasion, nick, corrosion, gasket material, and foreign matter from each flange and mating surface exposed by disassembly and removals
- 3.2.5 Remove each burr and high spot from each sliding surface, screw thread, key, and keyway exposed by disassembly and removals.
 - 3.2.5.1 Chase and tap each exposed threaded area.
- 3.2.6 Handwork and skim cut each machined, sealing, aligning, mating, and gasket surface, taking precautions to ensure minimum metal removal.
- 3.2.7 Remove existing and install each new component in place of those identified to be missing or defective.
- 3.2.8 Install retained hardware of 3.2.1.
- 3.2.9 Apply new sealant to each threaded hardware in accordance with Note One of 2.5.
- 3.2.10 Apply new grease conforming to NLGI#2 to each fan motor lubrication point, using paragraph 6.5.1.4 step b through d of 2.2 for guidance.
- 3.2.11 Tighten each fastener at each torque value stated in Paragraph 6.5 of 2.2.

3.3 Accomplish the requirements of 009-032 of 2.1 for each new and disturbed surface.

(V) (G) "OPERATIONAL TEST"

3.4 Accomplish the requirements of 2.10 (TP) for an operational test of each equipment listed in 1.3.1 through 1.3.11. -S246

4. NOTES:

4.1 Initial Condition - S/F DISCOVERED cooling system issues on LARC 936 due to corrosion to primary cooling system to include: fan hydraulic motors, fan blades, fan motor solenoids, fan motor relief valves, engine

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coolant radiator, hydraulic oil radiator, coolant heat exchanger, hydraulic oil heat exchanger, strainer baskets, seawater pump solenoids, and salt water pump assembly.

4.2 SRF Tool Room will issue each torque wrench and pressure gage when required.

4.3 Ship's Force will provide liquids for operation such as coolant, hydraulic oil, and fuel oil when refilling.

4.4 GFM listed in 5.1 have been ordered under ICN 38NBA51403-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	3 PG	SCREW,CAP,HEXAGO N H, P/N 93190A712, HHD, 1/2"-13	1-35-4	2.3	9B530501 5867692	3.2
2.	12 EA	WASHER,LOCK, P/N 92147A033, LOCK WASHER, 1/2", 316	1-35-5	2.3	9B531001 5602862	3.2
3.	1 PG	SCREW,CAP,HEXAGO N H, P/N 93190A622, 1 PG=10 EA, HH	1-35-6	2.3	9B530501 5290058	3.2
4.	1 PG	WASHER,SPLIT, P/N 92147A031, 1PG=50EA, LOCK WASHER	1-35-7	2.3	9B531001 5287188	3.2
5.	1 PG	SCREW,CA, P/N 93190A716, 1PG=5EA, HHCS, 1/2"-13 UNC	1-17-7	2.3	9B530501 5430481	3.2
6.	4 EA	NUT,SELF- LOCKING,SL, P/N 90715A165, NYLOCK NUT, 1/	1-17-8	2.3	9B531001 4664115	3.2
7.	1 PG	WASHER,FLAT, P/N 91950A033, 1/2", 316 SST, 1PG=25E	1-17-9	2.3	9B531001 5598231	3.2
8.	2 PG	SCREW,CAP,HEXAGO N, P/N 93190A678, HHCS, 7/16"-14 U	1-17-10	2.3	9B5305SA 2300181	3.2
9.	8 EA	WASHER,FLAT, P/N 90107A032, 7/16", 316 SST	1-17-11	2.3	9B531001 4764801	3.2
10.	1 PG	WASHER,LOCK, P/N 92147A032, LOCK WASHER, 7/16", 31	1-17-12	2.3	9B531001 5890368	3.2

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11.	2 EA	GASKET, BILRITE RED , GASKET, BILRITE RED SBR, 4"	1-17-13	2.3	9B5330SA 2300182	3.2
12.	4 PG	SCREW,CAP,HEXAGO N H, P/N 93190A714, HHCS, 1/2"-13	1-15-4 1-16-4	2.3	9B530501 5290063	3.2
13.	16 EA	WASHER,F, P/N 90107A033, FLAT, 1/2", SST 316	1-15-6 1-16-6	2.3	9B531001 4662744	3.2
14.	2 EA	COUPLING, P/N L600238 600- OHLA, SAE B MOUNT 12 SPL	51	2.4	9B3010SA 2300304	3.2
15.	2 EA	MOTOR,FAN, P/N 3799855	55	2.4	9B414001 6443365	3.2
16.	2 EA	FAN BLADE ASSY, NYLO, PN 27.56/9- 932.50/PAG/4ZL/4 0	3 (REF 2.7)	2.6	9B4140SA 2300305	3.2
17.	2 EA	FAN,CIRCULATING, P/N: 700/9- 9/32.5/PAG- 4ZL/40MMX8M	1-15-3 1-16-3	2.3	9B414001 5791955	3.2
18.	4 EA	VALVE,LINEAR,DIR ECT, P/N: DS163BD024W-A16T	65	2.5	9B481001 6523187	3.2
19.	1 EA	VALVE,SAFETY RELIEF, P/N: RAH161T30	1-10-19	2.3	9B482001 5457169	3.2
20.	2 KT	PACKING ASSORTMENT,, P/N SK16-2			9B533001 4816843	3.2
21.	8 EA	SCREW,CAP,HEXAGO N H, P/N 93190A624, 3/8"- 16 UNC X	1-13-2	2.3	9B530501 5090161	3.2
22.	1 PG	WASHER,SPLIT, P/N 92147A031, 1PG=50EA, LOCK WASHER	1-35-7	2.3	9B531001 5287188	3.2
23.	1 EA	RADIATOR,ENGINE COO, P/N: 00660001	1-15-1	2.3	9B293001 5745575	3.2
24.	1 EA	COOLER,LUBRICATI NG, P/N 00660002	1-16-1	2.3	9B293001 5742123	3.2
25.	16 EA	SCREW,CAP,HEXAGO N H, P/N 93190A624, 3/8"- 16 UNC X	1-13-2	2.3	9B530501 5090161	3.2
26.	1 EA	HEATER,FLUID,IND	1-14-3	2.3	9B442001	3.2

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		USTR, PN AB1602D6FP-CNT- B-2Z			6935989	
27.	20 EA	SCREW, CA, P/N 93190A626, 3/8"- 16 UNC X 1-1/4" LG,	1-14-42	2.3	9B530501 4858084	3.2
28.	1 PG	NUT, SELF LOCK, P/N; 90715A145, 3/8"-16 UNC X 3/4"	1-14-43	2.3	9B5310SA PHA0071	3.2
29.	1 EA	HEATER, FLUID, IND UST, OIL HEAT TRANSFER UNIT, P/N A	1-14-4	2.3	9B442000 1628964	3.2
30.	4 EA	STRAINER, SEDIMEN T, P/N 18029, ASSEMBLY (STRAINER B	1-1-22	2.3	9B473001 5458687	3.2
31.	2 EA	PUMP, CENTRIFUGAL , P/N: SE11/2ELHYC	1-1-23	2.3	3B432001 5461550	3.2
32.	10 OZ	CHEMICAL CORROSION, CHEMICAL CORROSION, TEF- GEL	107	2.5	9B685001 4504009	3.2
33.	2 TU	GREASE, AIRCRAFT, NLGI GRADE 2	-	2.2	9B915000 1817724	3.2
34.	4 EA	O-RING, P/N: 11803	-		9B533101 5458809	3.2
35.	1 CN	HYDRAULIC FLUID, PET, MIL- PRF-17672, 5.000 GALLONS	-		9B915000 9857237	3.1
36.	4 EA	PLASTIC BARBED HOSE , PLASTIC BARBED HOSE FITTING,	-		9B4730SA 2501313	3.2
37.	81 EA	O-RING, P/N: 3- 916	-		9B533100 4044619	3.2
38.	18 EA	O-RING, P/N: 3- 912	-		9B533101 4802366	3.2
39.	4 PG	CONNECTOR, PLUG, E LEC, P/N DT04-2P	-		9B593501 5065555	3.2
40.	4 FV	RETAINER, ELECTRI CAL, P/N W2P	-		9B593501 5191808	3.2
41.	8 EA	CONTACT, ELECTRIC AL, P/N 0460- 202-16141	-		9B599901 2163648	3.2
42.	4 EA	CONNECTOR BODY, PLUG, P/N DT06-2S	-		9B593501 6657216	3.2
43.	4 PG	RETAINER, ELECTRI	-		9B593501	3.2

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		CAL, P/N W2S- PO12			4800358	
44.	8 EA	CONTACT,ELECTRIC AL, P/N 0462- 201-16141	-		9B599901 2036687	3.2
45.	1 LG	INSULATION SLEEVEING, HEAT- SHRINK, SLEEVE, ADHESIVE	-		9B597001 2558990	3.2
46.	4 FT	INSULATE, SLEEVEING, HEAT- SHRINK, SLEEVE, ADHESIVE,	-		9B597001 1520397	3.2
47.	20 FT	HOSE, NONMETALLIC , FLEXIBLE HOSE, FC163/1509-24, 1.	-		9B472000 2895221	
48.	10 EA	ADAPTER,, REUSABLE FITTING, SAE 37 FLARE SWIVEL, 1	-		9B473000 7034393	
49.	5 EA	HOSE, FLEX HOSE, NOVAFLEX 375, FUEL FILL HOSE, USC	-		9B4730SA PTE0278	
50.	100 EA	HOSE CLAMP, TIGHT-SEAL BOLT CLAMP FOR FIRM HOSE AN	-		9B4730SA PTE0279	
51.	100 EA	HOSE CLAMP, TIGHT-SEAL BOLT CLAMP FOR FIRM HOSE AN	-		9B4730SA PTE0280	
52.	1 TU	SEALING COMPOUND, SEALG COMPOUND, MIL-S- 45180 TY.I	-		9B803000 2523391	
53.	1 PT	LUBRICATING OIL, GEN, LUBE OIL, HOSE FTG ASSY OIL P	-		9B915001 3596563	
54.	1 CO	GREASE, EXTREME PRES, HOSE FTG ASSY OIL FOR CRES &	-		9B915001 0379016	

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

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Catalog HY15-3503R1/US
Product Information

NEW

**Pilot Operated Relief Valve
 Series RAH161**

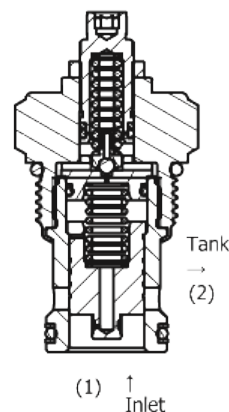
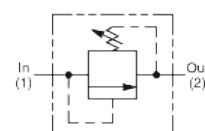
CV	Check Valves
SH	Shuttle Valves
LM	Load/Motor Controls
FC	Flow Controls
PC	Pressure Controls
LE	Logic Elements
DC	Directional Controls
MV	Manual Valves
SV	Safety Valves
PV	Proportional Valves
CE	Cats & Electronics
CB	Cartridge Bodies
BC	Bodies & Cavities
TD	Technical Data

General Description

Pilot Operated Spool-Type Relief Valve. For additional information see Technical Tips on pages PC1-PC6.

**Features**

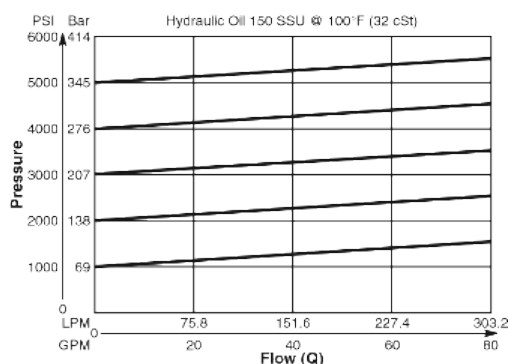
- Hardened, precision ground parts for durability
- Low profile adapter for minimal space requirements
- Fully guided poppet for more consistent reseal
- Internal screening protects pilot seat from debris
- Steel adapters are zinc plated

**Specifications**

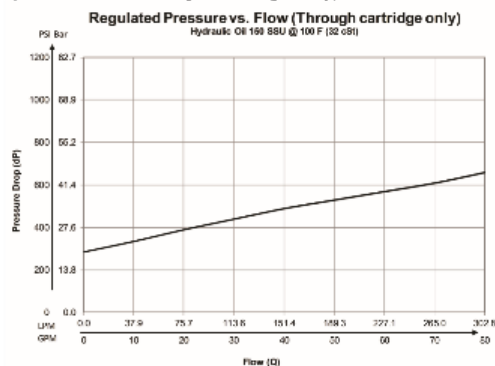
Rated Flow	302.8 LPM (80 GPM)
Maximum Inlet Pressure	380 Bar (5500 PSI)
Maximum Pressure Setting	350 Bar (5000 PSI)
Sensitivity: Pressure/Turn	10 19.6 Bar (285 PSI) 30 58.9 Bar (859 PSI) 50 131.7 Bar (1910 PSI)
Maximum Tank Pressure	350 Bar (5000 PSI)
Reseat Pressure	90% of crack pressure
Leakage at 150 SSU (32 cSt)	5 cc per 6.8 Bar (100 PSI) setting
Cartridge Material	All parts steel. All operating parts hardened steel.
Operating Temp. Range/Seals	-34°C to +121°C (Nitrile) (-30°F to +250°F) -26°C to +204°C (Fluorocarbon) (-15°F to +400°F)
Fluid Compatibility/Viscosity	Mineral-based or synthetic with lubricating properties at viscosities of 45 to 2000 SSU (6 to 420 cSt)
Filtration	ISO-4406 18/16/13, SAE Class 4
Approx. Weight	0.34 kg (0.75 lbs.)
Cavity	C16-2 (See BC Section for more details)

Performance Curve**Flow vs. Inlet Pressure**

(Pressure rise through cartridge only)

**Performance Curve****Regulated Pressure vs. Flow**

(Pressure rise through cartridge only)



PC29

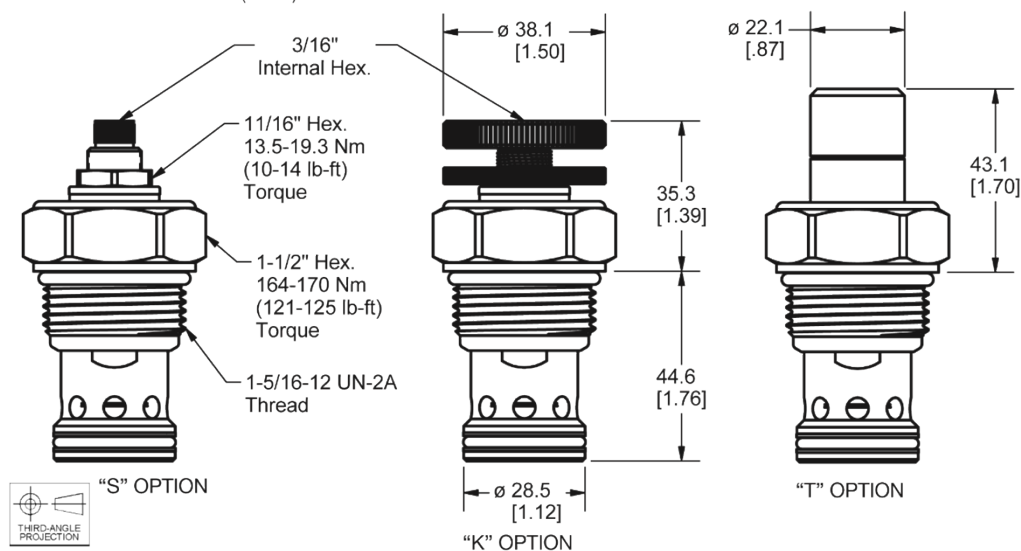
ENCLOSURE (A) 1/2

Parker Hannifin Corporation
 Hydraulic Cartridge Systems Division

Catalog HY15-3503R1/US
Product Information

Pilot Operated Relief Valve
Series RAH161

Dimensions Millimeters (Inches)



Ordering Information

RAH161

16 Size
Pilot Operated
Relief Valve



Adjustment
Style



Pressure
Range



Seals

Highlighted represents preferred options that offer the shortest lead times. Other options may be available, but at extended lead times.

Code	Adjustment Style / Kit No.
K	Knob Adjust (717784-15)
S	Screw Adjust
T	Tamper Resistant Cap (717785)

Code	Seals / Kit No.
Omit	Nitrile / (SK16-2)
V	Fluorocarbon / (SK16-2V)

Order Bodies Separately
See section BC

B16	-	2	-	
16 size		2-Way Cavity		Port Size

Code	Porting / Body Material
16T	SAE-16 / Steel (5000 PSI)
A16T	SAE-16 / Aluminium (3000 PSI)

Code	Pressure Range
10	6.9 - 69 Bar (100 - 1000 PSI) Standard Setting: 34.5 Bar (500 PSI) @ .95 LPM (.25 GPM)
30	20.7 - 207 Bar (300 - 3000 PSI) Standard Setting: 103.5 Bar (1500 PSI) @ .95 LPM (.25 GPM)
50	34.5 - 345 Bar (500 - 5000 PSI) Standard Setting: 172.4 Bar (2500 PSI) @ .95 LPM (.25 GPM)



PC30
ENCLOSURE (A) 2/2

Parker Hannifin Corporation
 Hydraulic Cartridge Systems Division

CV	Check Valves
SH	Shuttle Valves
LM	Load/Motor Controls
FC	Flow Controls
PC	Pressure Controls
LE	Logic Elements
DC	Directional Controls
MV	Manual Valves
SV	Solenoid Valves
PV	Proportional Valves
CE	Coils & Electronics
CB	Cartrp. Bodies
BC	Bodies & Cavities
TD	Technical Data

SHERWOOD Sea Water Strainers

Durable Non-Corrosive Strainers provide an excellent filtration for fresh and salt water applications.

Sherwood Sea Water strainers are designed to deliver full flow of liquid at ultra low pressure differential while providing effective filtration of foreign matter in supply line. Convenient ports feature stainless steel ferrule reinforcements. Strainer cap and bowl are connected by rugged National Buttress Threads (NBT). Buna-N gasket forms a tight fluid seal against bowl leakage.

Cleaning is accomplished easily without removing strainer from lines. Simply unscrew lower bowl from cap and lift out screen. Flush sediment from mesh with water and replace screen by first fitting to flange in cap. Align gasket to assure proper seal before tightening bowl. Replace gasket if it is cut or showing excessive wear. Turning bowl hand tight provides sufficient seal against cap. Clean strainers regularly for best system performance.

FEATURES:

ADDED FILTRATION.

Stainless steel screen provides a large ratio of screen area to port size.

NON-CORROSIVE CONSTRUCTION.

Nylon cap, polyamide bowl, stainless steel screen provides maximum protection against electrolysis.

EASY TO CLEAN.

Clear bowl enables quick

examination for cleaning without removing the strainer from the line.

LIGHTWEIGHT MOUNT. Offers simple in line mounting and eliminates the need for supporting brackets

Strainer should be mounted away from areas subjected to high engine heat transfer. Use available mounting bracket to position strainer away from engine for easy access.

For Pumps and Parts
go to PumpVendor.com



This illustration shows how flow is channeled uniformly through the screen in Sherwood line strainers. Sediments are trapped inside screen with ample open screen area to minimize clogging of the strainer.



MOUNTING BRACKETS FOR LINE STRAINERS

Brackets provide a sturdy mounting harness for Sherwood Line Strainers. Unique design enables easy trouble-free assembly without further use of clamps and brackets. Does not effect quick cleaning ability of strainer bowl.

Model No.	Port Size	Screen Mesh
13227	1/4" ID 1/2" OD	20 Mesh Stainless Steel
15935	1/2" (F)	20 Mesh Stainless Steel
18001	3/4" (F)	20 Mesh Stainless Steel
18005	1" (F)	20 Mesh Stainless Steel
18016	1 1/4" (F)	20 Mesh Stainless Steel
18029	1 1/2" (F)	20 Mesh Stainless Steel

Bracket #	Fits -
14240	3/4" Strainers
14239	1" Strainers
14238	1 1/4" Strainers
21930	1 1/2" Strainers

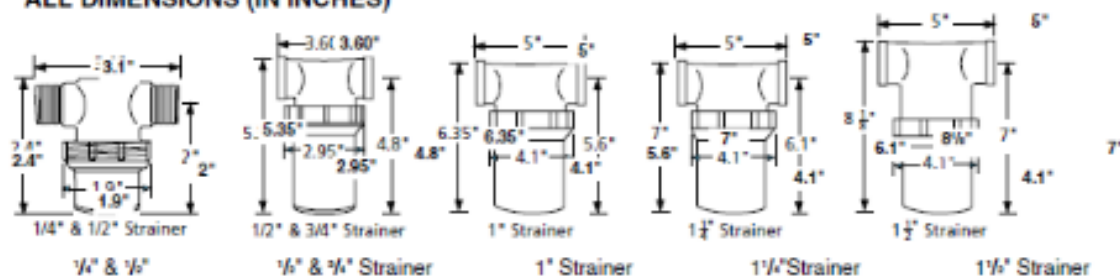
ENCLOSURE (B) 1/2

SHIP:

NBU-7 (RMC 57078)

SHERWOOD #	STRAINER COMPONENTS	
	DESCRIPTION	STRAINER SIZE
11800	Cap	1"
11801	Cap	3/4"
11802	Cap	1/2"
11803	O-ring-BunaN	1", 1-1/4", 1-1/2"
11804	O-ring-Viton	1", 1-1/4"
11805	O-ring-BunaN	1/2", 3/4"
11806	O-ring-Viton	1/2", 3/4"
11807	Screen-20 mesh	1"
11808	Screen-40 mesh	1"
11809	Screen-80 mesh	1"
11810	Screen-20 mesh	1/2", 3/4"
11811	Screen-40 mesh	1/2", 3/4"
11812	Screen-80 mesh	1/2", 3/4"
11814	Bowl-Solid Plastic	1", 1-1/4", 1-1/2"
11815	Bowl-Solid Plastic	1/2", 3/4"
13245	Cap	1-1/4"
14214	Screen-20 mesh	1-1/4"
14379	Bowl-Clear plastic	1/2", 3/4"
14380	Bowl-Clear plastic	1", 1-1/4", 1-1/2"
15929	O-ring-BunaN	1/4", 3/8"
15930	Screen-20 mesh	1/4", 3/8"
15931	Cap	1/4"
15932	Bowl-Solid plastic	1/4", 3/8"
16126	Screen-20 mesh	1-1/2"
18240	Cap	1-1/2"

ALL DIMENSIONS (IN INCHES)

A WICOR COMPANY
MARINE PRODUCTS GROUP375 Fifth Avenue NW • New Brighton, NY 11512-3288
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10/97 HC
Printed in USA

ENCLOSURE (B) 2/2

SSP: SASN-003-26

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ITEM NO: 514-11-001

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SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>561-11-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L008</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>AKAGI, TAKUYA</u> <u>NAGAMINE, YUUICHI</u> <u>SAITO, KAZUYUKI</u> <u>TERASHITA, SHUUJI</u>

1. SCOPE:

1.1 Title: Steering System Component; replace

1.2 Location of Work:

1.2.1 Operator's Cabin

1.2.2 Forward Compartment

1.2.3 Under Hull

1.2.4 Midships Compartment

1.3 Identification:

1.3.1 Quantity (One EA), Steering Wheel located in 1.2.1

Parts No. - 404266

Figure - Fig.1-55-21 of 2.3 / No.23 of 2.4

1.3.2 Quantity (One EA), Steering Colum located in 1.2.1

Parts No. - SK000014-3175

Figure - Fig.1-55-22 of 2.3 / No.24 of 2.4

1.3.3 Quantity (One EA), Upper Steering Arm located in 1.2.2

Parts No. - 1600904-2/04 / 1600904-2/63 (See Note 5 of 2.6)

Figure - Fig.1-25-3 of 2.3 / No.4 and No.63 of 2.6

1.3.4 Quantity (One EA), Steering Control Valve located in 1.2.2

Parts No. - HGA32120

Figure - Fig.3-115 of 2.2 / No.35 of 2.8

1.3.5 Quantity (One EA), Rudder Steering Arm located in 1.2.2

Parts No. - 1600904-2/30

Figure - Fig.1-25-29 of 2.3, No.30 of 2.6

1.3.6 Quantity (2 EA), Wheel Tie Rod located in 1.2.3

Parts No. - 1600904-2/2

Figure - Fig.1-25-2 of 2.3, No.2 of 2.6

1.3.7 Quantity (One EA), Steering Shaft located in 1.2.2

Parts No. - 1600904-2/08 / 1600904-2/64 (See Note 5 of 2.6)

Figure - Fig.1-25-7 of 2.3 / No.8 and No.64 of 2.6

1.3.8 Quantity (One EA), Rudder Angle Indicator Assembly located in 1.2.1, Driver Board (P/N: SCT0342) FIND No. 2 of 2.10, and Rudder Angle Indicator (P/N: SCT0580) FIND No. 123 of 2.10, MFR- Sentinal

Control Tech Inc and Backing Ring (4-3/8 Inch OD by 3-11/16 Inch ID by 1/8 Inch Thickness, Polyethylene (UHMW), Commercial Source)

1.3.9 Quantity (One EA), Potentiometer located in 1.2.2 for the Rudder Angle Indicator listed in 1.3.8, P/N CLP-250, MFR - Celesco Inc, FIND No. 61 of 2.10 and ITEM No. 23 of 2.7

1.3.10 Quantity (One EA), Hydraulic Flow Alert Switch located in 1.2.4, P/N H700A-020-F2, MFR - Hedland Flotech Inc, FIND No. 37 of 2.8

1.3.11 Quantity (One EA), Lower Steering Arm located in 1.2.3
Parts No. - 1600904-2/01 / 1600904-2/62 (See Note 5 of 2.6)

Figure - Fig.1-25-1 of 2.3, No.One and No.62 of 2.6

1.3.12 Quantity (One EA), Steering Cylinder located in 1.2.2
Parts No. - 4.00CKDD-2HCTS38A/38AC

Figure - Fig.1-25-23 of 2.3, No.24 of 2.6

1.3.13 Quantity (2 EA), Wheel Steering Arm located in 1.2.3
Parts No. - PDI-10020

Figure - Fig.1-23-56 of 2.3, No.9 of 2.6

2. REFERENCES:

2.1 Standard Items, (FY26)

2.2 F9YC0-AB-TMM-010 Rev 5, Change U, Systems Technical Manual for LARC-V A1/A2, Organizational and Intermediate Maintenance

2.3 F9YC0-AL-IPB-010, Systems Technical Manual for LARC-V A1/A2, Illustrated Parts Breakdown

2.4 1600909-1 Rev V, Operator's Console & Controls

2.5 1600909-1 Rev R, Operator's Console & Controls

2.6 1600904-2 Rev R, Land Steering System

2.7 1600904-2 Rev G, Land Steering System

2.8 1600911-2 Rev W, Hydraulic System Schematic

2.9 1600904-9 Rev F, Rudder Steering System

2.10 1600907-1 Rev W, Electrical System Installation & Details

2.11 1600907-2 Rev E, LARC V Vehicle Electronic Control System Layout & Details

2.12 1600907-3 Rev M, Electronic Transmission Control Schematic Diagram

2.13 Maintenance Requirement Card, MIP Control Number: 59A32/B02, MRC: 45 2UG6 N, Lubricate LARC V (Enclosed)

2.14 Test Procedure, Test No.58331-4-100_26128, Steering System Component

3. REQUIREMENTS:

3.1 Remove each existing and install each new steering system component listed in 1.3.1 through 1.3.12 installing each new seal, gasket and

packing, using Paragraphs 6.7 of 2.2, 2.3 through 2.12 and ENCLOSURE (A) for guidance.

- 3.1.1 Matchmark, identify and retain disconnected hardware.
- 3.1.2 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.
- 3.1.3 Inspect each part not specified for renewal for wear and defect.
- 3.1.4 Remove each high spot, burr, abrasion, nick, corrosion, gasket material, and foreign matter from each flange and mating surface exposed by disassembly and removals.
- 3.1.5 Remove each burr and high spot from each sliding surface, screw thread, key, and keyway exposed by disassembly and removals.
 - 3.1.5.1 Chase and tap each exposed threaded area.
- 3.1.6 Handwork and skim cut each machined, sealing, aligning, mating, and gasket surface, taking precautions to ensure minimum metal removal.
- 3.1.7 Remove existing and install each new component in place of those identified to be missing or defective.
- 3.1.8 Install retained hardware of 3.1.1.
- 3.1.9 Remove defective and install new electric connectors listed in paragraph 5.
- 3.1.10 Accomplish the calibration of the new equipment listed in 1.3.8 in accordance with paragraph 6.7.8.5 of 2.2.
- 3.1.11 Accomplish the calibration of the new equipment listed in 1.3.10 in accordance with paragraph 6.7.10.5 of 2.2.
- 3.1.12 Seal each thread fastener, electrical end connection, and electrical box and printed circuit board of the new equipment in 1.3.8, using note 4, 6 and 7 of 2.10 for guidance.
- 3.1.13 Adjust each tie rod, using Paragraph 6.7.5.4 step d of 2.2 for guidance.
- 3.1.14 Apply new grease conforming to A-A-50433A to each tie rod lubrication point, using Step h. of 2.13 for guidance.
- 3.1.15 Apply new sealant to each threaded hardware in accordance with Note 4 of 2.4 and Note one of 2.6.
- 3.1.16 Tighten each fastener at each torque value stated in Paragraph 6.7 of 2.2.

3.2 Accomplish the requirements of 009-032 of 2.1 for each new and disturbed surface.

(V) (G) "OPERATIONAL TEST"

3.3 Accomplish the requirements of 2.14 (TP) for an operational test of each steering system component listed in 1.3.1 through 1.3.13. -S246

4. NOTES:

4.1 Initial Condition - S/F DISCOVERED several steering system issues on LARC 936 due to corrosion to primary steering system components to include: steering wheel, steering column, upper steering arm, steering control valve, rudder steering arm, wheel tie rod, steering shaft, rudder angle indicator, rudder angle indicator potentiometer, hydraulic flow alert switch, lower steering arm, steering cylinder.

4.2 SRF Tool Room will issue each torque wrench and pressure gages when required.

4.3 Each reference listed in 2.5, 2.7 and 2.10 is applicable to this electrical work item only based on response on ESR MISC-301.

4.4 GFM listed in 5.1 have been ordered under ICN 38NBA56102-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	1 EA	STEERING WHEEL, P/N: 404266, 23 (REF 2.4)	1-55-21	2.3	9B253001 5456456	3.1
2.	1 EA	STEERING COLU, P/NSK000014- 3175, 24 (REF 2.4)	1-55-22	2.3	9B253001 5456447	3.1
3.	2 EA	SPACER, SLEEVE, P/N: 1600904-2 FN06	6	2.6	9B536501 6416448	3.1
4.	1 PG	NUT, 5/8"-11UNC, 10 (REF 2.6)	1-25-9	2.3	9B531001 5287648	3.1
5.	1 EA	PIN, P/N 1600904-2, FIND NO.18 (REF 2.6)	1-25-17	2.3	9B531501 6523743	3.1
6.	1 PG	SCREW, #10-24 UNC X 0.50" LG, 316, P/N: 92185A242	1-25-20	2.3	9B530501 6084222	3.1
7.	1 EA	SCREW, #10-24 UNC X 0.50" LG, 316, P/N: 92185A242	1-25-20	2.3	9B530501 6084222	3.1
8.	1 PG	SCREW, 10-24 UNC X 0.75" LG, 316, P/N:92185A245	1-25-21	2.3	9B530501 6543696	3.1
9.	1 EA	SCREW, 10-24 UNC X 0.75" LG, 316, P/N:92185A245	1-25-21	2.3	9B530501 6543696	3.1
10.	1 PG	WASHER, 3/8"DIA, 316SST, 2 7 (REF 2.6)	1-25-26	2.3	9B531001 5959268	3.1
11.	3 EA	SCREW, 3/8"-16 UNC X 1.50" LG, 316, 28 (REF 2.6)	1-25-27	2.3	9B530501 3898016	3.1

SHIP:

NBU-7 (RMC 57078)

12.	1 EA	NUT, 1-1/2"-6UNC, 29 (REF 2.6)	1-25-28	2.3	9B5310SA 2300306	3.1
13.	2 EA	NUT, SELF, 1- 1/2"-6 UNC, 316 SST, 32 (REF 2.6)	1-25-31	2.3	9B531001 6411401	3.1
14.	2 EA	BOLT, 1"-14 X 4" LG, P/N 91257A989, 33 (REF 2.6)	1-25-32	2.3	9B5305SA 2300307	3.1
15.	2 EA	NUT, P/N: 95030A320, 1"- 14, 34 (REF 2.6)	1-25-33	2.3	9B531001 5073998	3.1
16.	2 EA	PIN, COTTER, P/N: 98401A591, 5/16" X 2", 316 SST	1-25-34	2.3	9B531501 6158485	3.1
17.	1 EA	SERVOVALVE, HYDRA ULI, P/N: HGA32120	35	2.8	9B482001 5457116	3.1
18.	4 EA	NUT, P/N 1WU35, 1-1/4"-12UNF, CRES, 42 (REF 2.6)	1-25-39	2.3	9B531001 6112068	3.1
19.	4 EA	TIE ROD END, P/N L24SV8209B12, 43 (REF 2.6)	1-25-40	2.3	9B253000 9176776	3.1
20.	1 EA	BEARING, BRONZE, P/N 1600904- 2/15, 15 (REF 2.6)	1-25-14	2.3	9B3120SA 2300162	3.1
21.	2 EA	BEARING, WASHER, T HRU, P/N: 5906K413, 16 (REF 2.6)	1-25-15	2.3	9B312001 5788819	3.1
22.	1 EA	CYLINDER ASSEMBLY, 24 (REF 2.6)	1-25-23	2.3	9B304001 5456383	3.1
23.	2 EA	ROD EYE KNUCKL, PN 69096 (1600904- 2/25), 25 (REF 2.6)	1-25-24	2.3	9B5340SA 2300171	3.1
24.	1 EA	COUPLING, P/N: M050-1212, 26 (REF 2.6)	1-25-25	2.3	9B3010SA 2300310	3.1
25.	1 EA	STEERING ARM, P/N 1600904- 2/30, 30 (REF 2.6)	1-25-29	2.3	9B5340SA 2300166	3.1
26.	1 EA	BUSHING, LOWER, P/N PDI-10037, 39 (REF 2.6)	1-25-36	2.3	9B4730SA 2300163	3.1
27.	10 OZ	TEF-GEL, NOTE 4 (2.4) AND NOTE ONE (2.6)	-		9B685001 4504009	3.1

SSP: SASN-003-26

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ITEM NO: 561-11-001

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NBU-7 (RMC 57078)

28.	1 EA	BUSHING, SLEEVE, P/N: 1600904-2 FN 39	39	2.6	9B312001 6521574	3.1
29.	2 EA	SLEEVE, 4.005'' ODX3.495' 'ID, 2.500'' LG, SEE NOTE 5	65	2.6	9B312000 9853417	3.1
30.	2 EA	SEAL, PLAIN ENCASED, P/N: 53X6534R2	67	2.6	9B533000 9832544	3.1
31.	24 EA	O-RING, P/N: AS568-214	-		9B533100 6180490	3.1
32.	1 PM	PLATE, BRONZE, SIZE: 1/8" THK X 24" W X 24" LG, S	36	2.6	9B9530SA 2501338	3.1
33.	2 EA	BRONZE ROD, DIAMETER 2 1/2", LENGTH 6 1/2"	15, 17, 39	2.6	9B6680SA 2501337	3.1
34.	1 EA	STAINLESS STEEL ROD, P/N: 8936K34, FOR FABRICATION	18, 44	2.6	9B6680SA PSH0021	3.1
35.	1 EA	BRONZE ROD, OVERSIZED 932, DIA 4-1/2", LENGTH 13"	66	2.6	9B6680SA 2501339	3.1
36.	1 EA	UPPER ARM, P/N: 1600904-2/04, 4 (REF 2.6)	1-25-3	2.3	9B5340SA 2501885	3.1
37.	2 EA	TIE ROD, P/N 1600904-2/02, 2 (REF 2.6)	1-25-2	2.3	9B5340SA 2501886	3.1
38.	1 EA	SHAFT, STEERING, P/N 1600904- 2/08, 8 (REF 2.6)	1-25-7	2.3	9B3040SA 2501887	3.1
39.	1 EA	LOWER ARM, P/N: 1600904-2/01, 1 (REF 2.6)	1-25-1	2.3	9B2530SA 2501888	3.1
40.	1 EA	RESISTOR, VARIABLE, W, POTENTIOMETER P/N CLP-250, (M	61	2.10	9B590501 5460197	3.1
41.	1 EA	SWITCH, FLOW, HYDRAULIC FLOW ALERT SWITCH P/N H700A	37	2.8	9B593001 5456750	3.1
42.	1 EA	INDICATOR, ANGLE, RUDDER ANGLE INDICATOR P/N SCT034	2	2.10	9B632001 5820161	3.1
43.	1 EA	CONNECTOR, PLUG, E LEC, P/N DT04-3P	117	2.10	9B593501 5150573	3.1
44.	1 EA	CONNECTOR, PLUG, E	118	2.10	9B593501	3.1

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		LEC, DT06-3S			5238855	
45.	3 EA	CONTACT,ELECTRIC AL, P/N 0460- 202-16141	119	2.10	9B599901 2163648	3.1
46.	3 EA	CONTACT,ELECTRIC AL, P/N 0462- 201-16141	120	2.10	9B599901 2036687	3.1
47.	1 EA	SLEEVE,LOCKING,E LEC, P/N W3P	121	2.10	9B593501 4708337	3.1
48.	1 EA	POLARIZING KEY,ELEC, P/N W3S	122	2.10	9B593501 4708342	3.1
49.	1 EA	METER DRIVER BOARD, P/N SCT0358FOR RUDDER ANGLE IN	124	2.10	9B6320SA 2300247	3.1
50.	1 TU	INSULATING COMPOUND, 3140 RTV, ITEM 133END CONNECT	133	2.10	9B597001 4871943	3.1
51.	1 BX	INHIBITOR,CORROS ION, P/N VCI- 101, CORROSION INHIBI	134	2.10	9B685001 3381392	3.1
52.	1 CN	DESICCANT,ACTIVA TED, P/N 8790670-8, ITEM 135	135	2.10	9B685000 2646564	3.1
53.	1 CN	CONFORMAL COATING, P/N CTAR-12, ITEM 136S245 CONCU	136	2.10	9B597001 4174158	3.1
54.	1 EA	SCREW, 10-24 UNC X 0.75" LG, 316, P/N:92185A245			9B530501 6543696	3.1
55.	1 EA	SCREW, #10-24 UNC X 0.50" LG, 316, P/N: 92185A242			9B530501 6084222	3.1
56.	1 EA	INDICATOR, RUDDER ANGLE INDICATORP/N SCT0580, ITEM	123	2.10	9B6320SA 2300248	3.1
57.	1 HD	SCREW,CAP,HEXAGO N H, P/N 93190A540, 1/4" -20 UNC -	-		9B530501 3524397	3.1
58.	10 EA	WASHER,LOCK, P/N 91475A029, 1/4"- 20	-		9B531001 4835290	3.1
59.	2 EA	BOX CONNECTOR,ELECT, P/N DB-1478	-		9B597500 2959696	3.1
60.	4 EA	O-RING, P/N	-		9B533101	3.1

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SHIP:

NBU-7 (RMC 57078)

	ZMSCH70006			6587266	
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5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

ESR RESPONSE on MISC -301

This email serves as S245 official email response to ESR MISC-301.

BLUF: Purchase P/N 000580 to replace the rudder angle indicator.

ESR Title: Rudder Angle Indicator; procure

REF:

- (a) ESR MISC-301, Rudder Angle Indicator; procure
- (b) NAVFAC EXWC Dwg 1600907-1 Rev. W, Electrical System Installation & Details

Attachment:

- (1) PN 000580 – Part Information and Pictures
- (2) E-mail Correspondence – NAVFAC EXWC Code EX262, dated 6 June 2023

Background:

1. Reference (a) requested SRF-JRMC Det Sasebo Design Branch Code S245 to research the correct part number to replace the rudder angle indicator. Currently, Planner has ordered the indicator P/N SCT0580 per F/N 123 in reference (b). However, the supplier (KLB Enterprises) responded that P/N SCT0580 is not a part number for the rudder angle indicator. Per their inventory, P/N 000580 is the correct part number for the indicator. See Attachment (1) for details.
2. Design Branch Code S245 contacted NAVFAC EXWC Code EX262 to verify the correct part number for the replacement per Attachment (2).
3. Later revisions of reference (b) incorporate an alteration to install a different rudder angle indicator, but that will not be accomplished at this time.

Designer's Comments:

1. Purchase P/N 000580 to replace the rudder angle indicator.

ENCLOSURE (A) 1/1

DISTRIBUTION STATEMENT A:

Approved for public release: distribution unlimited.

Date: April 2025 **MIP** 59A3 **MRC:** 45 2UG6 N **Periodicity:** M-4R
Series:

Location:**Ship System:** Special Purpose Systems 59000**System:** Logistics Over The Shore (LOTS) Support 59A00**SubSystem:** Logistics Over The Shore (LOTS), Personnel Transport 59A30**Equipment:** Lighter, Amphibious, Resupply, Cargo V (LARC V) 59A32

Rates	ManHours	Rates	Man- Hours	Rates	ManHours
CM3	1.5	CMCN	0.7		
Total ManHours:	2.2	Elapsed Time:	1.5		

MAINTENANCE REQUIREMENT DESCRIPTION

1. [Lubricate LARC V.](#)

SAFETY PRECAUTIONS

1. Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.
2. To prevent personnel injury, Tag-Out procedures shall be in accordance with the Tag-out Users Manual (TUM), and/or local tagout instruction.
3. Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.

TOOLS, PARTS, MATERIALS, TEST EQUIPMENT**MATERIALS**

1. [\[000073\]](#)Grease, Termalene 2, water-wash resistant, CID A-A-50433 Hazardous Material User's Guide (HMUG) Group 13
2. [\[001144\]](#)Tag, safety
3. [\[018450\]](#)Grease, white lithium, (As required in Step [1.c.\(3\)](#).) (Contingency)
Hazardous Material User's Guide (HMUG) Group 13

TOOLS

1. [\[000771\]](#)Lubricating gun, hand, Lever-operated, 16 OZ 12" rub hose ext & adpt kit incl
2. [\[018451\]](#)Creeper, mechanics, W/C provide, (If desired) (Contingency)
3. [\[020301\]](#)Tool kit, CESE maintenance, basic

MISCELLANEOUS

1. [\[010949\]](#)Spectacles, industrial with side cover
2. [\[017934\]](#)Gloves, 4-mil Nitrile, Light use only
3. [\[001365\]](#)Technical manuals/drawings, W/C provide (F9YC0-AM-TMM-030 LARC V; Technical Manual Supplement for Lighter Amphibious Resupply Cargo, Five Ton (LARC-V A1/A2))
4. [\[011922\]](#)S9086-WK-STM-020/NSTM 670 Vol II, Hazardous Material Users Guide (HMUG)
5. [\[015244\]](#)S0400-AD-URM-010/TUM; TAG OUT USER'S MANUAL

NOTE: Numbers in brackets can be referenced to Standard PMS Materials Identification Guide (SPMIG) for identification.

HAZARDOUS MATERIALS CONTROL STATEMENT(U)

For all Navy commands, units, and activities regardless of warfare community, OPNAV M-5100.23 Navy Safety and Occupational Health Manual provides guidance for Hazardous Material Control and Management (HMC&M) and directs Safety Data Sheets (SDS) be maintained and accessible to review, for all HM at the command. For forces afloat, the Hazardous Material Users Guide (HMUG), S9086-WK-STM-020, NSTM Chapter 670 Volume II supplements the technical data found in SDSs by providing additional control measures, precautions, personal protective equipment (PPE), and spill controls for the hazardous material(s) identified in the Tools, Parts, Materials, Test, Equipment block. Maintenance personnel shall determine if additional PPE is necessary to accomplish the MRC and take appropriate action to obtain and wear such PPE to ensure the safety of maintenance personnel. Final disposition of all HAZMAT, both shipboard and ashore, shall be IAW local HAZMINCEN/CHIMP guidance as required by OPNAV-M 5090.1 series Table 35-3.

PROCEDURE

SCHEDULING CRITERIA

Accomplish this maintenance requirement when any of the following periodicities or situations occur.

- a. Monthly.
- b. After each days use.

WARNING: Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.

Preliminary

- a. Ensure that the LARC is in a maintenance-ready condition.

NOTE 1: The steering ball valve handle is accessible in the operator's cabin under the port passenger seat. During normal operation the ball valve handle should be in the OPEN position.

- b. Ensure the emergency steering ball valve is in the OPEN position, if applicable.

WARNING: To prevent personnel injury, Tag-Out procedures shall be in accordance with the Tag-out Users Manual (TUM), and/or local tag-out instruction.

- c. Attach safety tag in accordance with the Tag-out Users Manual (TUM) and/or local tag-out instruction.

WARNING: Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics. d. Put on eye protection and gloves.

1. Lubricate LARC V.

- a. Lubricate Propeller drive shaft lubrication points inside center bay (#1 on Figure 1 and Figure 2):⁽¹⁾
Inspect propeller drive shaft bearing lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.a.(2) through Step 1.a.(3) and proceed to Step 1.a.(4). If fresh or excessive grease is not present proceed to Step 1.a.(2).
 - (2) Clean propeller drive shaft bearing lubrication fittings.
 - (3) Using grease gun, apply grease three to four pumps to propeller drive shaft grease fitting. Do not over-grease to avoid damaging craft component seals.

- (4) Remove excess grease from bearing.
- (5) Repeat Step 1.a.(1) through Step 1.a.(4) for remaining propeller drive shaft grease fittings.
- b. Lubricate Rudder post. Inside Aft Stbd engine bay near hatch with Lubrication Access improvements (#2 on Figure 1 and Figure 4), without Lubrication Access improvements (#2 on Figure 6):
 - (1) For craft without the Lubrication Access improvements proceed to Step 1.b.(2). Inspect rudder post remote lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.b.(3) through Step 1.b.(4) and proceed to Step 1.b.(5). If fresh or excessive grease is not present proceed to Step 1.b.(3).
 - (2) Inspect rudder post cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.b.(3) through Step 1.b.(4) and proceed to Step 1.b.(5). If fresh or excessive grease is not present proceed to Step 1.b.(3).
 - (3) Clean rudder post remote lubrication fitting.
 - (4) Using grease gun, apply grease to rudder post remote lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (5) Remove excess grease from remote lubrication fitting.
- c. Lubricate Aft Port/Stbd rudder cable lube points (Located either side of engine) with Lubrication Access improvements (#3 on Figure 1 and Figure 4), without Lubrication Access improvements (#3 on Figure 6):
 - (1) Inspect Aft Port/Stbd rudder cables lube points for fresh or excessive lubricant. If fresh or excessive lubricant is present omit Step 1.c.(2) through Step 1.c.(3) and proceed to Step 1.c.(4). If fresh or excessive lubricant is not present proceed to Step 1.c.(2).
 - (2) Clean Aft Port/Stbd rudder cables and linkages.
 - (3) Apply lithium grease to Aft Port/Stbd rudder cables and linkages.
 - (4) Remove excess grease from Aft Port/Stbd rudder cables and linkages.
 - (5) For craft without the Lubrication Access improvements proceed to Step 1.c.(6). Inspect remote Port rudder cable lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.c.(7) through Step 1.c.(8) and proceed to Step 1.c.(9). If fresh or excessive grease is not present proceed to Step 1.c.(7).
 - (6) Inspect Aft Port/Stbd rudder cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.d.(8) through Step 1.d.(9) and proceed to Step 1.d.(10). If fresh or excessive grease is not present proceed to Step 1.d.(8).
 - (7) Clean rudder cable lubrication fitting.
 - (8) Using grease gun, apply grease to remote Port rudder cable lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (9) Remove excess grease from remote Aft Port rudder cable lubrication fitting.
 - (10) Repeat Step 1.c.(7) through Step 1.c.(9) for remote Stbd rudder cable lubrication fitting.

- d. Lubricate Fwd Port/Stbd steering cable lube points (Located inside hull) with Lubrication Access improvements (#4 on Figure 1 and Figure 4), without Lubrication Access improvements (#4 on Figure 5):
- (1) Inspect Fwd Port/Stbd steering cables lube points for fresh or excessive lubricant. If fresh or excessive lubricant is present omit Step 1.d.(2) through Step 1.d.(3) and proceed to Step 1.d.(4).
 - (2) Clean Fwd Port/Stbd steering cables and linkages.
 - (3) Apply white lithium grease to Fwd Port/Stbd steering cables and linkages.
 - (4) Remove excess lubricant from Fwd Port/Stbd steering cables and linkages.
 - (5) For craft without the Lubrication Access improvements proceed to Step 1.d.(6). Inspect remote Fwd Port/Stbd steering cables lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.d.(7) through Step 1.d.(8) and proceed to Step 1.d.(9).
 - (6) Inspect Fwd Port/Stbd steering cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present,omit Step 1.d.(7) through Step 1.d.(8) and proceed to Step 1.d.(9). If fresh or excessive grease is not present proceed to Step 1.d.(7).
 - (7) Clean steering cable lubrication fitting.
 - (8) Using grease gun, apply grease to remote Port steering cable lubrication fitting. Do not overgrease to avoid damaging craft component seals.
 - (9) Remove excess grease from remote Fwd Port steering cable lubrication fitting.
 - (10) Repeat Step 1.d.(7) through Step 1.d.(9) for remote Stbd steering cable lubrication fitting.
- e. Lubricate hatch dogging mechanism (6 ea) with Lubrication Access improvements (#5 on Figure 6), without Lubrication Access improvements (#5 on Figure 2):
- (1) Inspect hatch hinges and dogging mechanisms for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.e.(2) through Step 1.e.(3) and proceed to Step 1.e.(4). If fresh or excessive grease is not present proceed to Step 1.e.(2).
 - (2) Clean hatch dogging mechanism.
 - (3) Using grease gun, apply grease to hatch hinge and dogging mechanism lubrication fitting until grease extrudes from hinge and pivot point.
 - (4) Remove excess grease.
 - (5) Repeat Step 1.e.(1) through Step 1.e.(4) for remaining hatch hinge and dogging mechanisms.

NOTE 2: For craft with Steering System Modification completed proceed to Step 1.e.(6). For craft without the Steering System Modification completed omit Step 1.e.(6) through Step 1.e.(9). and proceed to Step 1.f.

NOTE 3: For craft with Lubrication Access Improvements complete proceed to Step 1.f. (6)

Inspect cabin floor bulkhead fittings for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.e.(7) through Step 1.e.(8) and proceed to Step 1.e.(9). If fresh or excessive grease is not present proceed to Step 1.e.(7).

- (7) Clean cabin floor bulkhead fitting between center seat and port seat.
- (8) Using grease gun, apply grease to cabin floor bulkhead fitting until grease extrudes from bulkhead fitting if required.
- (9) Remove excess grease.
- f. Lubricate Engine/bow/top hatch (16 points) (#6 on Figure 2):
 - (1) Inspect Engine/bow/top hatch for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.f.(2) through Step 1.f.(3) and proceed to Step 1.f.(4). If fresh or excessive grease is not present proceed to Step 1.f.(2).
 - (2) Clean hatch hinge and lubrication fitting.
 - (3) Using grease gun, apply grease to hatch hinge and lubrication fitting until grease extrudes from hinge and pivot point.
 - (4) Remove excess grease from lubrication fitting.
 - (5) Repeat Step 1.f.(1) through Step 1.f.(4) for remaining hatch hinge lubrication fittings.
- g. Lubricate Steering post inside Fwd bilge area with Lubrication Access improvements (#7 on Figure 1), without Lubrication Access improvements (#7 on Figure 6):
 - (1) For craft without the Lubrication Access improvements proceed to Step 1.g.(2). Inspect remote steering post inside Port Fwd bilge area for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.g.(3) through Step 1.g.(4) and proceed to Step 1.g.(5). If fresh or excessive grease is not present proceed to Step 1.g.(3).
 - (2) Inspect steering post inside Port Fwd bilge area for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.g.(3) through Step 1.g.(4) and proceed to Step 1.g.(5). If fresh or excessive grease is not present proceed to Step 1.g.(3).
 - (3) Clean steering post and remote lubrication fitting, if applicable.
 - (4) Using grease gun, apply grease to remote steering post lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (5) Remove excess grease from steering post and lubrication fitting.
- h. Lubricate Tie rod ends underneath Fwd hull between front wheels (4 points) (#8 on Figure 3):
 - (1) Inspect tie rod ends underneath Fwd hull between front wheels for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.h.(2) through Step 1.h.(3) and proceed to Step 1.h.(4). If fresh or excessive grease is not present proceed to Step 1.h.(2).
 - (2) Clean tie rod ends.
 - (3) Using grease gun, apply grease to tie rod ends until grease extrudes from ends.
 - (4) Remove excess grease from tie rod ends.
 - (5) Repeat Step 1.h.(2) through Step 1.h.(4) for remaining tie rod ends.
- i. Lubricate Aft Propeller bearing bushing (2 points each) (#9 on Figure 3):
 - (1) Inspect Aft Port/Stbd Propeller bearing bushing for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.i.(2) through Step 1.i.(3) and proceed to Step 1.i.(4). If fresh or excessive grease is not present proceed to Step 1.i.(2).

- (2) Clean Aft Port/Stbd Propeller bearing bushing.

CAUTION: Do not over-grease to avoid damaging craft component seals.

- (3) Using grease gun, apply two to three pumps of grease to Aft Port/Stbd Propeller bearing bushing.
- (4) Remove excess grease from bearing bushing.
- (5) Repeat Step 1.i.(1) through Step 1.i.(4) for remaining propeller bearing bushings.

j. Wheel motor steering bearings (4 points) (#11 on Figure 1):

NOTE 4: There are two fittings on each steering bearing cover plate. Craft have one lubrication fitting and a relief valve.

CAUTION: Do not grease wheel motor lubrication fittings.

- (1) Inspect wheel motor steering bearing cover plate grease fittings for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.j.(2) through Step 1.j.(3) and proceed to Step 1.j.(4). If fresh or excessive grease is not present proceed to Step 1.j.(2).
- (2) Clean lubrication fitting and relief valve; remove relief valve.
- (3) Using grease gun, apply grease to the steering bearing cover plate until grease exits the open hole.
- (4) Reinstall relief valve.
- (5) Using grease gun, apply 3 more pumps of grease.
- (6) Remove excess grease from lubrication fitting.
- (7) Repeats Step 1.j.(2) through Step 1.j.(6) for remaining three steering bearing cover plates.
- k. Lubricate Fwd propeller bearing bushing and shaft universal joint (2 points) (#10 on Figure 2):
 - (1) Inspect Fwd propeller bearing bushing and shaft universal joint for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.k.(2) through Step 1.k.(3) and proceed to Step 1.k.(4). If fresh or excessive grease is not present proceed to Step 1.k.(2).
 - (2) Clean Fwd propeller bearing bushing and shaft universal joint lubrication fitting.
 - (3) Using grease gun, apply grease to Fwd propeller drive shaft bearing bushing and shaft universal joint lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (4) Remove excess grease from Fwd bearing bushing and universal joint lubrication fitting.

NOTE 5: For craft with Lubrication Access Improvements and Hydraulic Cover Modification proceed to Step 1.l. For craft without Lubrication Access Improvements and Hydraulic Cover Modification omit Step 1.l through Step 1.o.(4) proceed to Step 1.p.

- l. Lubricate hydraulic fill cover hinge and slide-bolts (10 points) (#12 on Figure 2):
 - (1) Inspect Port hydraulic fill cover hinge and slide-bolts for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.l.(2) through Step 1.l.(3) and proceed to Step 1.l.(4). If fresh or excessive grease is not present proceed to Step 1.l.(2).
 - (2) Clean cover hinge and slide-bolts.
 - (3) Using hand, apply grease to cover hinge and slide-bolts. Remove excessive grease.

- (4) Repeat Step 1.m.(1) through Step 1.l.(4) for Starboard cover hinge and slide-bolts.
- m. Lubricate steering cylinder rod pin (1 point) (#13 on Figure 1):
 - (1) Inspect steering cylinder rod pin for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.m.(2) through Step 1.m.(3) and proceed to Step 1.m.(4). If fresh or excessive grease is not present proceed to Step 1.m.(2).
 - (2) Clean steering cylinder rod pin and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote steering cylinder rod pin lubrication fitting. Do not over-grease to avoid craft damage.
 - (4) Remove excessive grease from steering cylinder rod pin and remote lubrication fitting.
- n. Lubricate linear alignment coupling (1 point) (#14 on Figure 1):
 - (1) Inspect linear alignment coupling for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.n.(2) through Step 1.n.(3) and proceed to Step 1.n.(4). If fresh or excessive grease is not present proceed to Step 1.n.(2).
 - (2) Clean linear alignment coupling and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote linear alignment coupling lubrication fitting. Do not over-grease to avoid craft damage.
 - (4) Remove excessive grease from linear alignment coupling and remote lubrication fitting.
- o. Lubricate land steering shaft (1 point) (#15 on Figure 1):
 - (1) Inspect steering shaft for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.o.(2) through Step 1.o.(3) and proceed to Step 1.o.(4). If fresh or excessive grease is not present proceed to Step 1.o.(2).
 - (2) Clean land steering shaft and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote steering shaft lubrication fitting. Do not over-grease to avoid craft damage.
 - (4) Remove excessive grease from steering shaft and remote lubrication fitting.
- p. Remove safety tag.
- q. Take off gloves and then eye protection.
- r. Return LARC to readiness condition.

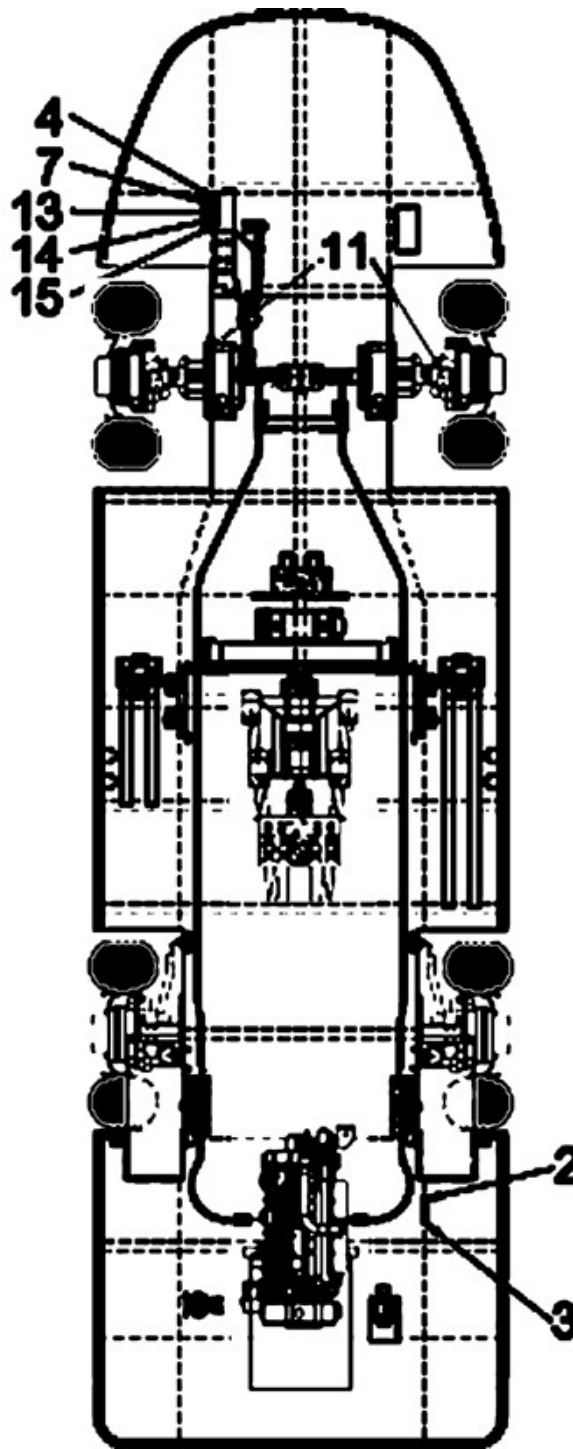


Figure 1 Lube Points

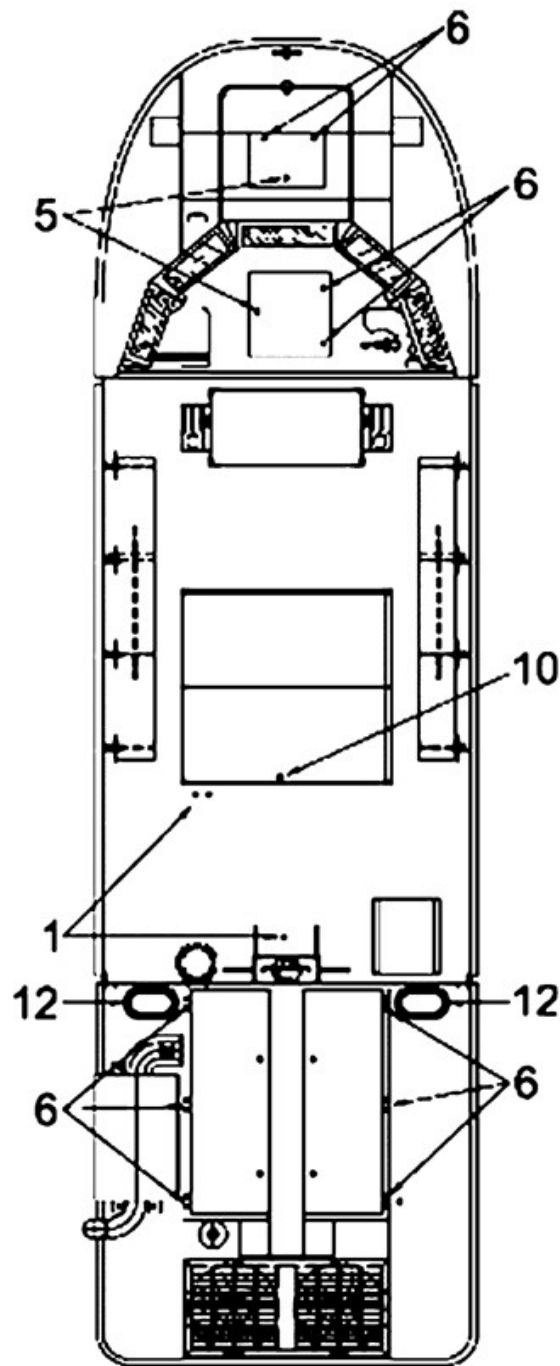


Figure 2 Lube Points

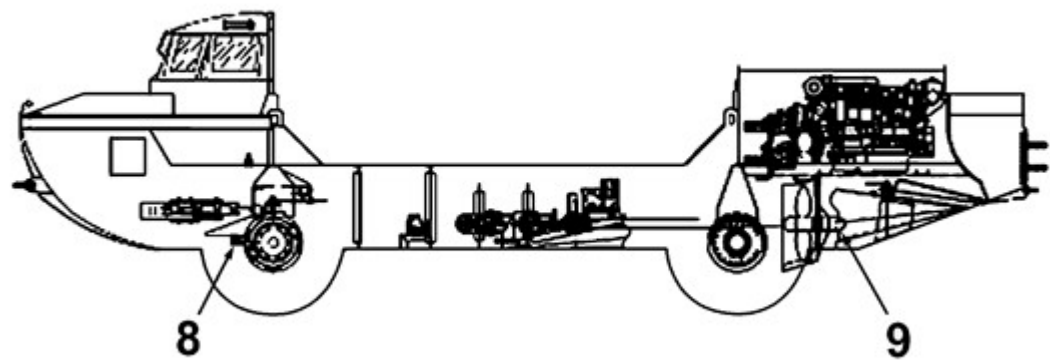


Figure 3 Lube Points

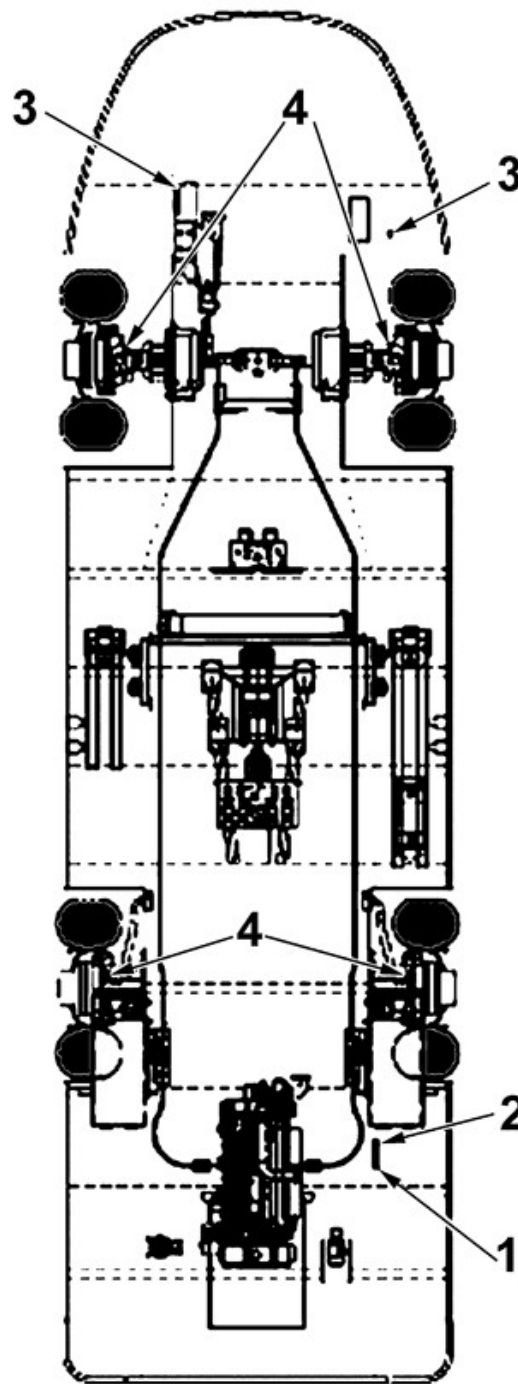


Figure 4 Lube Points

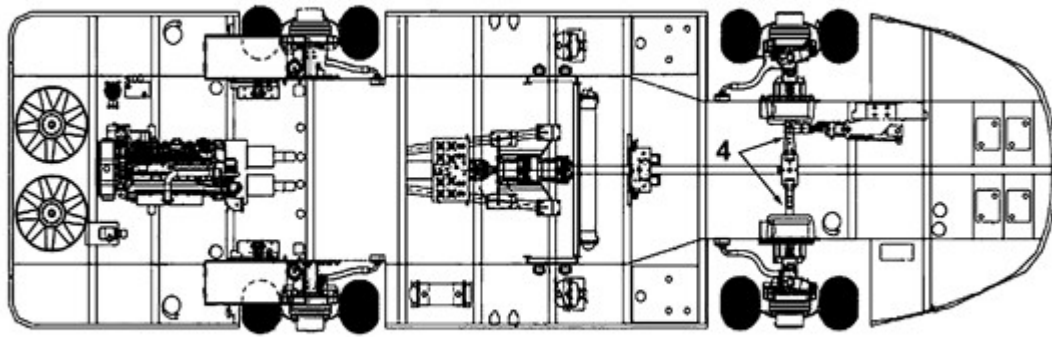


Figure 5 Lube Points

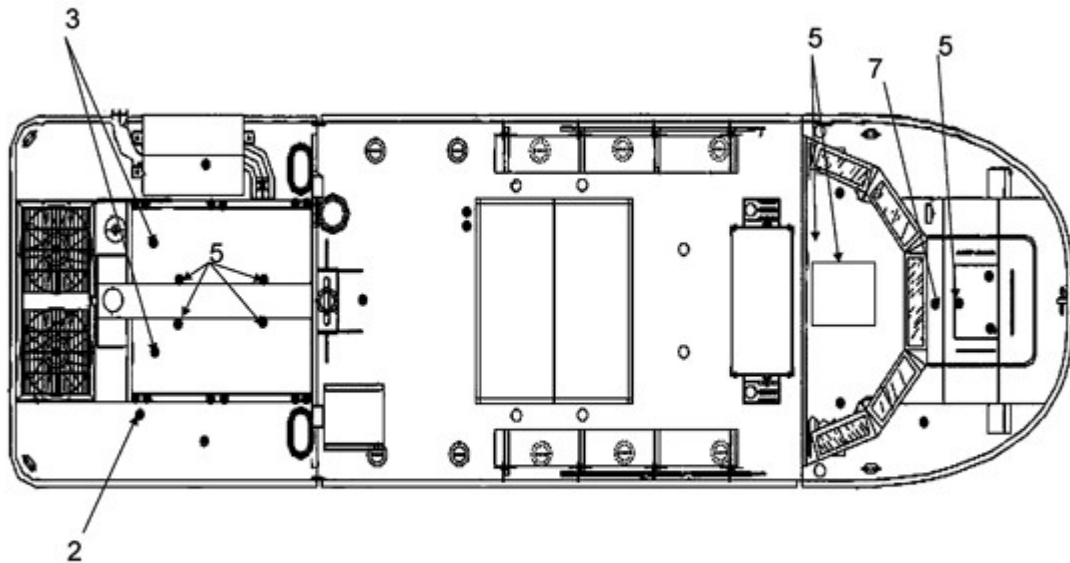


Figure 6 Lube Points

Item Number	Nomenclature
1	Propeller drive shaft lubrication points inside center bay
2	Rudder post (1 Aft remote lubrication fitting)
3	Aft Port/Stbd rudder cable lube points (2 Aft remote lubrication fittings)
4	Forward Port/Stbd steering cable lube points (2 Fwd remote lubrication fittings)
5	Hatch dogging handles
6	Engine/bow/top hatch (16 points)
7	Steering post inside forward bilge area centerline (1 Fwd remote lubrication fitting)
8	Tie rod ends underneath forward hull (between front wheels) (4 points)
9	Aft propeller bearing (2 points)
10	Forward propeller bearing and shaft universal joint (2 points)
11	Wheel motor steering bearing cover plate (4 points)
12	Hydraulic fill cover hinge and slide-bolts (10 points)
13	Land Steering cylinder rod pin (1 Fwd remote lubrication fitting)
14	Land steering linear alignment coupling (1 Fwd remote lubrication fitting)
15	Land steering shaft (1 Fwd remote lubrication fitting)

Table 1 LARC Lubrication Point Identification

SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>589-53-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L009</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>SAITO, KAZUYUKI</u> <u>NAGAMINE, YUUICHI</u> <u>TERASHITA, SHUUJI</u>

1. SCOPE:

1.1 Title: Hydraulic System Component; replace and flash

1.2 Location of Work:

1.2.1 Midships Compartment

1.2.2 Engine Compartment

1.2.3 Under Hull

1.2.4 Through the Craft

1.3 Identification:

1.3.1 Quantity (One AY), Hydraulic System located in 1.2.4

1.3.1.1 Quantity (2 EA), FWD and AFT High-Low Displacement Shift Control Solenoid Valve located in 1.2.1, MFR Parker-Hannifin Co., P/N DS123B/D024WD-A8T, Figure 1-20 Index No.One of 2.3

1.3.1.2 Quantity (2 EA), STBD Upper and PORT Lower Auxiliary (AUX) Gear Pump located in 1.2.2, MFR Parker-Hannifin Co., Model PGP511B0160AA1H2ND7D5C-511A0070, P/N 3349122216, FIND No.59 and 62 of 2.4

1.3.1.3 Quantity (One EA), STBD Lower AUX Gear Pump located in 1.2.2, MFR Parker-Hannifin Co., Model PGP511B0230AA1H2ND7D6C-511A0160, P/N 3349122215, FIND No No.60 of 2.4

1.3.1.4 Quantity (One EA), PORT Upper AUX Gear Pump located in 1.2.2, MFR Parker-Hannifin Co., Model PGP511B0310AA1H2ND7D6C-511A0080, P/N 3349122217, FIND No.61 of 2.4

1.3.1.5 Quantity (4 EA), Boost Pump Hydraulic Oil Filter located in 1.2.2, MFR Parker-Hannifin Co., Model 15CN210QH150N4N411, FIND No.7 of 2.4

1.3.1.6 Quantity (4 EA), Boost Pump Proportional Relief Valve located in 1.2.2, MFR Hydra Force INC., Model TS12-2712-0-V-24DL, FIND No.23 of 2.4

1.3.1.7 Quantity (2 EA), Hydraulic Oil Return Filter located in 1.2.2, MFR Parker-Hannifin Co., Model ILP110QBMP35Y9Y91, FIND No.6 of 2.4

1.3.1.8 Quantity (One EA), Propeller Shaft Drive Motor located in 1.2.2, MFR Linde Hydraulics, Corporation, P/N H MV165T (HMA 165-02), FIND No.One of 2.5

1.3.1.9 Quantity (One EA), Hydraulic Oil Moisture Sensor located in 1.2.1, MFR Parker-Hannifin Co., P/N MS2202510, FIND No.36 of 2.4

1.3.1.10 Quantity (4 EA), Main Transmission Hydraulic Pump

located in 1.2.2, MFR Linde Hydraulics Corporation, Model HPV-02-135, P/N 2650104R, FIND No.3 of 2.4

1.3.1.11 Quantity (2 EA), FWD Rotary Swivel for FWD Wheel Motor located in 1.2.3, MFR Rotary Systems INC., P/N D0454-S, Figure 1-56 Index No.2 of 2.3

1.3.1.12 Quantity (2 EA), AFT Rotary Swivel for FWD Wheel Motor located in 1.2.3, MFR Rotary Systems INC., P/N D0454, Figure 1-56 Index No.One of 2.3

1.3.1.13 Quantity (2 EA), STDB and PORT Hydraulic Oil Tank located in 1.2.2, Capacity Approx. 45 GL

2. REFERENCES:

- 2.1 NAVSEA Standard Items, (FY26)
- 2.2 F9YC0-AB-TMM-010 Rev 5, Change U, System Technical Manual for LARC-V A1/A2, Organizational and Intermediate Maintenance
- 2.3 F9YC0-AL-IPB-010, System Technical Manual for LARC-V A1/A2, Illustrated Parts Breakdown
- 2.4 1600911-2 Rev W, Hydraulic System Schematic
- 2.5 1600905-1 Rev G, Sea Drive Motor Installation and Foundation
- 2.6 1600911-1 Rev F, LARC V Vehicle Hydraulic System Layout & Details
- 2.7 1600911-12 Rev E, Hydraulic Tank Assembly LARC-V
- 2.8 1600907-1 Rev AA, Electrical System Installation & Details
- 2.9 1600907-3 Rev M, Electronic Transmission Control Schematic Diagram
- 2.10 ELM 225086, LARC-V Hydraulic Flush Filter Guidance
- 2.11 S9086-H7-STM-010/CH-262 Rev 12, Lubricating Oils Greases Specialty Lubricants and Lubrication Systems
- 2.12 S9086-S4-STM-010/CH-556 Rev 10, Hydraulic Equipment (Power Transmission and Control)
- 2.13 Maintenance Requirement Card, MRC Control Number: 55 2UH2 N (Provide Hydraulic Oil Sample For Testing) (enclosed)
- 2.14 Test Procedure, Test No.58331-4-100_26129, Hydraulic System Component

3. REQUIREMENTS:

3.1 Remove and pump out existing hydraulic oil from each hydraulic oil tank listed in 1.3.1.13, using Paragraph 6.3.10.3 of 2.2 and 2.7 for guidance.

3.1.1 Wipe and clean each hydraulic oil tank interior, free of oil, rust, corrosion and foreign materials using clean lint-free cloth.

3.2 Remove existing and install each new component for hydraulic system listed in 1.3.1.1 through 1.3.1.4, 1.3.1.6, 1.3.1.8 through 1.3.1.12 installing each new gasket and packing, using Paragraph 6.3 of 2.2, 2.3 through 2.9, ENCLOSURE (A), and ENCLOSURE (C) for guidance.

3.2.1 Matchmark, identify and retain disconnected hardware.

- 3.2.2 Clean each part exposed by disassembly and removal free of oil and foreign matter leaving no residue or injurious effects.
- 3.2.3 Inspect each part not specified for renewal for wear and defect.
- 3.2.4 Chase and tap each exposed threaded area.
- 3.2.5 Remove each high spot, burr, abrasion, nick, corrosion, gasket material, and foreign matter from each flange and mating surface exposed by disassembly and removal.
- 3.2.6 Handwork and skim cut each machined, sealing, aligning, mating, and gasket surface, taking precautions to ensure minimum metal removal.
- 3.2.7 Remove existing and install each new component in place of those identified to be missing or defective.
- 3.2.8 Remove defective and install each new electric connector listed in material list of Paragraph 5.
- 3.2.9 Apply new sealant compound to each threaded hardware in accordance with Note One of 2.5.
- 3.2.10 Apply new anti-seize compound to each threaded hardware for each rotary swivel in accordance with Note of detail for the swivel of 2.6.
- 3.2.11 Accomplish the calibration of each new equipment listed in 1.3.1.6 and 1.3.1.9 in accordance with Paragraph 6.3.7.5 and 6.3.4.8 of 2.2.
- 3.2.12 Tighten each fastener at each torque value stated in Paragraph 6.3 of 2.2.
- 3.3 Disassemble and assemble each filter listed in 1.3.1.5 and 1.3.1.7 to replace each element installing each new element, gasket, and packing, using Paragraph 6.3.3 and 6.3.5 of 2.2, ENCLOSURE (B) and ENCLOSURE (D) for guidance.
 - 3.3.1 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.
 - 3.3.2 Inspect each part not specified for renewal for wear and defect.
 - 3.3.3 Remove each high spot, burr, abrasion, nick, corrosion, gasket material, and foreign matter from each flange and mating surface exposed by disassembly and removal.
- 3.4 Fill each hydraulic tank listed in 1.3.1.13 with new oil, using Paragraph 6.3.10.4 of 2.2 and Paragraph 556-5.5.3 of 2.12 for guidance.
- 3.5 Accomplish the hydraulic oil flushing for hydraulic system listed in 1.3.1, in accordance with Paragraph 6.3.11 of 2.2 and 2.10, using 2.4 for guidance.
 - 3.5.1 Disconnect each existing hydraulic hose and install each temporary flushing equipment to accomplish the hydraulic oil flushing, using Paragraph 6.3.11.4 of 2.2 for guidance.
 - 3.5.2 Fill each hydraulic oil tank listed in 1.3.1.13 with new oil

to maintain the normal operating tank level during the flushing, using Paragraph 6.3.10.4 of 2.2 and Paragraph 556-5.5 of 2.12 for guidance.

3.5.3 Verify each disturbed joint free of leaks.

(V) (G) "CLEAR AND BRIGHT"

3.5.4 Visual inspect the oil sample, using 262-5.5.2 of 2.11 and 556-8.7.1.1 of 2.12 for guidance. Test criteria are "clear and bright" with no visible water or particulate contamination. -S130

3.5.4.1 Sample the oil at the end of hydraulic oil flushing, using 2.13 for guidance.

(V) (G) "CLEANLINESS INSPECTION"

3.5.5 Analyze and verify oil sample meets acceptance criteria. Oil cleanliness must be in accordance with 2.13. -S130

3.5.5.1 Sample the oil at the end of hydraulic oil flushing, using 2.13 for guidance.

3.5.5.2 Provide a contamination kit for analysis.

3.5.5.3 Continue the hydraulic oil flushing until the cleanliness meets criteria of 2.13.

(V) "REPORT"

3.5.6 Submit one legible copy of condition report listing results of the requirement of 3.5.4 and 3.5.5 to SUPERVISOR.

3.5.7 Upon completion of the hydraulic oil flushing, remove each temporary flushing equipment and connect each existing hydraulic hose disconnected in 3.5.1.

3.6 Accomplish the requirements of 009-032 of 2.1 for each new and disturbed surface.

(V) (G) "OPERATIONAL TEST"

3.7 Accomplish the requirements of 2.14 (TP) for an operational test of each equipment listed in 1.3.1.1 through 1.3.1.12. -S246

3.8 Return each temporary flushing equipment provided as GFM to NBU-7.

4. NOTES:

4.1 Initial Condition - S/F discovered hydraulic system issues on LARC 936 due to corrosion on primary hydraulic components to include: shift control solenoid valves, auxiliary hydraulic gear pumps, boost pump hydraulic oil filters, boost pump proportional relief valves, return hydraulic oil filters, propeller shaft motors, hydraulic moisture sensor, main hydraulic pump, rotary swivels, hydraulic system flush.

4.2 SRF Tool Room will issue each torque wrench and pressure gages when required.

4.3 Ship's Force will provide fluids for operation such as hydraulic oil when refilling.

4.4 The use of the temporary hydraulic system flush filter, model

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KF51KZ10VF24D5, was accepted by 2.10 (ELM) instead of model KF81K10PF24D5 stated in Paragraph 6.3.11.2 of 2.2.

4.5 GFM listed in 5.1 have been ordered under ICN 38NBA58903-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
2.	2 EA	VALVE, SOLENOID, P/N; DS123B/D024WD- A8T	1-20-1	2.3	9B481001 5457728	3.2
3.	2 EA	O-RING, P/N; 2- 014	1-20-3	2.3	9B533101 1459056	3.2
4.	4 KT	PARTS KIT, SEAL REPL, P/N; SK 12-3	-		9B533001 5786698	3.2
5.	4 KT	PARTS KIT, SEAL REPL, P/N; SK 12-3	-		9B533001 5786698	3.2
6.	2 EA	PUMP UNIT, ROTARY, P/N; PGP511B0160AA1H2 ND7D5C-511A	59, 62	2.4	3B432001 5457027	3.2
7.	2 EA	PUMP UNIT, ROTARY, P/N; PGP511B0160AA1H2 ND7D5C-511A	59, 62	2.4	3B432001 5457027	3.2
8.	1 EA	PUMP UNIT, ROTARY, P/N; PGP511B0230AA1H2 ND7D6C-511A	60	2.4	3B432001 5457018	3.2
9.	1 EA	PUMP UNIT, ROTARY, P/N; PGP511B0310AA1H2 ND7D6C-511A	61	2.4	3B432001 5457032	3.2
10.	1 EA	PUMP UNIT, ROTARY, P/N; PGP511B0310AA1H2 ND7D6C-511A	61	2.4	3B432001 5457032	3.2
11.	8 EA	FILTER ELEMENT, FLUI, P/N; 932618Q (FILTER ASSY; 15	7	2.4	9B294001 3634377	3.3
12.	16 EA	O-RING, P/N; N72142	-		9B533101 5472046	3.3
13.	4 EA	VALVE, SAFETY	23	2.4	9B482001	3.2

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SHIP:

NBU-7 (RMC 57078)

		RELIEF, P/N; TS12-2712-0-V- 24DL			6417039	
14.	8 KT	SEAL KIT, P/N; SK12-2X-B FOR RELIEF VALVE "TS12-27	-		9B5331SA 2300218	3.2
15.	8 EA	O-RING, P/N; 3- 912	1-59-30	2.3	9B533101 6229886	3.2
16.	4 EA	FILTER ELEMENT, FLUI, P/N; 937395Q (FILTER ASSY; IL	6	2.4	9B294001 6190899	3.3
17.	4 EA	O-RING, P/N; 2650104RN72350	-		9B533101 1016076	3.3
18.	4 EA	O-RING, P/N; N72251	-		9B533101 6007262	3.3
19.	4 EA	O-RING, P/N; 937410	-		9B5331SA 2300173	3.3
20.	1 EA	MOTOR, HYDRAULIC, P/N; HMV165T	1	2.5	7H432001 5462442	3.2
21.	4 PG	SCREW, P/N; 93190A718, 1/2"- 13UNC X 1-3/4" LG, SST	9	2.5	9B5305SA PHI0412	3.2
22.	2 PG	WASHER, FLAT, P/N; 91950A033, 25 EA/PG	10	2.5	9B531001 5598231	3.2
23.	16 EA	NUT, SELF, P/N; 90715A165, 1/2"- 13UNC, SST 316	11	2.5	9B531001 4664115	3.2
24.	1 PG	SCREW, CA, P/N; 93190A716 , 1/2"-13UNC X 1- 1/2" LG,	12	2.5	9B530501 5430481	3.2
25.	4 EA	WASHER, FLAT, P/N; 91860A033, 1/2", SST 316	13	2.5	9B531001 5775628	3.2
26.	2 EA	SCREW, CAP, HEXAGO N H, P/N; 93190A846, 3/4"- 10UNC X	14	2.5	9B530501 5867428	3.2
27.	2 EA	WASHER, FLAT, P/N; 91950A036, 3/4", SST 316	15	2.5	9B531001 5400898	3.2
28.	2 EA	NUT, SELF, P/N; 90715A036, 3/4"- 10UNC	16	2.5	9B531001 5265282	3.2
29.	1 EA	HYGROMETER, DIGIT AL, P/N; MS100-5 (MS2202510)	36	2.4	9B668501 5455973	3.2
30.	1 EA	HYGROMETER, DIGIT AL, P/N; MS100-5 (MS2202510)	36	2.4	9B668501 5455973	3.2

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31.	4 EA	ELBOW, SWIVEL FLANGE, P/N; D0454 (D0454-MT)	29	2.6	9B473001 5593653	3.2
32.	4 EA	ELBOW, SWIVEL FLANGE, P/N; D0454-S (D0454S- MT), FWD	30	2.6	9B473001 5464938	3.2
33.	4 EA	FLANGE ASSY, P/N; W44-20-20- M14SS	33	2.6	9B4730SA 2300312	3.2
34.	16 EA	SCREW, HEX HD, P/N; OS2002- 0922-1, 14MM X 2.0MM X	34	2.6	9B5305SA 2300169	3.2
35.	1 PG	WASHER, SPRING, P/N; 92153A439, 14MM, SST 316, 25	35	2.6	9B5310SA 2300313	3.2
36.	8 EA	O-RING, P/N; 2- 222 N90	36	2.6	9B533100 2979990	3.2
37.	4 OZ	CHEMICAL CORROSION, ANTICORROSIVE COMPOUND, TEF-GE	25	2.5	9B685001 4504009	3.2
38.	1 BT	LUBRICANT, ANTISE IZE, THREAD LOCKER 80078	-	2.6	9B915001 3116473	3.2
39.	4 EA	FILTER ELEMENT, FLUI, P/N; 936704Q	PARA3.19 .2	2.2	9B433001 6188249	3.5
40.	2 EA	FILTER ELEMENT, FLUI, P/N; 932688Q	PARA3.19 .1	2.2	9B294001 5707012	3.5
41.	4 EA	FILTER, P/N; KF51KZ10VF24D5	PARA6.3. 11.2.a	2.2	9B4330SA 2300223	3.5
42.	8 EA	FILTER ELEMENT, FLUI, P/N; KZ10	PARA6.3. 11.2.a	2.2	9B433001 5049506	3.5
43.	8 EA	ELEMENT, FOR K5 SERIES, P/N; KZ10V	-		9B4330SA 2300224	3.5
44.	8 EA	FILTER ELEMENT, FLUI, P/N; KZ3	-		9B433001 3464729	3.5
45.	4 EA	PUMP, HYDRAULIC, P/N; 2650104, MODEL HPV-02-135	3	2.4	9B4320SA 2600635	3.2
46.	6 EA	O-RING, P/N F37030250, FOR 5700 SEARIES PUMP DRIVE	-		9B5331SA 2501817	3.2
47.	6 EA	O-RING, P/N F37030276, FOR 5700 SEARIES	-		9B5331SA 2501818	3.2

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SHIP:

NBU-7 (RMC 57078)

		OUMP DRIVE				
48.	6 EA	O-RING, P/N F37030258, FOR 5700 SEARIES OUMP DRIVE	-		9B5331SA 2501819	3.2
49.	6 EA	O-RING, P/N F37030260, FOR 5700 SEARIES PUMP DRIVE	-		9B5331SA 2501814	3.2
50.	6 EA	O-RING, P/N 000 963 08 40, 84X2.5-V80, FOR HYDRAUL	-		9B5331SA 2501815	3.2
51.	1 EA	HYGROMETER, DIGIT AL, P/N; MS100-5 (MS2202510)	36	2.4	9B668501 5455973	3.2
52.	4 PG	CONNECTOR, PLUG, E LEC, P/N DT04-2P	-		9B593501 5065555	3.2
53.	4 EA	CONNECTOR BODY, PLUG, P/N DT06-2S	-		9B593501 6657216	3.2
54.	14 EA	CONTACT, ELECTRIC AL, P/N 0460- 202-16141	-		9B599901 2163648	3.2
55.	14 EA	CONTACT, ELECTRIC AL, P/N 0462- 201-16141	-		9B599901 2036687	3.2
56.	4 FV	RETAINER, ELECTRI CAL, P/N W2P	-		9B593501 5191808	3.2
57.	4 PG	RETAINER, ELECTRI CAL, P/N W2S- PO12	-		9B593501 4800358	3.2
58.	2 EA	CONNECTOR, PLUG, E LEC, P/N DT04-3P	-		9B593501 5150573	3.2
59.	2 EA	CONNECTOR, PLUG, E LEC, P/N DT06-3S	-		9B593501 5238855	3.2
60.	2 EA	LOCK, SECONDARY, P/N W3P-PO12	-		9B534001 4793023	3.2
61.	2 EA	POLARIZING KEY, ELEC, P/N W3S	-		9B593501 4708342	3.2
62.	1 LG	INSULATION SLEEVING, HEAT- SHRINK, SLEEVE, ADHESIVE	-		9B597001 2558990	3.2
63.	4 FT	INSULATE, HEAT- SHRINK, SLEEVE, ADHESIVE, BLK, 1"	-		9B597001 1520397	3.2
64.	20 FT	HOSE, NONMETALLIC , FLEXIBLE HOSE, FC234-12, 0.63"ID	-		9B472001 2802307	3.2
65.	10 TE	ELBOW, TUBE TO	-		9B473000	3.2

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SHIP:

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		HOSE, 90 ELBOW, HOSE FITTING, 190261			8524309	
66.	20 EA	ADAPTER, STRAIGHT , TU, REUSABLE FITTING, SAE37 FLARE	-		9B473000 8815536	3.2
67.	1 TU	SEALING COMPOUND, SEALG COMPOUND, MIL-S- 45180 TY.I	-		9B803000 2523391	3.2
68.	1 CO	GREASE, EXTREME PRES, HOSE FTG ASSY OIL FOR CRES &	-		9B915001 0379016	3.2
69.	1 PT	LUBRICATING OIL, GEN, LUBE OIL, HOSE FTG ASSY OIL	-		9B915001 3596563	3.2
70.	10 FT	TUBING, PLASTIC, S PIR, PLASTIC PROTECTIVE COIL SLEEV	-		9B933001 0855187	3.2

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

Catalog 3105/USA Technical Information

Directional Control Valves Series DS123B

General Description

The DS123B, 3-way spool valves are used as low flow directional valves, pilot valves, or selector valves. They can be used in normally open or normally closed applications.

Operation

In the de-energized mode, the spool is positioned by spring force. The solenoid, when energized, directly shifts the spool against the spring, thus changing the flow path through the valve.

Features

- Hardened, precision ground parts for long life
- Cost effective - replaces two cartridges
- High flow capacity with reduced space requirements
- Low leakage design
- Fully guided poppet for smooth operation
- Standard valve body and common cavity

Specifications

Rated Flow	45 LPM (12 GPM)
Max. Oper. Press.	210 Bar (3000 PSI)
Max. Leakage at Rated Pressure	82 cc/min. (5 cu. in./min.)
Operating Temp. Range (Ambient)	-40°C to +93.3°C (Nitrile) (-40°F to +200°F) -31.7°C to +121.1°C (Fluorocarbon) (-25°F to +250°F)
Cartridge Material	All parts steel. All operating parts hardened steel.
Body Material	Steel or Aluminum
Filtration	ISO Code 16/13, SAE Class 4 or better
Mounting	No restrictions
Cavity	Common Cavity No. C12-3

Solenoid Coil Specifications

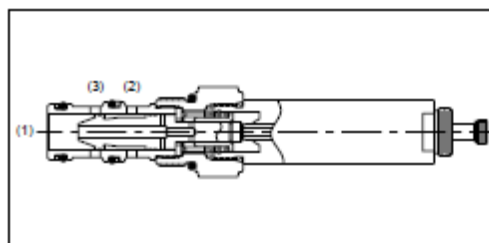
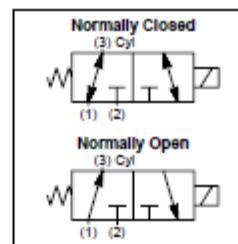
Voltage	12 VDC	24 VDC
Duty	STD.	STD.
Watts	30	30
Amps	2.5	1.3
Response Time (at nominal flow and max. pressure)		
Normally Closed	Up to 90ms	
Normally Open	Up to 100ms	
Wire Class	"H" for all voltages	
A.C. Rectifiers	Integral full wave bridge	

ds123b pr05.5, bl, ll



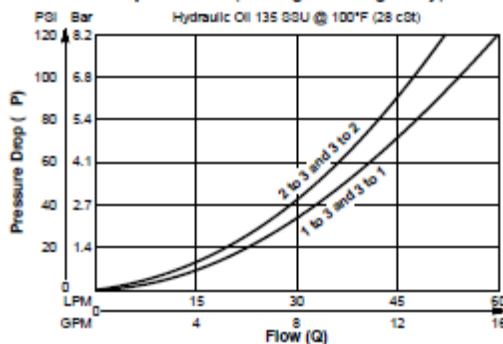
ENCLOSURE (A) - 1/3
DC104

Parker Hannifin Corporation
Hydraulic Valve Division
Elyria, Ohio 44035 USA

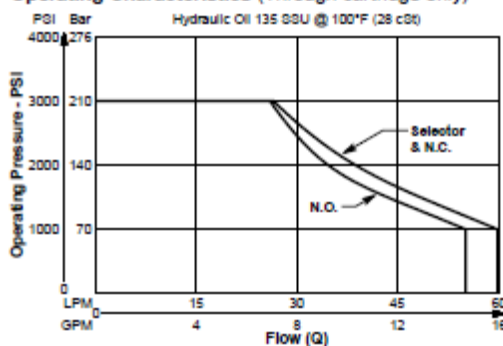


Performance Curves

Pressure Drop vs. Flow (Through cartridge only)



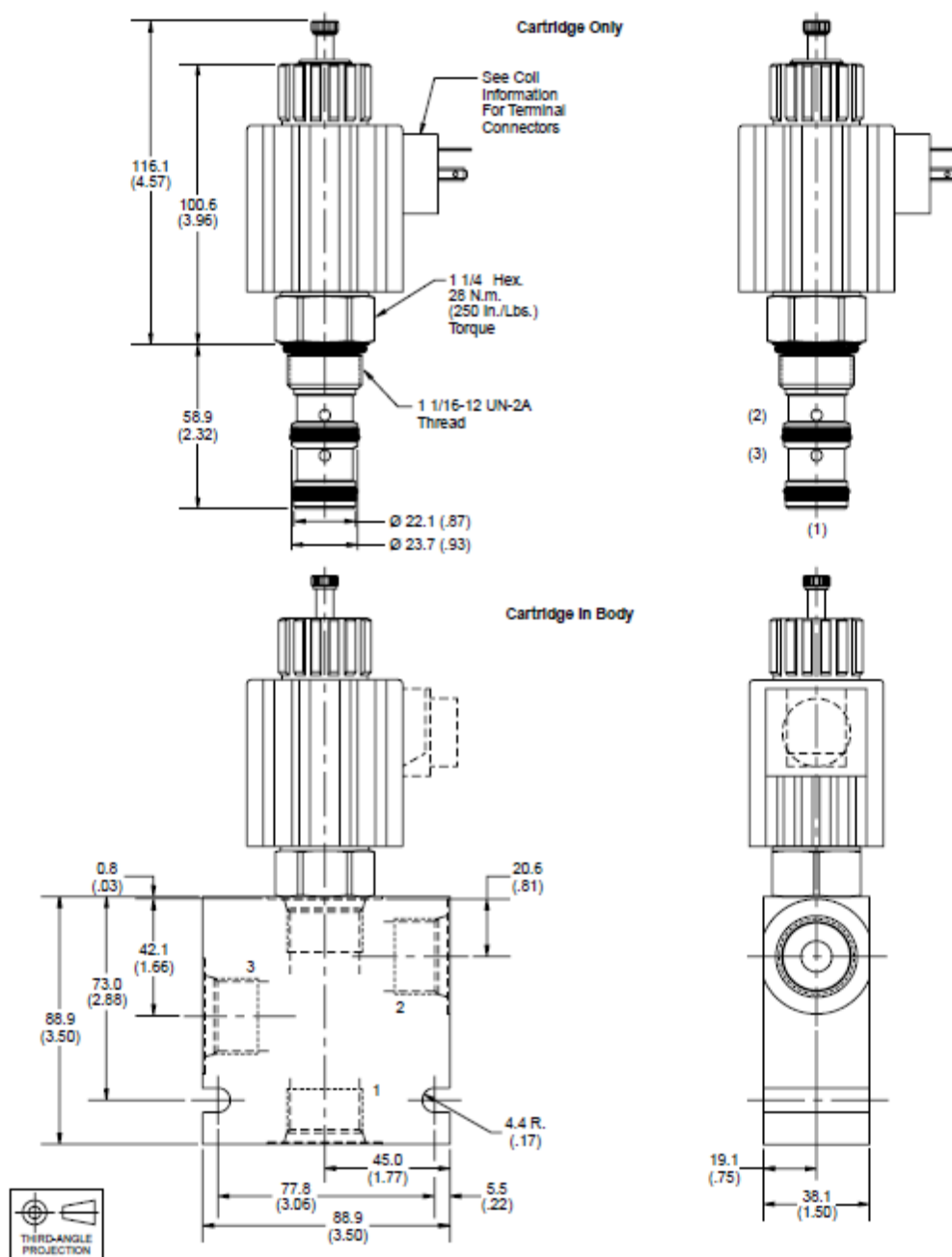
Operating Characteristics (Through cartridge only)



Catalog 3105/USA
Dimensions

Directional Control Valves
Series DS123B

*Inch equivalents for millimeter dimensions are shown in ("")



ds123b.pmf.5, bl. II

Parker Hydraulics

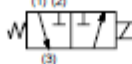
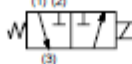
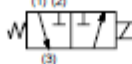
ENCLOSURE (A) - 2/3
DC105

Parker Hannifin Corporation
Hydraulic Valve Division
Elyria, Ohio 44035 USA

SHIP:

NBU-7 (RMC 57078)

Catalog 3105/USA
Ordering InformationDirectional Control Valves
Series DS123B

DS	12	3	B																																											
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						<table border="1"><thead><tr><th>Code</th><th>Port Size & Material</th></tr></thead><tbody><tr><td>Omit</td><td>Cartridge Only</td></tr><tr><td>6P</td><td>3/8" NPTF (B12-3-6P) Steel</td></tr><tr><td>A6P</td><td>3/8" NPTF (B12-3-A6P) Aluminum</td></tr><tr><td>8P</td><td>1/2" NPTF (B12-3-8P) Steel</td></tr><tr><td>A8P</td><td>1/2" NPTF (B12-3-A8P) Aluminum</td></tr><tr><td>12P</td><td>3/4" NPTF (B12-3-12P) Steel</td></tr><tr><td>A12P</td><td>3/4" NPTF (B12-3-A12P) Aluminum</td></tr><tr><td>6T</td><td>SAE-6 (B12-3-6T) Steel</td></tr><tr><td>A6T</td><td>SAE-6 (B12-3-A6T) Aluminum</td></tr><tr><td>8T</td><td>SAE-8 (B12-3-8T) Steel</td></tr><tr><td>A8T</td><td>SAE-8 (B12-3-A8T) Aluminum</td></tr><tr><td>12T</td><td>SAE-12 (B12-3-12T) Steel</td></tr><tr><td>A12T</td><td>SAE-12 (B12-3-A12T) Aluminum</td></tr><tr><td>6B</td><td>3/8" BSPG (B12-3-6B) Steel</td></tr><tr><td>A6B</td><td>3/8" BSPG (B12-3-A6B) Aluminum</td></tr><tr><td>8B</td><td>1/2" BSPG (B12-3-8B) Steel</td></tr><tr><td>A8B</td><td>1/2" BSPG (B12-3-A8B) Aluminum</td></tr><tr><td>12B</td><td>3/4" BSPG (B12-3-12B) Steel</td></tr><tr><td>A12B</td><td>3/4" BSPG (B12-3-A12B) Aluminum</td></tr></tbody></table>	Code	Port Size & Material	Omit	Cartridge Only	6P	3/8" NPTF (B12-3-6P) Steel	A6P	3/8" NPTF (B12-3-A6P) Aluminum	8P	1/2" NPTF (B12-3-8P) Steel	A8P	1/2" NPTF (B12-3-A8P) Aluminum	12P	3/4" NPTF (B12-3-12P) Steel	A12P	3/4" NPTF (B12-3-A12P) Aluminum	6T	SAE-6 (B12-3-6T) Steel	A6T	SAE-6 (B12-3-A6T) Aluminum	8T	SAE-8 (B12-3-8T) Steel	A8T	SAE-8 (B12-3-A8T) Aluminum	12T	SAE-12 (B12-3-12T) Steel	A12T	SAE-12 (B12-3-A12T) Aluminum	6B	3/8" BSPG (B12-3-6B) Steel	A6B	3/8" BSPG (B12-3-A6B) Aluminum	8B	1/2" BSPG (B12-3-8B) Steel	A8B	1/2" BSPG (B12-3-A8B) Aluminum	12B	3/4" BSPG (B12-3-12B) Steel	A12B	3/4" BSPG (B12-3-A12B) Aluminum
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SERVICE PARTS

DIN Connector: Grey — 692914 Black — 692915	Coil Assemblies 853494***** Conduit Coil 853495***** DIN (Hirschmann) Coil 853496***** Double Spade Vertically-Oriented Coil 853497***** Double Wire Coil 853498***** Double Screw Vertically-Oriented Coil
Coil Nut: 118378-00	(***** Calls out voltage. For example: 853496-012VDC is a 12 volt, 30 watt Double Spade Coil.)
Nitrile Seal Kit: SK12-3	
Fluorocarbon Seal Kit: SK12-3V	

Shipping Weight
Cartridge Only .59 kg (1.3 lbs.)
Cartridge In Body 1.4 kg (3.0 lbs.)

ds123b.pmf.5, bl, il

ENCLOSURE (A) - 3/3
DC106Parker Hannifin Corporation
Hydraulic Valve Division
Elyria, Ohio 44035 USA

SSP: SASN-003-26

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ITEM NO: 589-53-001

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Medium Pressure Filters

15/40/80CN Series

Specifications: CN Series

Pressure Ratings:

Maximum Allowable Operating Pressure (MAOP): 1000 psi (69 bar)
 Rated Fatigue Pressure: 800 psi (55.2 bar)
 Design Safety Factor: 2.5:1

Operating Temperatures:

Buna: -40°F (-40°C) to 225°F (107°C)
 Fluorocarbon: -15°F (-26°C) to 250°F (135°C)

Element Collapse Rating:

Standard: 150 psi (10.3 bar)

Materials:

Head and Bowl: Aluminum
 Indicators: Aluminum body, plastic connectors
 Bypass: Nylon

Element Condition Indicators:

Visual 360° green/red auto reset
 Electrical/Visual

5A@240VAC, 3A@28VDC



Electrical-Heavy Duty

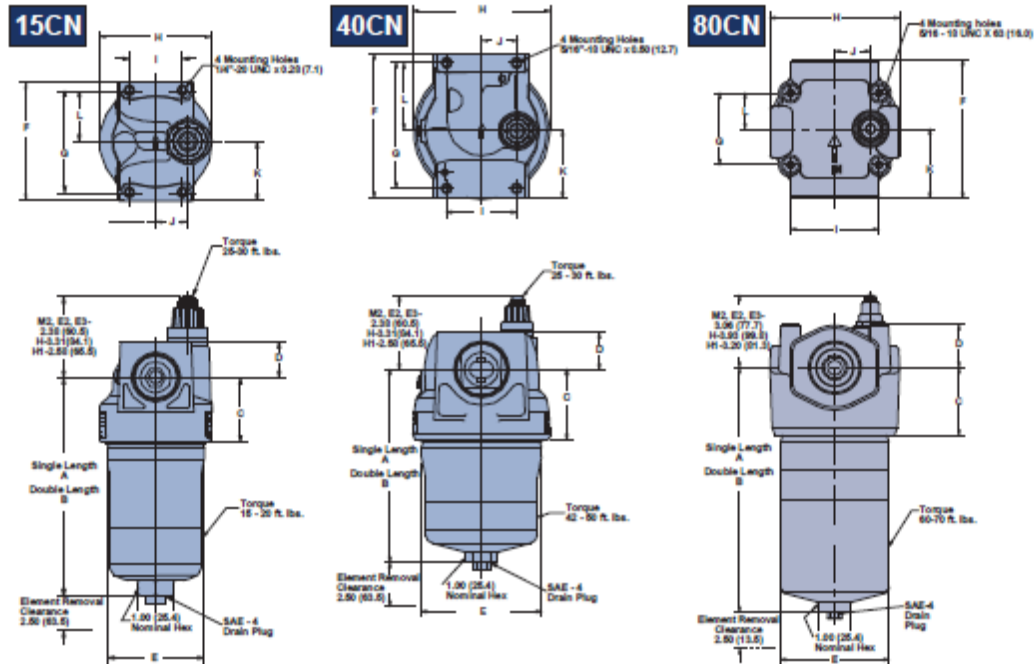
.25A(resistive) MAX 5 watts
 12 to 28 VDC & 110 to 175 VAC

Color code: White (common), Black (normally open),
 Blue (normally closed)

Weights (approximate):

Model	Single length	Double length
15CN	2.5 lb. (1.13 kg)	3.5 lb. (1.6 kg)
40CN	4.5 lb. (2.00 kg)	5.5 lb. (2.49 kg)
80CN	12.4 lb. (5.62 kg)	15.2 lb. (6.89 kg)

Dimensions are in (mm)	A	B	C	D	E	F	G	H	I	J	K	L
15CN	6.17(156.6)	9.87(250.7)	1.83(46.5)	1.09(25.4)	2.80(71.1)	3.38(85.9)	2.88(73.2)	3.25(82.6)	1.50(38.1)	.90(22.9)	1.09(42.9)	1.44(36.6)
40CN	6.73(170.8)	10.33(262.4)	2.44(62.0)	1.28(32.6)	4.22(107.2)	5.00(127.0)	4.37(111.0)	4.80(121.9)	2.44(62.0)	1.25(31.8)	2.32(58.8)	2.37(60.2)
80CN	11.06(280.9)	15.81(401.6)	3.06(77.7)	1.95(49.5)	4.91(124.8)	6.25(158.7)	3.25(82.6)	5.96(151.4)	4.00(101.6)	1.62(41.1)	3.12(79.4)	1.63(41.3)



ENCLOSURE (B) - 1/3
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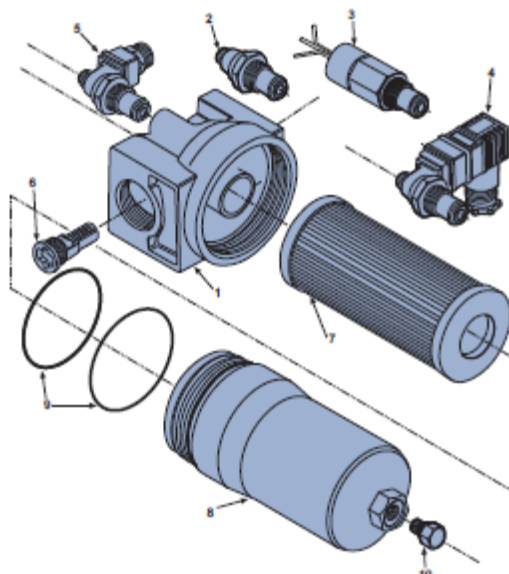
Parker Hannifin Corporation
 Hydraulic Filter Division
 Metamora, OH

Medium Pressure Filters

15/40/80CN Series

Element Service

- Stop the system's power unit.
- Relieve any system pressure in the filter line.
- Drain the filter bowl if drain port option is provided.
- Loosen and remove bowl.
- Remove element by pulling downward with a slight twisting motion and discard.
- Check bowl o-ring for damage and replace if necessary.
- Lubricate element o-ring with system fluid and place on post in filter head.
- Install bowl and tighten to specified torque.
 - 15CN - 15-20 ft. lbs
 - 40CN - 42-50 ft. lbs
 - 80CN - 60-70 ft. lbs
- Confirm there are no leaks after powering the system.



Parts List

Index	Description	15CN	40CN	80CN
1	Head			
	3/4" NPT bypass/ indicator ready	933865	N/A	N/A
	3/4" NPT no bypass/ no indicator	933877	N/A	N/A
	3/4" NPT no bypass/ indicator ready	933869	N/A	N/A
	3/4" NPT no indicator/ bypass ready	933873	N/A	N/A
	SAE-12 bypass/ indicator ready	933863	N/A	N/A
	SAE-12 no bypass/ no indicator	933875	N/A	N/A
	SAE-12 no bypass/ indicator ready	933867	N/A	N/A
	SAE-12 no indicator/ bypass ready	933871	N/A	N/A
	1" NPT bypass/ indicator ready	933864	932950	N/A
	1" NPT no bypass/ no indicator	933876	932986	N/A
	1" NPT no bypass/ indicator ready	933868	932962	N/A
	1" NPT no indicator/ bypass ready	933872	932974	N/A
	SAE-16 bypass/ indicator ready	933862	932947	N/A
	SAE-16 no bypass/ no indicator	933874	932983	N/A
	SAE-16 no bypass/ indicator ready	933866	932959	N/A
	SAE-16 no indicator/ bypass ready	933870	932971	N/A
	1 1/2" NPT bypass/ indicator ready	N/A	932948	934012
	1 1/2" NPT no bypass/ no indicator	N/A	932984	934018
	1 1/2" NPT no bypass/ indicator ready	N/A	932960	934016
	1 1/2" NPT no indicator/ bypass ready	N/A	932972	934014
	SAE-24 bypass/ indicator ready	N/A	932945	934027
	SAE-24 no bypass/ no indicator	N/A	932981	934033
	SAE-24 no bypass/ indicator ready	N/A	932957	934031
	SAE-24 no indicator/ bypass ready	N/A	932969	934029
	2" NPT bypass/ indicator ready	N/A	N/A	934020
	2" NPT no bypass/ no indicator	N/A	N/A	934026
	2" NPT no bypass/ indicator ready	N/A	N/A	934024
	2" NPT no indicator/ bypass ready	N/A	N/A	934022
	SAE-32 bypass/ indicator ready	N/A	N/A	934035
	SAE-32 no bypass/ no indicator	N/A	N/A	934042
	SAE-32 no bypass/ indicator ready	N/A	N/A	934040
	SAE-32 no indicator/ bypass ready	N/A	N/A	934037
	Flange face, SAE 2" bypass/ indicator ready	N/A	N/A	934103
	Flange face, SAE 2" no bypass/ no indicator	N/A	N/A	934109
	Flange face, SAE 2" no bypass/ indicator ready	N/A	N/A	934107
	Flange face, SAE 2" no indicator/ bypass ready	N/A	N/A	934105
2	Indicators			
	M2-Visual auto reset/ 25 psi	932026	932026	932026
	M2-Visual auto reset/ 50 psi	932027	932027	932027
3	H-Electrical/ 25 psi w/ 1/2" conduit connection	933053	933053	933053
	H-Electrical/ 50 psi w/ 1/2" conduit connection	932905	932905	932905
	H1-Electrical/ 25 psi w/ wire leads	933054	933054	933054
	H1-Electrical/ 50 psi w/ wire leads	932906	932906	932906
4	E2-Electrical/Visual 25 psi w/ DIN connection	931153	931153	931153
	E2-Electrical/Visual 50 psi w/ DIN connection	929599	929599	929599
5	E3-Electrical/Visual 25 psi w/ 3-pin connection	932773	932773	932773
	E3-Electrical/Visual 50 psi w/ 3-pin connection	929596	929596	929596
6	Bypass Valve			
	25 psid assembly	928979	930507	933628
	50 psid assembly	928981	933424	933630
7	Element (see model code page)			
8	Bowl			
	Single length w/o drain	933931	934248	933936
	Double length w/o drain	933932	934249	933937
	Single length w/ drain	933933	934250	933938
	Double length w/ drain	933934	934251	933939
9	Bowl and Dust Seal			
	Buna N (Nitrile)	Nr2142	Nr2239	Nr2244
	Fluorocarbon	Vr2142	Vr2239	Vr2244
10	Drain Plug - SAE-4			
	Buna N (Nitrile)	921088	921088	921088
	Fluorocarbon	928882	928882	928882



ENCLOSURE (B) - 2/3
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Parker Hannifin Corporation
Hydraulic Filter Division
Metamora, OH

Medium Pressure Filters

15/40/80CN Series

HOW TO ORDER:

Select the desired symbol (in the correct position) to construct a model code.

Example:

BOX 1	BOX 2	BOX 3	BOX 4	BOX 5	BOX 6	BOX 7	BOX 8	BOX 9
	40CN	1	10Q	M2	25	N4N4	1	Design number assigned by Parker

BOX 1: Seals	
Symbol	Description
None	Buna (N) nitrile
E5	EPR
F3	Fluorocarbon

BOX 2: Model Number	
Symbol	Description
15CN	In-line Filter
40CN	In-line Filter
80CN	In-line Filter

BOX 3: Element Length	
Symbol	Description
1	Single
2	Double

BOX 4: Element Media	
Symbol	Description
10C	Cellulose
20Q	Microglass III
10Q	Microglass III
05Q	Microglass III
02Q	Microglass III
WR	Water Removal

BOX 5: Indicators	
Symbol	Description
M2	Visual/Auto reset
H	Electrical (w/1/2" npt conduit connection and wire leads)
H1	Electrical (w/1/2" leads only)
E	Electrical/Visual (w/ 1/2" NPT conduit connection and wire leads)
E2	Electrical/Visual (DIN 43650 Hirschman style connection)
E3	Electrical/Visual (ANSI/B93.55M 3-Pin Brad Harrison style connection)
P	Indicator port plugged
N	No Indicator port

BOX 6: Bypass	
Symbol	Pressure Setting
25	25 PSI (1.7 bar) setting
50	50 PSI (3.5 bar) setting
	If "no bypass" option (-11) and an indicator is selected, above symbols (25,50) denote indicator setting

BOX 7: Ports	
Symbol	Description
	15CN
B1B1	1/2" NPT
C1C1	1" NPT
M4M4	SAE-12 straight thread
N4N4	SAE-16 straight thread
	40CN
C1C1	1" NPT
E1E1	1 1/2" NPT
N4N4	SAE-16 straight thread
P4P4	SAE-24 straight thread
	80CN
E1E1	1 1/2" NPT
F1F1	2" NPT
P4P4	SAE-24 straight thread
R4R4	SAE-32 straight thread
Y9Y9	Flange face, SAE-2"

BOX 8: Options	
Symbol	Description
1	None
11	No Bypass
19	Drain port on bowl
21	No bypass and drain port

BOX 9: Design Number	
Applied to the filter by Parker Hydraulic Filter Division. Use the full model code, including the design number when ordering replacement parts, elements and cartridges.	

Replacement Elements (Fluorocarbon)

MEDIA	15CN-1	15CN-2	40CN-1	40CN-2	80CN-1	80CN-2
20Q	930369Q	930370Q	930100Q	930119Q	929903Q	929927Q
10Q	932612Q	932618Q	932651Q	932655Q	932661Q	932667Q
05Q	932611Q	932617Q	932650Q	932654Q	932660Q	932666Q
02Q	932610Q	932616Q	932649Q	932653Q	932659Q	932665Q
10C	925385	925394	930096	930115	929912	929936
WR	N/A	N/A	931412	931414	931416	931418
74W	N/A	N/A	931317	931881	931895	931887
40W	N/A	N/A	930092	930111	929918	929942

Please note the bolded options reflect standard options with a reduced lead-time. Consult factory on all other lead-time options.

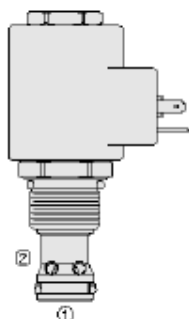


ENCLOSURE (B) - 3/3
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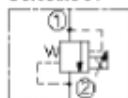
Parker Hannifin Corporation
Hydraulic Filter Division
Metamora, OH

TS12-27

hydraforce.com

PRESSURE RELIEVING PROPORTIONAL CARTRIDGE VALVE U.S. PATENT # 6,267,350 and 7,137,406**Overview****Symbol**

USASMSO:

**Description**

TS12-27X is a pilot-operated, spool-type, proportional, pressure relieving valve which CAN be infinitely adjusted across a prescribed range using a variable electric input. The regulated pressure is inversely proportional to the input electrical current. The valve is intended for use as a pressure limiting device in demanding applications

Operation

The TS12-27X blocks flow from port 1 to port 2 until sufficient pressure is present at port 1 to open the pilot section by overcoming the induced spring force. With no current applied, the valve will relieve at maximum set pressure. Applying current to the coil proportionally decreases the pressure required to open the valve from port 1 to port 2

Features

- Several adjustable pressure ranges plus factory preset fixed pressure model.
- Hardened parts for long life.
- 12 and 24 volt coils standard.
- Industry common cavity.
- Optional waterproof E-Coils rated up to IP69K.

Ratings**Pressure Ratings**

Pressure rating	275.8 bar (4000 psi)	- Note: Port 1
	103.4 bar (1500 psi)	- Note: Port 2
Proof pressure	413.7 bar (6000 psi)	- Note: Port 1
	206.8 bar (3000 psi)	- Note: Port 2
Burst pressure	813.6 bar (11800 psi)	
Operating pressure range	6.9 to 207 bar (100 to 3000 psi)	- Note: Tested at range A 7.5 lpm (2 gpm)
	6.9 to 138 bar (100 to 2000 psi)	- Note: Tested at range b 7.5 lpm (2 gpm)
	6.9 to 69 bar (100 to 1000 psi)	- Note: Tested at range c 7.5 lpm (2 gpm)
Note	Minimum pressure CAN be factory set to 6.2 to 10.3 bar (90 to 150 psi)	
Maximum tank port pressure	103.4 bar (1500 psi)	- Note: Dynamic condition
Pressure rise	0.29 bar/lpm (16 psi/gpm)	- Note: A
	0.24 bar/lpm (13 psi/gpm)	- Note: B
	0.18 bar (10 psi/gpm)	- Note: C

Flow Ratings

Flow rating	189 lpm (50 gpm)	- Note: Delta-p (cartridge only) port 1-2 (coil de-energized) 15.1 to 18.9 bar (220 to 275 psi)
Maximum pilot flow	0.76 lpm (0.2 gpm)	
Hysteresis per cent	2.5 %	- Note: (a)
	3.4 %	- Note: (b)
	6 %	- Note: (c)

Temperature Ratings

Operating fluid temperature	-40 to 100 °C (-40 to 212 °F)	- Note: With buna N seals
	-26 to 204 °C (-15 to 400 °F)	- Note: With fluorocarbon seals
	-54 to 107 °C (-65 to 225 °F)	- Note: With polyurethane seals
Storage temperature	-40 to 70 °C (-40 to 160 °F)	
Ambient temperature	-40 to 70 °C (-40 to 160 °F)	

Other Ratings

Corrosion protection	960 hr salt spray per ASTM b117
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ENCLOSURE (C) - 1/3

hydraforce.com

09/02/2011 (021)

TS12-27

TS12-27

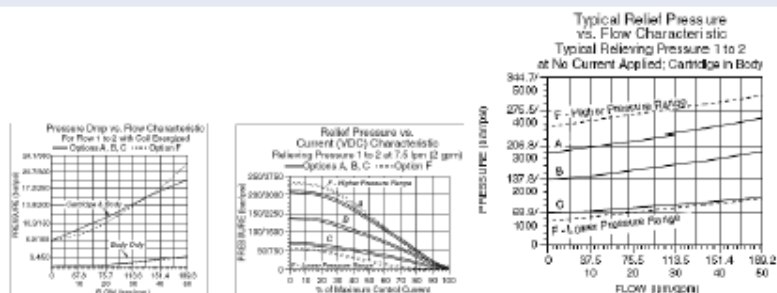
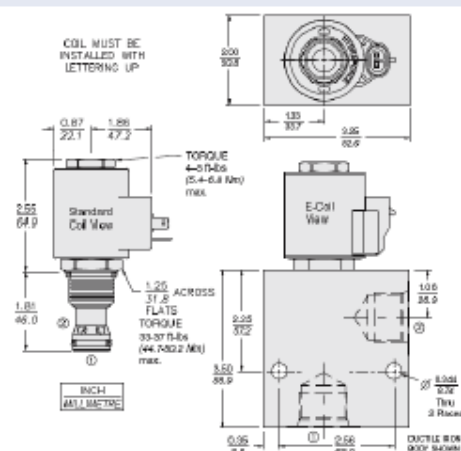
hydraforce.com

PRESSURE RELIEVING PROPORTIONAL CARTRIDGE VALVE U.S. PATENT # 6,267,350 and 7,137,406**Operating Parameters**

Fluids	Mineral based or synthetic with lubricating properties
Fluid viscosity range	7.4 to 420 cSt
Maximum operating contamination level	18/16/13 per ISO 4406

Properties

Unit weight	0.25 kg (0.56 lb)
Internal wetted surface area	244 cm ² (37.8 in ²)

Performance**Dimensions****Materials****Materials**

Cartridge: Weight: 0.25 kg (0.56 lb); Steel with hardened work surfaces. Zinc-plated exposed surfaces; Buna O-rings and TFE back-ups standard.

Standard Ported Body: Weight: 0.36 kg (0.8 lb); Anodized high-strength aluminum alloy, rated to 207 bar (3000 psi); Ductile iron bodies available; dimensions may differ.

D-Coil: Weight each: 0.32 kg (0.7 lb); Utilized thermoplastic encapsulated, Class H high temperature magnet wire.

E-Coil: Weight: 0.41 kg (0.9 lb); Perfect wound, fully encapsulated with rugged external metal shell; Rated up to IP69K with integral connectors.

ENCLOSURE (C) - 2/3

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09/04/2021 (021)

TS12-27

SHIP:

NBU-7 (RMC 57078)

TS12-27

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PRESSURE RELIEVING PROPORTIONAL CARTRIDGE VALVE U.S. PATENT # 6,267,350 and 7,137,406



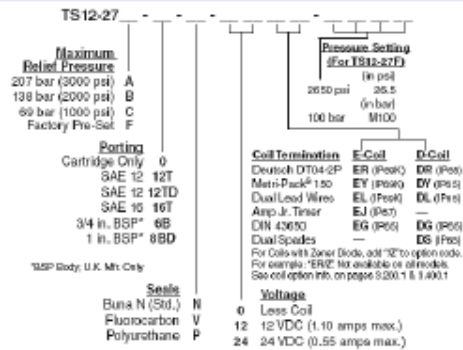
Installation Specifications

Cavity	VC12-2
Cartridge installation torque	44.7 to 50.2 N-m (33 to 37 ft-lb)
Coil nut torque	5.4 to 6.8 N-m (4 to 5 ft-lb)
Orientation restriction	None

Accessories

Seal kit	SK12-2X-B	- Note: X = seal option
Housings		
Coils		

Order Code



POSITION	CODE	DESCRIPTION
		TS12-27D-H-J-L
D		Regulating Pressure Range
D	A	206.8 to 6.9 bar (3000 to 100 psi) Pressure Range
D	B	137.9 to 6.9 bar (2000 to 100 psi) Pressure Range
D	C	68.9 to 6.9 bar (1000 to 100 psi) Pressure Range
D	F	248.2 to 55.2 bar (3600 to 800 psi) Crack Pressure at Zero Current
H		Line Body
H	0	No Body
H	10T	Aluminum SAE 10
H	12T	Aluminum SAE 12
H	16T	Aluminum SAE 16
H	12TD	Ductile Iron SAE 12
H	16TD	Ductile Iron SAE 16
H	8B	Aluminum BSPP 1" (8)
H	8BD	Ductile Iron BSPP 1" (8)
J		Seal
J	N	Buna-N
J	V	Fluorocarbon
J	P	Polyurethane
J	U	PPDI Urethane
L		Coil
L	0	No Coil

ENCLOSURE (C) - 3/3

hydraforce.com

09/04/2021 (021)

TS12-27

SSP: SASN-003-26

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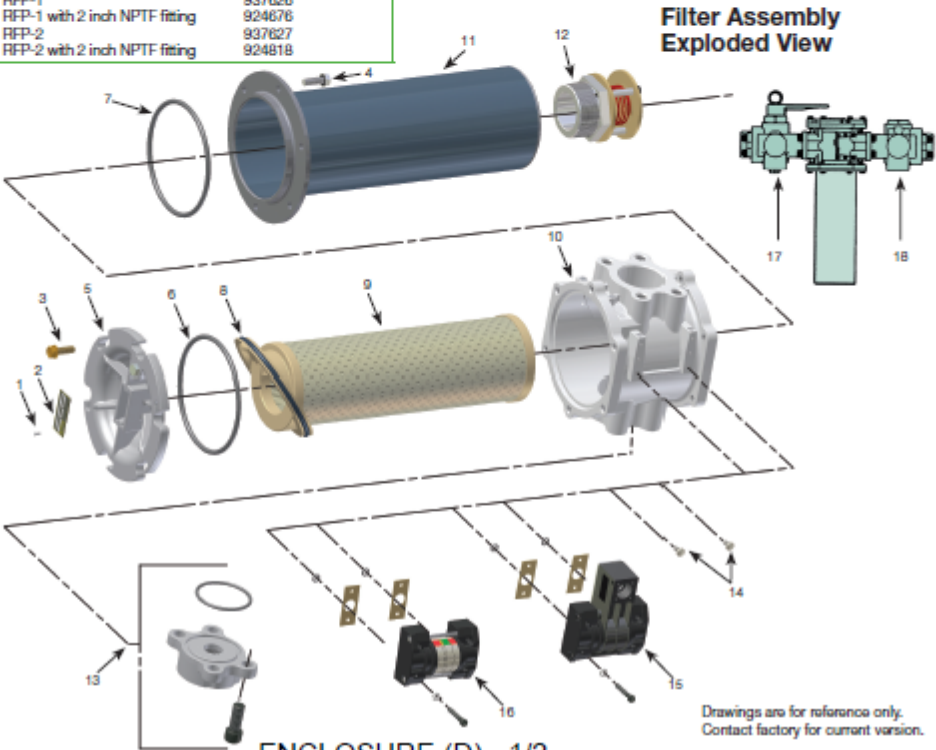
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Moduflow™ Plus Series

Parts List

Index	Description	Part No.	Quantity	Index	Description	Part No.	Quantity
1	Screws, Nameplate	900028	2	12	Check Valve Assy.	925120	1
2	Name Plate, Unstamped	920928	1	13	Flange Kits O-Ring	Refer to Table V72228	1 1
3	Cover Screws, 5/16-18 UNC x 1"	926633	6	14	Plug Kit, Fastener, self-sealing, o-ring seal included with fastener	925974	2
4	Bowl Screws, 5/16-18 UNC x 1"	926633	6	15	Indicator Electrical 35 psid 35 psid, 3-pin male receptacle Gasket O-Ring	926643 926753 926126 V72010	Optional 2 2 2
5	Cover, Without nameplate	924634	1	16	Indicator Visual 35 psid 4-band Bracket, Inline mounting Indicator Kit, Remote mount	926748 924904 924894	Optional Optional Optional
6	Cover O-Ring Nitrile Fluorocarbon	N72350 V72350	1 1	17	Changeover Valve Assy., Duplex	926758	Optional
7	Bowl O-Ring Nitrile Fluorocarbon	N72251 V72251	1 1	18	Check Valve Assy., Duplex	926757	Optional
8	Element Seal Nitrile Fluorocarbon	937410 937411	1 1	Not Shown	Drain Plug, SAE-24 for RFP model Nitrile Fluorocarbon	909992 928363	1 1
9	Element	Refer to Table	1	Not Shown	Check Valve Assy., Duplex	N72265	1
10	Head, Machined only 2" SAE Flange 1 1/2" SAE Flange 1 1/2" NPTF	925972 926146 925949	1 1 1				
11	Bowl, Select desired model ILP-1 ILP-2 RFP-1 RFP-1 with 2 inch NPTF fitting RFP-2 RFP-2 with 2 inch NPTF fitting	925916 924816 937626 924676 937627 924818	1				



Moduflow™ Plus Series

Low pressure filters

How To Order

Select the desired symbol (in the correct position) to construct a model code.

Example:

BOX 1	BOX 2	BOX 3	BOX 4	BOX 5	BOX 6	BOX 7	BOX 8
ILP	1	10Q	B	MP	35	Y9Y9	1

BOX 1: Filter Series	
Symbol	Description
RFP	Return-In filter, Inlet on side outlet on bottom
ILP	In-line filter
DILP	In-line filter

BOX 2: Element Length	
Symbol	Description
1	Single
2	Double

BOX 3: Media Code	
Symbol	Description
02Q	Microglass, 2 micron
05Q	Microglass, 5 micron
10Q	Microglass, 10 micron
20Q	Microglass, 20 micron
WR	Water Removal

BOX 4: Seals	
Symbol	Description
B	Nitrile
E	EPR
V	Fluorocarbon

BOX 5: Indicator	
Symbol	Description
P	Plugged Ports
M	Visual Indicator w/ "no element" warning
E	Electrical Indicator w/ 12" leads
D	Electrical Indicator w/ 3-pin male quick disconnect

Note: First letter of indicator code = left side of filter head when looking into inlet with bowl down; second letter = right side of filter head when looking into inlet with bowl down.

BOX 6: Bypass	
Symbol	Description
35	35 psid (2.4 bar)

BOX 7: Ports		
Filter Model	Inlet Symbol/Description	Outlet Symbol/Description
RFP	Y9 2" flange face	00 No fitting
	P9 SAE-24 Integral threads	P9 2" NPTF
		FB External check valve
ILP	Y9 2" flange face	Y9 2" flange face
	P9 SAE-24 Integral threads	P9 SAE-24 Integral threads
DILP	Y9 2" flange face	Y9 2" flange face

BOX 8: Options	
Symbol	Description
1	None

1. First pair of symbols denotes inlet for all filter styles; second pair of symbols denotes outlet.
2. Four symbols required: two for inlet, two for outlet.
3. Unused ports in filters come plugged with a blank flange.
4. See Flange Kits table for port flange options. Flange Kits are ordered separately.

Please note the bolded options reflect standard options with a reduced lead time.

RFP/ILP/ DILP Replacement Elements

Media	Nitrile Seals		Fluorocarbon Seals	
	Single	Double	Single	Double
02Q	937393Q	937397Q	937401Q	937405Q
05Q	937394Q	937398Q	937402Q	937406Q
10Q	937395Q	937399Q	937403Q	937407Q
20Q	937396Q	937400Q	937404Q	937408Q
WR	940733	940734	940735	940736

ENCLOSURE (D) - 2/2

SHIP: NBU-7 (RMC 57078)

DISTRIBUTION STATEMENT A:

Approved for public release: distribution unlimited.

Date: May 2025 **MIP** 59A3 **MRC:** 55 2UH2 N **Periodicity:** S-8
Series:

Location:

Ship System: Special Purpose Systems 59000
System: Logistics Over The Shore (LOTS) Support 59A00
SubSystem: Logistics Over The Shore (LOTS) Support, Personnel Transport 59A30
Equipment: Lighter, Amphibious, Resupply, Cargo V (LARC V) 59A32

Rates	Man-Hours	Rates	Man-Hours	Rates	Man-Hours
2CMCN	1.0				
Total Man-Hours:	1.0	Elapsed Time:	0.5		

MAINTENANCE REQUIREMENT DESCRIPTION

1. [Provide Hydraulic Oil Sample For Testing.](#)

SAFETY PRECAUTIONS

1. Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.
 2. Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.
 3. Rotating fan blades and moving belts are hazardous when contacted. Remain clear of rotating and moving engine parts.
 4. Exercise caution when handling hot liquid.
-

TOOLS, PARTS, MATERIALS, TEST EQUIPMENT

MATERIALS

1. [\[000307\]](#)Container (all types), W/C provide
2. [\[000828\]](#)Hydraulic fluid, petroleum base, MS2135TH
Hazardous Material User's Guide (HMUG) Group 12
3. [\[015351\]](#)Sampling kit, oil
4. [\[020040\]](#)Microbore Hose Assemblies, Test 20 (M16x2)

TOOLS

1. [\[020301\]](#)Tool kit, CESE maintenance, basic

MISCELLANEOUS

1. [\[010949\]](#)Spectacles, industrial with side cover
2. [\[017934\]](#)Gloves, 4-mil Nitrile, Light use only
3. [\[001365\]](#)Technical manuals/drawings, W/C provide (F9YC0-AM-TMM-010 LARC V; Technical Manual Operating Instructions)
4. [\[001365\]](#)Technical manuals/drawings, W/C provide (F9YC0-AM-TMM-020 LARC V; Technical Manual Organizational and Intermediate Maintenance)
5. [\[011922\]](#)S9086-WK-STM-020/NSTM 670 Vol II, Hazardous Material Users Guide (HMUG)
6. [\[030149\]](#)S9086-H7-STM-010/NSTM 262 LUBRICATING OILS, GREASES, SPECIALTY LUBRICANTS AND LUBRICATION SYSTEMS

NOTE: Numbers in brackets can be referenced to Standard PMS Materials Identification Guide (SPMIG) for identification.

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HAZARDOUS MATERIALS CONTROL STATEMENT(U)

For all Navy commands, units, and activities regardless of warfare community, OPNAV M-5100.23 Navy Safety and Occupational Health Manual provides guidance for Hazardous Material Control and Management (HMC&M) and directs Safety Data Sheets (SDS) be maintained and accessible to review, for all HM at the command. For forces afloat, the Hazardous Material Users Guide (HMUG), S9086-WK-STM-020, NSTM Chapter 670 Volume II supplements the technical data found in SDSs by providing additional control measures, precautions, personal protective equipment (PPE), and spill controls for the hazardous material(s) identified in the Tools, Parts, Materials, Test, Equipment block. Maintenance personnel shall determine if additional

PPE is necessary to accomplish the MRC and take appropriate action to obtain and wear such PPE to ensure the safety of maintenance personnel. Final disposition of all HAZMAT, both shipboard and ashore, shall be IAW local

HAZMINCEN/CHIRMP guidance as required by OPNAV-M 5090.1 series Table 35-3.

PROCEDURE

WARNING: Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.

Preliminary

- a. Ensure that the LARC is in a maintenance-ready condition.

NOTE 1: The steering ball valve handle is accessible in the operator's cabin under the port passenger seat. During normal operation the ball valve handle should be in the OPEN position.

- b. Ensure the emergency steering ball valve is in the OPEN position, if applicable.
c. Run engine to normal operating temperature.

NOTE 2: Engine must be running to perform the following maintenance requirement.

- d. Leave the engine running and place the transmission in NEUTRAL.

WARNING: Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.

- e. Put on eye protection and gloves.

1. Provide Hydraulic Oil Sample For Testing.

WARNING: Rotating fan blades and moving belts are hazardous when contacted. Remain clear of rotating and moving engine parts.

WARNING: Exercise caution when handling hot liquid.

- a. Place drain pan adjacent to the GA1 port of the AFT transmission manifold.
b. Clean area around the GA1 port of the AFT transmission manifold.
c. Remove GA1 port fitting cap.

NOTE 3: Port fitting has an integrated ball check valve which closes automatically when the port fitting cap is removed. The sampling hose fitting must be threaded onto the port fitting far enough to open the ball check valve in order to obtain oil sample.

- d. Screw the sampling hose onto the GA1 port fitting and placing the other end of the sampling hose in the drain pan. Allow fluid to run through the sampling hose for 5 seconds prior to taking sample.
e. Fill sample bottle with oil; cap bottle immediately to prevent contamination of sample.
f. Remove sample hose from GA1 port fitting, and replace GA1 port fitting cap.
g. Tag bottle with date, equipment, sampling point, oil type, or specification.

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- h. Reposition drain pan to under hydraulic oil drain.

NOTE 4: Take sample from hydraulic oil drain within 2 minutes of stopping engine. Sampling more than two minute after stopping the engine will not provide a representative sample.

- i. Stop engine using standard operating procedures.
j. Open hydraulic oil drain ball valve. Allow fluid to drain for 5 seconds prior to taking sample.
k. Fill sample bottle with oil; cap bottle immediately to prevent contamination of sample.
l. Close hydraulic oil drain ball valve.
m. Tag bottle with date, equipment, sampling point, oil type, or specification.
n. Clean mid-ship bilge compartment.
o. Remove and empty drain pan.
p. Ensure port and starboard hydraulic tank oil levels are equal and visible at midway point in sight glass; add fresh oil if required.
q. Take off gloves and then eye protection.
r. Return LARC to readiness condition.

NOTE 5: Upon completion of analysis, compare test results with limits from Table [1](#) or Table [2](#) and NSTM Chapter 262 depending on results provided by testing facility. If results indicate fluid properties are outside allowable limits, refer to F9YC0-AM-TMM-010 and F9YC0-AM-TMM-020 for hydraulic flush procedure.

NOTE 6: Oil analysis results should be archived and maintained on file for the service life of the vehicle in order document hydraulic system trend data. A copy of the results should be sent to the In Service Engineer Activity (ISEA), Naval Facilities Engineering and Expeditionary Warfare Center, Code EX262, LARC-V A1/A2 ISEA.

- s. Forward hydraulic fluid sample(s) to NOAP laboratory for analysis. Refer to NSTM Chapter 262 (S9086-H7-STM-010), Section 4, for specific handling instructions and deployment procedures. Refer to Table 262-4-2 for location of nearest NOAP laboratory.

Table 1 Fluid Cleanliness Standards.

ISO Code	Particles/Milliliter			Disavowed SAE Level (1963)
	≥2 Micrometers	≥4 Micrometers	≥14 Micrometers	
18/16/13	2,600	640	80	3
Viscosity			Water Content	
Between 17 - 300 cSt at 40°C			<800 ppm (<.08%)	

Table 2 NAS 1638 Fluid Cleanliness Standards.

					NAS 1638
5-15	15-25	25-50	50-100	>100	
<128K	<22.8K	<2025	<360	<64	<8

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Viscosity	Water Content
Between 17 - 300 cSt at 40°C	<800 ppm (<.08%)

SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>589-53-002</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L020</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>OSHIRO, YUKI</u> <u>SHIMOMURA, KENGO</u>

1. SCOPE:

1.1 Title: Flexible Hose Assembly for Hydraulic and Cooling System; replace

1.2 Location of Work:

1.2.1 See ENCLOSURE (A)

1.3 Identification:

1.3.1 Quantity (18 EA), Flexible Hose Assembly for Hydraulic System, Item No.21 through Item No.28, Item No.45 through Item No.48, Item No.142, Item No.143 and Item No.147 through Item No.150 of 2.1, 37-Degree Flare clamped Hose Fitting Ends, See ENCLOSURE (A) 1/4 through 3/4

1.3.2 Quantity (8 EA), Flexible Hose Assembly (NOVAFLEX 375-24) for Sea Water Cooling System, Item No.24 of 2.2, Barbed Hose Fitting Ends, See ENCLOSURE (A) 4/4

1.3.3 Quantity (2 EA), Flexible Hose Assembly (NOVAFLEX 375-32) for Sea Water Cooling System, Item No.9 and Item No.7 of 2.2, Barbed Hose Fitting Ends, See ENCLOSURE (A) 4/4

2. REFERENCES:

2.1 1600911-3 Rev J, Hydraulic Hose Schedule LARC-V

2.2 1600906-1 Rev M, Cooling System Inst & Dets

2.3 S6430-AE-TED-010 Rev 7, Piping Devices and Flexible Hose Assemblies

3. REQUIREMENTS:

3.1 Remove each existing and install each new flexible hose assembly to replace each flexible hose assembly listed in 1.3.1, using 2.1 for guidance. (See 4.2.)

3.1.1 Install a new identification tag on each new flexible hose assembly engraved, using Paragraph 7-4 of 2.3. Tag must include following information.

3.1.1.1 Ship's Name.

3.1.1.2 Hose Type.

3.1.1.3 Hydrostatic test pressure.

3.1.1.4 Installation Date.

3.1.1.5 Drawing Number and Item No.

3.2 Remove each existing and install each new flexible hose assembly to replace each flexible hose assembly listed in 1.3.2 and 1.3.3, using 2.2 for guidance.

3.2.1 Clean and reuse each existing barbed hose fitting for reuse.

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(V) (G) "OPERATING PRESSURE TEST"

3.3 Witness an operating pressure test to be conducted by Ship's Force to verify tightness of each mechanical joint affected by this work item, until the specified criteria is verified. Allowable leakage: None. - S130

4. NOTES:

4.1 Initial Condition - Each flexible hose assembly need to be replace.

4.2 Each new flexible hose assembly with clamed hose fitting ends is assembled and tested by manufacturer. Manufacturer will provide certificate of hydrostatic test for each new flexible hose assembly.

4.3 GFM listed in 5.1 have been ordered under ICN 38NBA58911-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	3 EA	FLEX HOSE, FLEX HOSE, NOVAFLEX 375, FUEL FILL HOSE			9B4730SA PTE0278	3.2
2.	25 EA	HOSE CLAMP, TIGHT-SEAL BOLT CLAMP FOR FIRM HOSE AN			9B4730SA PTE0279	3.2
3.	25 EA	HOSE CLAMP, TIGHT-SEAL BOLT CLAMP FOR FIRM HOSE AN			9B4730SA PTE0280	3.2
4.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0026	3.1
5.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0027	3.1
6.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0028	3.1
7.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0029	3.1
8.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0030	3.1
9.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH			9B4730SA POS0031	3.1

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ITEM NO: 589-53-002

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		CRIMP HOSE				
10.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0032	3.1
11.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0033	3.1
12.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0034	3.1
13.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0035	3.1
14.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0036	3.1
15.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0037	3.1
16.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0038	3.1
17.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0039	3.1
18.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0040	3.1
19.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0041	3.1
20.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0042	3.1
21.	1 EA	HOSE ASSY, FLEXIBLE HOSE ASSEMBLY WITH CRIMP HOSE			9B4730SA POS0043	3.1
22.	2 EA	HOSE, FLEX HOSE, NOVAFLEX 375, FUEL FILL HOSE, USC			9B4730SA PTE0315	3.2
23.	10 EA	CLAMP, TIGHT-			9B4730SA	3.2

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ITEM NO: 589-53-002

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SHIP:

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		SEAL BOLT CLAMP FOR FIRM HOSE AND TUB			POS0044	
24.	1 PG	PLATE, ID, LABEL PLATE, HOSE IDENTIFICATION, MIL-DT			9Q990501 1933701	3.1

5.2 PUSH MATERIAL:

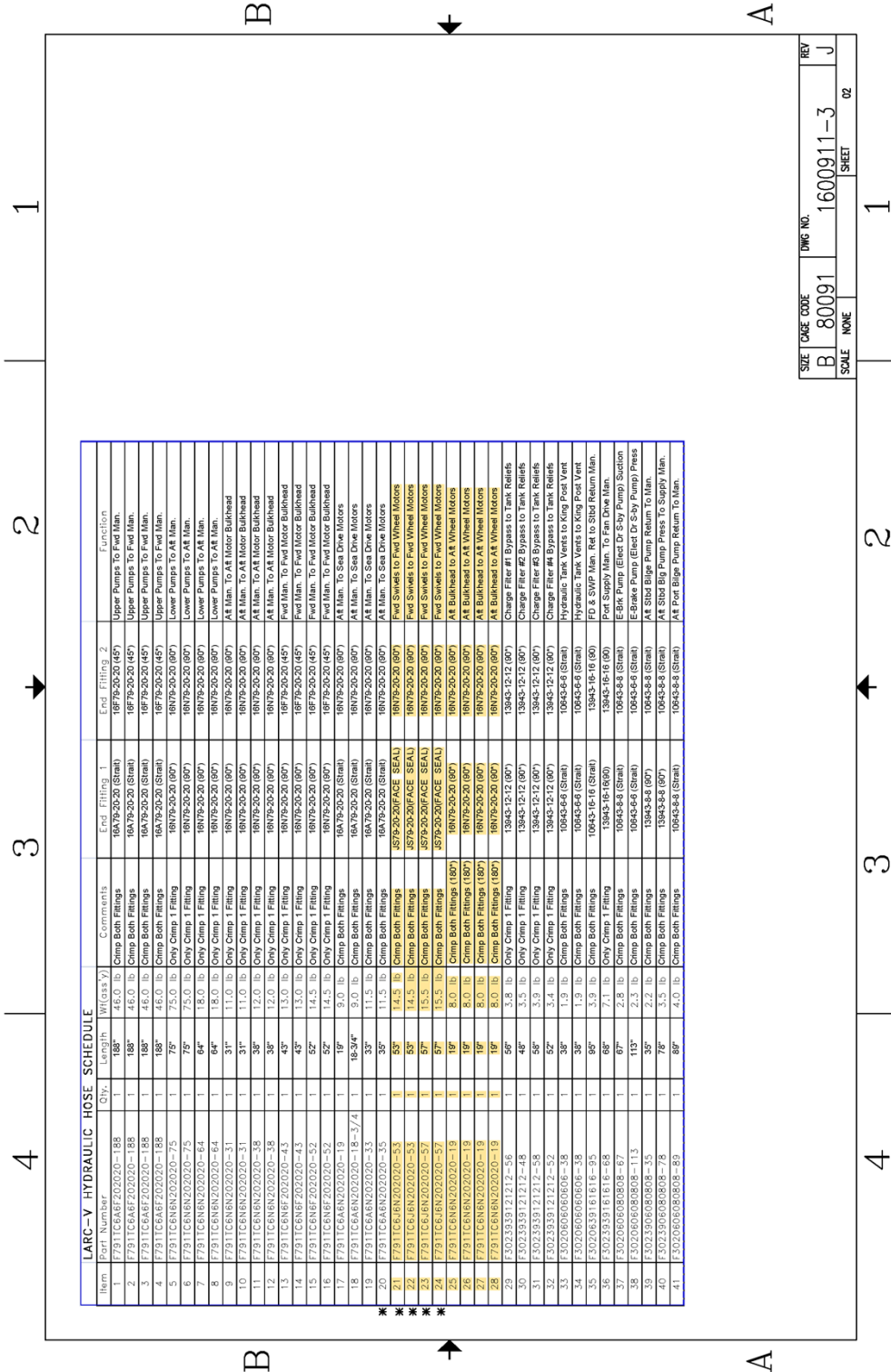
1. None.

5.3 KITTED MATERIAL:

1. None.

SHIP:

NBU-7 (RMC 57078)



ENCLOSURE (A) 1/4

SIZE	CAGE CODE	DWG NO.	REV
B	80091	1600911-3	J
SCALE	NONE	SHEET	02

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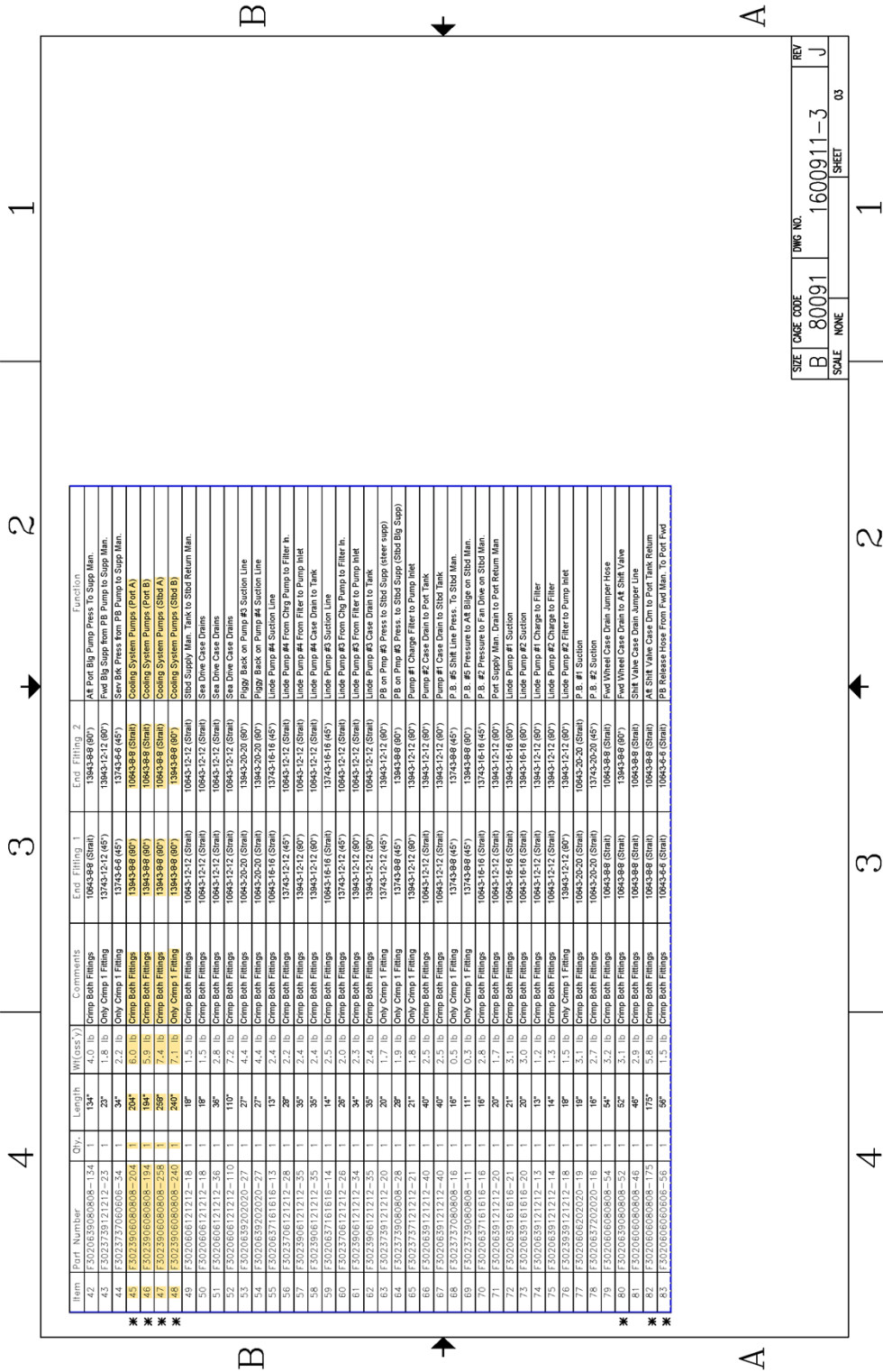
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ITEM NO: 589-53-002

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SHIP:

NBU-7 (RMC 57078)

4				3				2				1					
Item	Port Number	Qty	Length	Wt (kg)	Comments	Est. Filling 1	Est. Filling 2	Function	Item	Port Number	Qty	Length	Wt (kg)	Comments	Est. Filling 1	Est. Filling 2	Function
126	F3020606060608-30	1	30"	1.4 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Case Oil From Bld to Stbd Fwd Vln Mtr	127	F302060606060404-30	1	30"	0.8 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr
128	NOT USED								129	F302060606060404-30	1	30"	0.8 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr
130	F302060606060606-30	1	30"	1.2 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr	131	F302060606060808-30	1	30"	1.4 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr
132	F302060606060404-30	1	30"	0.8 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr	133	F302060606060606-30	1	30"	1.2 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr
134	F302060606060404-30	1	30"	0.8 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr	135	F302060606060606-30	1	30"	1.2 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr
136	F302060606060808-30	1	30"	1.4 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr	137	F302060606060808-318	1	318"	10.7 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr
138	F302060606060606-371	1	371"	5.6 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr	139	F302060606060404-132	1	132"	2.9 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr
140	F302060606060404-80	1	80"	1.4 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr	141	F302060606060606-275	1	275"	1.4 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr
142	F302060606060404-193	1	193"	2.6 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr	143	F302060606060404-215	1	215"	3.5 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr
144	F302060606060404-18	1	18"	7.4 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr	145	F302060606060404-18	1	18"	7.4 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr
146	F302060606060606-15	1	15"	3.0 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr	147	F302060606060606-36	1	36"	2.8 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr
148	F302060606060606-34	1	34"	2.5 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr	149	F302060606060606-34	1	34"	2.5 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr
150	F302060606060606-30	1	30"	2.8 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr	151	F302060606060606-30	1	30"	2.8 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr
152	F302060606060808-35	1	34.5"	1.7 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr	153	F302060606060808-35	1	34.5"	1.7 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr
154	F302060606060404-58	1	58"	1.3 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr	155	F302060606060808-40	1	40"	1.9 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr
156	F302060606060808-28	1	27.5"	3.4 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr	157	F302060606060808-28	1	27.5"	3.4 lb.	Comp Both Filings	10643-8-8 (Steel)	10643-8-8 (Steel)	Shift From Bld to Port Aft Vln Mtr
158	F302060606060808-25	1	25"	3.2 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr	159	F302060606060808-25	1	25"	3.2 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr
160	F302060606060808-79	1	79"	4.1 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr	161	F302060606060808-254	1	254"	15.6 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr
162	F302060606060808-165	1	165"	15.7 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr	163	F302060606060808-165	1	165"	15.7 lb.	Comp Both Filings	10643-16-16 (Steel)	10643-16-16 (Steel)	Shift From Bld to Port Aft Vln Mtr
164	F302060606060404-64	1	64"	1.6 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr	165	F302060606060404-54	1	54"	1.3 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr
166	F302060606060404-54	1	54"	1.3 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr	167	F302060606060404-159	1	159"	3.5 lb.	Comp Both Filings	10643-4-4 (Steel)	10643-4-4 (Steel)	Shift From Bld to Port Aft Vln Mtr

ENCLOSURE (A) 3/4

SSP: SASN-003-26

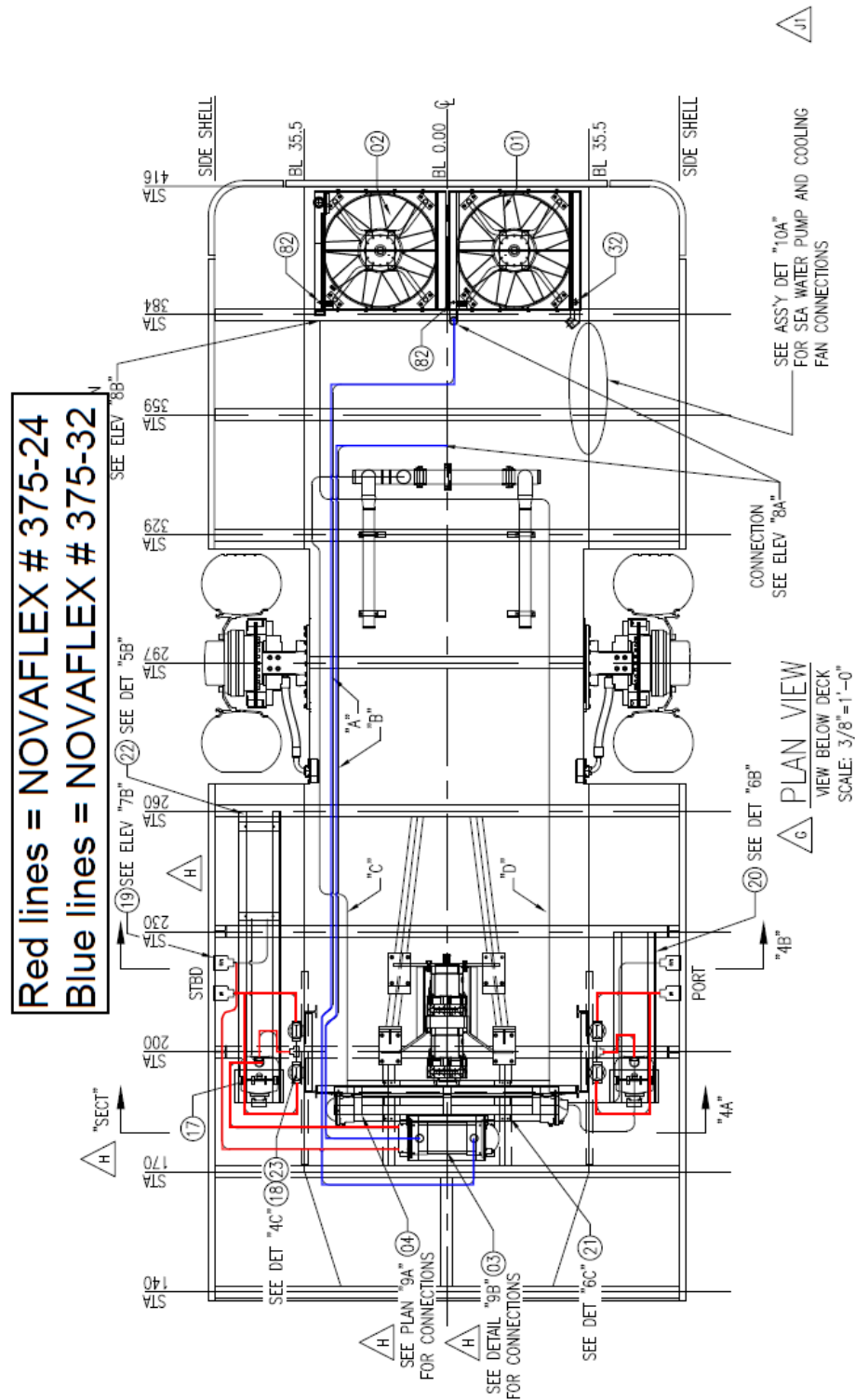
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ITEM NO: 589-53-002

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SHIP:

NBU-7 (RMC 57078)



ENCLOSURE (A) 4/4

SSP: SASN-003-26

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ITEM NO: 589-53-002

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SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>589-54-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L013</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>SAITO, KAZUYUKI</u> <u>NAGAMINE, YUUICHI</u> <u>TERASHITA, SHUUJI</u>

1. SCOPE:

1.1 Title: Wheel and Wheel Braking System Component; replace

1.2 Location of Work:

1.2.1 Forward Compartment

1.2.2 Under Hull

1.2.3 Engine Compartment

1.3 Identification:

1.3.1 Quantity (One EA), Brake Pedal Control Valve located in 1.2.1, MFR MICO INC., P/N 06-466-201, FIND No.38 of 2.4

1.3.2 Quantity (One EA), Service Brake Tail Light Hydraulic Pressure Switch located in 1.2.1, MFR The Nason Company INC., P/N WX6A-0150J-WD, Figure 1-57 Index No.2 of 2.3

1.3.3 Quantity (One EA), Service Brake Hydraulic Pressure Transducer located in 1.2.1, MFR Parker-Hannifin Co., P/N 2820009, FIND No.154 of 2.4

1.3.4 Quantity (One EA), Accumulator Control Valve located in 1.2.1, MFR MICO INC., P/N 06-463-202, FIND No.39 of 2.4

1.3.5 Quantity (2 EA), Accumulator for FWD and AFT Service Brake located in 1.2.1, MFR Parker-Hannifin Co., P/N A3N0058D1K, Figure 3-134 of 2.2

1.3.6 Quantity (4 EA), Wheel Gasket for FWD and AFT Wheel located in 1.2.2, Material PTFE, FIND No.7 of 2.5

1.3.7 Quantity (4 EA), FWD Wheel Motor Upper and Lower Drive Assembly Mounting Bracket located in 1.2.2, MFR Power Dynamics L.L.C., P/N 1600902-5, Figure 1-23 Index No.4 of 2.3

1.3.7.1 Quantity (4 EA), Upper and Lower Bearing Cover Plate for Upper and Lower Drive Assembly Mounting Bracket listed in 1.3.7, P/N 1600904-6, Figure 1-23 Index No.14 of 2.3

1.3.8 Quantity (4 EA), AFT Wheel Motor Upper and Lower Drive Mounting Bracket located in 1.2.2, MFR Power Dynamics L.L.C., P/N 1600903-8, Figure 1-24 Index No.4 of 2.3

1.3.9 Quantity (8 EA), FWD and AFT Wheel Brake Caliper located in 1.2.2, MFR MICO INC., P/N 03-520-091, Figure 1-23 Index No.3 and Figure 1-24 Index No.3 of 2.3

1.3.9.1 Quantity (8 EA), Brake Pad for FWD and AFT Wheel Brake Caliper listed in 1.3.9, P/N 20-060-119, ITEM No.52 of 2.6 and ITEM No.52 of 2.7

1.3.10 Quantity (4 EA), FWD and AFT Service Brake Rotor located in

1.2.2, MFR Knott Brake Company, P/N 800721-010, Figure 1-23 Index No.11 and Figure 1-24 Index No.11 of 2.3

1.3.11 Quantity (One AY), Service Brake Emergency Hydraulic Pump (P/N 1300380, NSN 4320-01-628-1111) and Motor (P/N 1300622, NSN 6105-01-628-1110) and Starting Solenoid (P/N 1300938, NSN 5930-01-628-6593) Assembly located in 1.2.3, MFR Haldex Barnes

2. REFERENCES:

2.1 Standard Items, (FY26)

2.2 F9YC0-AB-TMM-010 Rev 5, Change U, System Technical Manual for LARC-V A1/A2, Organizational and Intermediate Maintenance

2.3 F9YC0-AL-IPB-010, System Technical Manual for LARC-V A1/A2, Illustrated Parts Breakdown

2.4 1600911-2 Rev W, Hydraulic System Schematic

2.5 1600901-7 Rev A, Wheel Assembly & Details

2.6 1600902-1 Rev H, LARC V Vehicle Ground Drive System FWD Drive Assembly

2.7 1600903-1 Rev J, LARC V Vehicle Ground Drive System AFT Drive Assembly

2.8 1600907-1 Rev AA, Vehicle Electrical System Installation & Detail

2.9 1600907-3 Rev M, LARC V Electronic Transmission Control Schematic Diagram

2.10 Maintenance Requirement Card, MRC Control Number: 45 2UG6 N (Lubricate LARC V) (enclosed)

2.11 Test Procedure, Test No.58331-4-100_26130, Wheel & Wheel Braking System Component

3. REQUIREMENTS:

3.1 Remove existing and install each new component for wheel and wheel braking system listed in 1.3.1 through 1.3.11 installing each new seal, bearing, gasket and packing, using Paragraph 6.8 of 2.2, and 2.3 through 2.9 for guidance.

3.1.1 Matchmark, identify and retain disconnected hardware.

3.1.2 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.

3.1.3 Inspect each part not specified for renewal for wear and defect.

3.1.4 Chase and tap each exposed threaded area.

3.1.5 Remove each high spot, burr, abrasion, nick, corrosion, gasket material, and foreign matter from each flange and mating surface exposed by disassembly and removals.

3.1.6 Handwork and skim cut each machined, sealing, aligning, mating, and gasket surface, taking precautions to ensure minimum metal removal.

3.1.7 Remove existing and install each new component in place of

those identified to be missing or defective.

3.1.8 Remove defective and install each new electric connectors listed in material list of Paragraph 5.

3.1.9 Fabricate each wheel gasket listed in 1.3.6, using 2.5 for guidance.

3.1.10 Apply new sealant to each threaded hardware in accordance with Note One of 2.6 and 2.7.

3.1.11 Apply new thread locking compound to each threaded hardware in accordance with Note One of 2.6 and 2.7.

3.1.12 Tighten each fastener at each torque value stated in Paragraph 6.8 of 2.2.

3.1.13 Apply new sealant to the head of each threaded hardware in accordance with Note 8 of 2.6 and Note 6 of 2.7.

3.1.14 Apply new grease conforming to A-A-50433A to each wheel and wheel braking system lubrication point, using 2.10 for guidance.

3.1.15 Bleed each service brake, using Paragraph 6.8.15 of 2.2 for guidance.

3.2 Accomplish the requirements of 009-032 of 2.1 for each new and disturbed surface.

(V) (G) "OPERATIONAL TEST"

3.3 Accomplish the requirements of 2.11 (TP) for an operational test of each component listed in 1.3.1 through 1.3.5, 1.3.10, and 1.3.11. -S246

4. NOTES:

4.1 Initial Condition - S/F discovered wheel and wheel braking system equipment damage on LARC 936 due to corrosion to primary wheel and braking system components to include: brake pedal control valves, service brake tail light pressure switch, service brake hydraulic pressure transducers, accumulator control valves, accumulators, wheel gasket installations, forward wheel motor upper and lower drive assembly mounting brackets, AFT wheel motor upper and lower drive assembly mounting brackets, upper and lower bearing cover plates, brake calipers, brake pads, brake bleeding, service brake rotors, service brake emergency hydraulic motor starting solenoids, service brake emergency hydraulic pump and motor assembly.

4.2 SRF Tool Room will issue each torque wrench and pressure gages when required.

4.3 Ship's Force will provide liquids for operation such as hydraulic oil when refilling.

4.4 The removal and filling of hydraulic oil from each tank and the hydraulic system flushing will be conducted under the job specification, SSP#SASN-003-26 WN#57078-RS01-L009 WI# 589-53-001.

4.5 GFM listed in 5.1 have been ordered under ICN 38NBA58904-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM):

SHIP:

NBU-7 (RMC 57078)

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	1 EA	ACCUMULATOR, HYDR AUL, P/N; 06- 466-201, FOOT PEDAL A	38	2.4	9B253001 6512860	3.1
2.	1 EA	SWITCH, PRESSURE, P/N; WX6A-0150J- WD	155	2.4	9B593001 5456770	3.1
3.	1 EA	CABLE ASSEMBLY, SPEC, TRUSDUCER, P/N; 2820009	154	2.4	9B599501 5661554	3.1
4.	1 EA	ACCUMULATOR, HYDR AUL, P/N; 06- 463-202	39	2.4	9B209001 6512291	3.1
5.	2 EA	ACCUMULATOR, P/N; A3N0058D1K	PARA3.34 4.3	2.2	9B3040SA 2300174	3.1
6.	2 EA	SHEET, PTFE, P/N; 8545K247, FOR 1600901-7 FN 07),	PARA6.8. 7.2.b	2.2	9B5365SA 2300314	3.1
7.	2 EA	V-BOLT, P/N; PDI-10020, FOR STBD STEERING ARM.	1-23-56	2.2	9B304001 5895481	3.1
8.	1 EA	V-BOLT, STEERING ARM, FOR LARC V, FOR PORT SIDE IS	1-23-56	2.2	9B3040SA 2600636	3.1
9.	8 EA	SCREW, CAP, HEXAGO N H, P/N; 91286A322, 1/2"- 13 UNC X	1-23-32	2.2	9B530501 5169271	3.1
10.	8 EA	WASHER, FLAT, P/N; 98180A150, 1/2"	1-23-33	2.2	9B531001 5166034	3.1
11.	4 EA	FORWARD DRIVE MOUNT, P/N 1600902-5, MFR DWG 160090	1-23-4	2.2	9B304001 5895695	3.1
12.	4 EA	FORWARD DRIVE MOUNT, P/N 1600902-5, MFR DWG 160090	1-23-4	2.2	9B304001 5895695	3.1
13.	3 PG	NUT, PLAIN, HEXAGO N, P/N; 94804A351, 5/8- 11 UNC, 316	1-23-19	2.2	9B531001 5287648	3.1
14.	24 EA	SCREW, CA, P/N; 92245A717 (MS35307-413), 1/2 -13UNC	1-23-15	2.2	9B530500 0213806	3.1

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ITEM NO: 589-54-001

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15.	1 EA	WIRE,NONELECTRIC AL, P/N; 8860K65, 0.051 DIA	1-23-24	2.2	9B952501 6423046	3.1
16.	4 EA	STEERING STOP, LARC V WHEEL MOTOR BEARING COVER PL	1-23-14	2.2	9B304001 5895478	3.1
17.	4 EA	STEERING STOP, LARC V WHEEL MOTOR BEARING COVER PL	1-23-14	2.2	9B304001 5895478	3.1
18.	4 EA	SEAL,PLAIN ENCASED, P/N; 8561S (8561SG)	1-23-16	2.2	9B533000 9636887	3.1
19.	4 EA	CONE AND ROLLERS,TA, P/N; 39590 (39590/39520/Q)	1-23-17	2.2	9B311000 1437538	3.1
20.	4 EA	SEAL,PLAIN ENCASED, P/N; A- 1205-Z-650	1-23-18	2.2	9B533000 8917826	3.1
21.	10 EA	O-RING, P/N; 2- 349	1-23-21	2.2	9B5331SA 2300175	3.1
22.	16 EA	SPACER,RING, P/N 3421, INDEX NO.23 OF FIGURE 1-24	1-24-23	2.2	9B536501 6421910	3.1
23.	16 EA	SPACER,RING, P/N 3421, INDEX NO.23 OF FIGURE 1-24	1-24-23	2.2	9B536501 6421910	3.1
24.	8 EA	CALIPER,DISC BRAKE, P/N; 03- 520-091	1-23- 3,1-24-3	2.2	9B253001 5791045	3.1
25.	8 EA	SCREW,CAP, SOCKET HE, P/N; 92185A714, 1/2- 13 UNC X	1-23-12	2.2	9B530501 5607483	3.1
26.	1 PG	NUT,SELF- LOCKING,HE, P/N; 91831A137, 1/2"-13 UNC,	1-23-28	2.2	9B531001 4759155	3.1
27.	8 EA	BOLT,MACHINE, P/N; 93190A724, 1/2-13 UNC X 3" LG 3	1-24-12	2.2	9B530601 5610755	3.1
28.	8 EA	NUT,SELF, P/N; 90715A165, 1/2"- 13 UNC, 316 SST	1-24-18	2.2	9B531001 4664115	3.1
29.	4 KT	BRAKE LINING KIT, PAD, P/N; 20-060-119 (20- 060-120	52	2.6	9B253001 6280537	3.1

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ITEM NO: 589-54-001

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SHIP:

NBU-7 (RMC 57078)

30.	4 EA	ROTOR, DISC BRAKE, P/N; 800721-010	1-23- 11, 1-24- 11	2.2	9B253001 5561213	3.1
31.	48 EA	BOLT, SOCKET HD, M12 X 1.75 SHCS X 30MM LG, 316 SS	1-23- 8, 1-24- 10	2.2	9B5305SA 4829463	3.1
32.	1 EA	PUMP, ROTARY, P/N; 1300380	1-7-15	2.2	9B432001 6281111	3.1
33.	1 EA	MOTOR, DIRECT CURREN, P/N; 1300622	1-7-17	2.2	9B610501 6281110	3.1
34.	1 PG	NUT, SELF LOCK, P/N; 90715A145, 3/8"-16 UNC X 3/4"	1-7-18	2.2	9B5310SA PHA0071	3.1
35.	1 PG	WASHER, SPLIT, P/N; 92147A031, 3/8", 50 EA/PG	1-7-19	2.2	9B531001 5287188	3.1
36.	1 EA	U BOLT, P/N; 29605T8, 3/8"-16 UNC X 5", 316 SST	1-7-20	2.2	9B5310SA 2300315	3.1
37.	4 EA	BEARING, TAPERED ROLLER, P/N; 39590/39520/Q	1-23-17	2.2	9B3110SA 2300316	3.1
38.	10 OZ	CHEMICAL CORROSION, TEF- GEL	60	2.6	9B685001 4504009	3.1
39.	2 CO	THREAD LOCKER, THREAD LOCKER, BLUE, PN 24240, 38ML	20	2.6	9B8030LL N002457	3.1
40.	4 EA	AFT DRIVE MAOUN BRAK, AFT MOUNT MOUNT BRACKET, JER	1-24-4	2.2	9B5331SA 2400159	3.1
41.	40 EA	SCREW, HEX HEAD, M22 X 1.5 X 45MM LG, 10.9 STL ALL	10	2.6	9B5306SA 2300170	3.1
42.	3 KT	SEALING COMPOUND, BELZONA 1111 SUPERMETAL	-		9B803001 2992171	3.1
43.	8 EA	INSERT, SCREW THREAD, PN MC122165, SIZE 1/2-13, CRE	-		9B532500 9907158	3.1
44.	24 EA	BOLT, MACHINE, MCMaster PN P/N; 92245A717, 1/2 - 13	-		9B5306SA 2301388	3.1
45.	2 EA	ROD, CONTINUOUS THRE, 5/8-11-	-		9B530600 9033538	3.1

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		UNC-2A, STEEL ASTM A19				
46.	1 EA	RUBBER, SHEET, MCMASTER P/N 8525T32, 24IN WD X 24I	For1- 23,24	2.3	9B5330SA 2502013	3.1
47.	1 EA	RUBBER, SHEET, MCMASTER P/N 8525T34, 24IN WD X 24I	For1- 23,24	2.3	9B5330SA 2500880	3.1
48.	4 TU	SEALING COMPOUND, STAR BRITE, P/N; 082103, 2.8 OZ/	-		9B803001 6755311	3.1
49.	10 EA	PIN, STRAIGHT, HEA DLE, P/N 98381A712, 1/2" DIA X 1"	1-23-31	2.3	9B531501 6038933	3.1
50.	1 EA	FORWARD STARBOARD, P/N 1600902-7, FOR STBD, FIGURE	1-23-5	2.3	9B304001 5895452	3.1
51.	1 EA	MNTG PLATE, FORWARD , MNTG PLATE, FORWARD MOTOR, P	1-23-5	2.3	9B5315SA 2600637	3.1
52.	1 EA	SWITCH, TOGGLE, P/N 1300938MFR HALDEX BARNES	65	2.8	9B593001 6286593	3.1
53.	1 EA	MOTOR, DIRECT CURREN, P/N; 1300622	1-7-17	2.2	9B610501 6281110	3.1
54.	1 EA	RETAINER, LENS, P/N M5939712ITEM NO.110	-		9B622000 7526020	3.1
55.	1 EA	STOP LIGHT- TAILLIGHT, P/N 8378785ITEM NO. 110	-		9B622000 6695623	3.1
56.	2 OZ	CHEMICAL CORROSION, TEF- GEL	60	2.6	9B685001 4504009	3.1
57.	1 TU	INSULATING COMPOUND, P/N 3140 RTV	-		9B597001 4871943	3.1
58.	1 PG	SCREW, CAP, HEXAGO N H, P/N 93190A6223/8" - 16 UNC, 3/	-		9B530501 5290058	3.1
59.	8 EA	SCREW, CAP, HEXAGO N H, , P/N 93190A6243/8" - 16 UNC C	-		9B530501 5090161	3.1
60.	1 HD	WASHER, LOCK, P/N	-		9B531001	3.1

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ITEM NO: 589-54-001

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NBU-7 (RMC 57078)

		98437A1153/8"-16 CRES			5616188	
61.	1 PG	NUT, SELF- LOCKING, HE, P/N 91831A127NUT3/8" -16 CRES	-		9B531001 4759244	3.1
62.	1 EA	SWITCH, PRESSURE, P/N; WX6A-0150J- WD	155	2.4	9B593001 5456770	3.1
63.	1 EA	PRESSURE SWITCH, P/N WX6C- 0150J/WPCONTINGE NT MTLSP	-		9B5930SA 2300317	3.1
64.	2 EA	CONNECTOR BODY, PLUG, P/N 12010717 3WAY MALE	-		9B593501 3366409	3.1
65.	6 EA	CONTACT, ELECTRIC AL, P/N 12124580 MALE TERMINAL 16-	-		9B599901 5986472	3.1
66.	6 EA	BOOT, DUST AND MOIST, P/N 12010293 CABLE SEAL 16-14	-		9B534001 2268078	3.1
67.	1 EA	CONNECTOR, PLUG, E LEC, P/N DT06- 4S-C015	-		9B593501 6730034	3.1
68.	1 EA	RETAINER, ELECTRI CAL, P/N W4S- P012	-		9B593501 5176022	3.1
69.	4 EA	CONTACT, ELECTRIC AL, P/N 1062-16- 0122	-		9B599901 5195307	3.1
70.	4 EA	PLUG, PROTECTIVE, DUS, P/N 114017	-		9B534001 6708236	3.1

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

SHIP: NBU-7 (RMC 57078)

DISTRIBUTION STATEMENT A:

Approved for public release: distribution unlimited

Date:	April 2025	MIP	59A3	MRC:	45 2UG6 N	Periodicity:	M-4R
		Series:					

Location:

Ship System:	Special Purpose Systems 59000
System:	Logistics Over The Shore (LOTS) Support 59A00
SubSystem:	Logistics Over The Shore (LOTS), Personnel Transport 59A30
Equipment:	Lighter, Amphibious, Resupply, Cargo V (LARC V) 59A32

Rates	Man-Hours	Rates	Man-Hours	Rates	Man-Hours
CM3	1.5	CMCN	0.7		
Total Man-Hours:	2.2	Elapsed Time:	1.5		

MAINTENANCE REQUIREMENT DESCRIPTION

1. Lubricate LARC V.

SAFETY PRECAUTIONS

1. Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.
2. To prevent personnel injury, Tag-Out procedures shall be in accordance with the Tag-out Users Manual (TUM), and/or local tag-out instruction.
3. Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.

TOOLS, PARTS, MATERIALS, TEST EQUIPMENT

MATERIALS

1. [000073]Grease, Termalene 2, water-wash resistant, CID A-A-50433
Hazardous Material User's Guide (HMUG) Group 13
2. [001144]Tag, safety
3. [018450]Grease, white lithium, (As required in Step 1.c.(3).) (Contingency)
Hazardous Material User's Guide (HMUG) Group 13

TOOLS

1. [000771]Lubricating gun, hand, Lever-operated, 16 OZ 12" rub hose ext & adpt kit incl
2. [018451]Creeper, mechanics, W/C provide, (If desired) (Contingency)
3. [020301]Tool kit, CESE maintenance, basic

MISCELLANEOUS

1. [010949]Spectacles, industrial with side cover
2. [017934]Gloves, 4-mil Nitrile, Light use only
3. [001365]Technical manuals/drawings, W/C provide (F9YC0-AM-TMM-030 LARC V; Technical Manual Supplement for Lighter Amphibious Resupply Cargo, Five Ton (LARC-V A1/A2))
4. [011922]S9086-WK-STM-020/NSTM 670 Vol II, Hazardous Material Users Guide (HMUG)
5. [015244]S0400-AD-URM-010/TUM; TAG OUT USER'S MANUAL

NOTE: Numbers in brackets can be referenced to Standard PMS Materials Identification Guide (SPMIG) for identification.

HAZARDOUS MATERIALS CONTROL STATEMENT(U)

For all Navy commands, units, and activities regardless of warfare community, OPNAV M-5100.23 Navy Safety and

SSP: SASN-003-26

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ITEM NO: 589-54-001

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Occupational Health Manual provides guidance for Hazardous Material Control and Management (HMC&M) and directs Safety Data Sheets (SDS) be maintained and accessible to review, for all HM at the command. For forces afloat, the Hazardous Material Users Guide (HMUG), S9086-WK-STM-020, NSTM Chapter 670 Volume II supplements the technical data found in SDSs by providing additional control measures, precautions, personal protective equipment (PPE), and spill controls for the hazardous material(s) identified in the Tools, Parts, Materials, Test, Equipment block. Maintenance personnel shall determine if additional PPE is necessary to accomplish the MRC and take appropriate action to obtain and wear such PPE to ensure the safety of maintenance personnel. Final disposition of all HAZMAT, both shipboard and ashore, shall be IAW local HAZMINCEN/CHIMP guidance as required by OPNAV-M 5090.1 series Table 35-3.

PROCEDURE

SCHEDULING CRITERIA

Accomplish this maintenance requirement when any of the following periodicities or situations occur

- a. Monthly.
- b. After each days use.

WARNING: Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.

Preliminary

- a. Ensure that the LARC is in a maintenance-ready condition.

NOTE 1: The steering ball valve handle is accessible in the operator's cabin under the port passenger seat. During normal operation the ball valve handle should be in the OPEN position.

- b. Ensure the emergency steering ball valve is in the OPEN position, if applicable.

WARNING: To prevent personnel injury, Tag-Out procedures shall be in accordance with the Tag-out Users Manual (TUM), and/or local tag-out instruction.

- c. Attach safety tag in accordance with the Tag-out Users Manual (TUM) and/or local tag-out instruction.

WARNING: Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.

- d. Put on eye protection and gloves.

1. Lubricate LARC V.

- a. Lubricate Propeller drive shaft lubrication points inside center bay (#1 on Figure 1 and Figure 2):
 - (1) Inspect propeller drive shaft bearing lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.a.(2) through Step 1.a.(3) and proceed to Step 1.a.(4). If fresh or excessive grease is not present proceed to Step 1.a.(2)
 - (2) Clean propeller drive shaft bearing lubrication fittings.
 - (3) Using grease gun, apply grease three to four pumps to propeller drive shaft grease fitting. Do not over-grease to avoid damaging craft component seals.
 - (4) Remove excess grease from bearing.
 - (5) Repeat Step 1.a.(1) through Step 1.a.(4) for remaining propeller drive shaft grease fittings.
- b. Lubricate Rudder post. Inside Aft Stbd engine bay near hatch with Lubrication Access improvements (#2 on Figure 1 and Figure 4), without Lubrication Access improvements (#2 on Figure 6):
 - (1) For craft without the Lubrication Access improvements proceed to Step 1.b.(2). Inspect rudder post remote lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.b.(3) through Step 1.b.(4) and proceed to Step 1.b.(5). If fresh or excessive grease is not present proceed to Step 1.b.(3).
 - (2) Inspect rudder post cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.b.(3) through Step 1.b.(4) and proceed to Step 1.b.(5). If fresh or excessive grease is not present proceed to Step 1.b.(3).

- (3) Clean rudder post remote lubrication fitting.
- (4) Using grease gun, apply grease to rudder post remote lubrication fitting. Do not over-grease to avoid damaging craft component seals.
- (5) Remove excess grease from remote lubrication fitting.
- c. Lubricate Aft Port/Stbd rudder cable lube points (Located either side of engine) with Lubrication Access improvements (#3 on Figure 1 and Figure 4), without Lubrication Access improvements (#3 on Figure 6):
 - (1) Inspect Aft Port/Stbd rudder cables lube points for fresh or excessive lubricant. If fresh or excessive lubricant is present omit Step 1.c.(2) through Step 1.c.(3) and proceed to Step 1.c.(4). If fresh or excessive lubricant is not present proceed to Step 1.c.(2).
 - (2) Clean Aft Port/Stbd rudder cables and linkages.
 - (3) Apply lithium grease to Aft Port/Stbd rudder cables and linkages.
 - (4) Remove excess grease from Aft Port/Stbd rudder cables and linkages.
 - (5) For craft without the Lubrication Access improvements proceed to Step 1.c.(6). Inspect remote Port rudder cable lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.c.(7) through Step 1.c.(8) and proceed to Step 1.c.(9). If fresh or excessive grease is not present proceed to Step 1.c.(7).
 - (6) Inspect Aft Port/Stbd rudder cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.d.(8) through Step 1.d.(9) and proceed to Step 1.d.(10). If fresh or excessive grease is not present proceed to Step 1.d.(8).
 - (7) Clean rudder cable lubrication fitting.
 - (8) Using grease gun, apply grease to remote Port rudder cable lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (9) Remove excess grease from remote Aft Port rudder cable lubrication fitting.
 - (10) Repeat Step 1.c.(7) through Step 1.c.(9) for remote Stbd rudder cable lubrication fitting.
- d. Lubricate Fwd Port/Stbd steering cable lube points (Located inside hull) with Lubrication Access improvements (#4 on Figure 1 and Figure 4), without Lubrication Access improvements (#4 on Figure 5):
 - (1) Inspect Fwd Port/Stbd steering cables lube points for fresh or excessive lubricant. If fresh or excessive lubricant is present omit Step 1.d.(2) through Step 1.d.(3) and proceed to Step 1.d.(4).
 - (2) Clean Fwd Port/Stbd steering cables and linkages.
 - (3) Apply white lithium grease to Fwd Port/Stbd steering cables and linkages.
 - (4) Remove excess lubricant from Fwd Port/Stbd steering cables and linkages.
 - (5) For craft without the Lubrication Access improvements proceed to Step 1.d.(6). Inspect remote Fwd Port/Stbd steering cables lubrication fitting on the starboard side of the engine compartment for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.d.(7) through Step 1.d.(8) and proceed to Step 1.d.(9).
 - (6) Inspect Fwd Port/Stbd steering cables lubrication fitting for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.d.(7) through Step 1.d.(8) and proceed to Step 1.d.(9). If fresh or excessive grease is not present proceed to Step 1.d.(7).
 - (7) Clean steering cable lubrication fitting.
 - (8) Using grease gun, apply grease to remote Port steering cable lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (9) Remove excess grease from remote Fwd Port steering cable lubrication fitting.
 - (10) Repeat Step 1.d.(7) through Step 1.d.(9) for remote Stbd steering cable lubrication fitting.
- e. Lubricate hatch dogging mechanism (6 ea) with Lubrication Access improvements (#5 on Figure 6), without Lubrication Access improvements (#5 on Figure 2):
 - (1) Inspect hatch hinges and dogging mechanisms for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.e.(2) through Step 1.e.(3) and proceed to Step 1.e.(4). If fresh or excessive grease is not present proceed to Step 1.e.(2).
 - (2) Clean hatch dogging mechanism.
 - (3) Using grease gun, apply grease to hatch hinge and dogging mechanism lubrication fitting until grease extrudes from hinge and pivot point.
 - (4) Remove excess grease.
 - (5) Repeat Step 1.e.(1) through Step 1.e.(4) for remaining hatch hinge and dogging mechanisms.

NOTE 2:

For craft with Steering System Modification completed proceed to Step 1.e.(6). For craft without the Steering System Modification completed omit Step 1.e.(6) through Step 1.e.(9). and proceed to Step 1.f.

NOTE 3:

- For craft with Lubrication Access Improvements complete proceed to Step 1.f.
- (6) Inspect cabin floor bulkhead fittings for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.e.(7) through Step 1.e.(8) and proceed to Step 1.e.(9). If fresh or excessive grease is not present proceed to Step 1.e.(7).
 - (7) Clean cabin floor bulkhead fitting between center seat and port seat.
 - (8) Using grease gun, apply grease to cabin floor bulkhead fitting until grease extrudes from bulkhead fitting if required.
 - (9) Remove excess grease.
- f. Lubricate Engine/bow/top hatch (16 points) (#6 on Figure 2):
- (1) Inspect Engine/bow/top hatch for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.f.(2) through Step 1.f.(3) and proceed to Step 1.f.(4). If fresh or excessive grease is not present proceed to Step 1.f.(2).
 - (2) Clean hatch hinge and lubrication fitting.
 - (3) Using grease gun, apply grease to hatch hinge and lubrication fitting until grease extrudes from hinge and pivot point.
 - (4) Remove excess grease from lubrication fitting.
 - (5) Repeat Step 1.f.(1) through Step 1.f.(4) for remaining hatch hinge lubrication fittings.
- g. Lubricate Steering post inside Fwd bilge area with Lubrication Access improvements (#7 on Figure 1), without Lubrication Access improvements (#7 on Figure 6):
- (1) For craft without the Lubrication Access improvements proceed to Step 1.g.(2). Inspect remote steering post inside Port Fwd bilge area for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.g.(3) through Step 1.g.(4) and proceed to Step 1.g.(5). If fresh or excessive grease is not present proceed to Step 1.g.(3).
 - (2) Inspect steering post inside Port Fwd bilge area for fresh or excessive grease. If fresh or excessive grease is present, omit Step 1.g.(3) through Step 1.g.(4) and proceed to Step 1.g.(5). If fresh or excessive grease is not present proceed to Step 1.g.(3).
 - (3) Clean steering post and remote lubrication fitting, if applicable.
 - (4) Using grease gun, apply grease to remote steering post lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (5) Remove excess grease from steering post and lubrication fitting.
- h. Lubricate Tie rod ends underneath Fwd hull between front wheels (4 points) (#8 on Figure 3):
- (1) Inspect tie rod ends underneath Fwd hull between front wheels for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.h.(2) through Step 1.h.(3) and proceed to Step 1.h.(4). If fresh or excessive grease is not present proceed to Step 1.h.(2).
 - (2) Clean tie rod ends.
 - (3) Using grease gun, apply grease to tie rod ends until grease extrudes from ends.
 - (4) Remove excess grease from tie rod ends.
 - (5) Repeat Step 1.h.(2) through Step 1.h.(4) for remaining tie rod ends.
- i. Lubricate Aft Propeller bearing bushing (2 points each) (#9 on Figure 3):
- (1) Inspect Aft Port/Stbd Propeller bearing bushing for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.i.(2) through Step 1.i.(3) and proceed to Step 1.i.(4). If fresh or excessive grease is not present proceed to Step 1.i.(2).
 - (2) Clean Aft Port/Stbd Propeller bearing bushing.
- CAUTION:** Do not over-grease to avoid damaging craft component seals.
- (3) Using grease gun, apply two to three pumps of grease to Aft Port/Stbd Propeller bearing bushing.
 - (4) Remove excess grease from bearing bushing
 - (5) Repeat Step 1.i.(1) through Step 1.i.(4) for remaining propeller bearing bushings.
- j. Wheel motor steering bearings (4 points) (#11 on Figure 1):

CAUTION:

Do not grease wheel motor lubrication fittings.

- (1) Inspect wheel motor steering bearing cover plate grease fittings for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.j.(2) through Step 1.j.(3) and proceed to Step

- 1.j.(4). If fresh or excessive grease is not present proceed to Step 1.j.(2).
- (2) Clean lubrication fitting and relief valve; remove relief valve.
- (3) Using grease gun, apply grease to the steering bearing cover plate until grease exits the open hole.
- (4) Reinstall relief valve.
- (5) Using grease gun, apply 3 more pumps of grease..
- (6) Remove excess grease from lubrication fitting.
- (7) Repeats Step 1.j.(2) through Step 1.j.(6) for remaining three steering bearing cover plates.
- k. Lubricate Fwd propeller bearing bushing and shaft universal joint (2 points) (#10 on Figure 2):
 - (1) Inspect Fwd propeller bearing bushing and shaft universal joint for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.k.(2) through Step 1.k.(3) and proceed to Step 1.k.(4). If fresh or excessive grease is not present proceed to Step 1.k.(2).
 - (2) Clean Fwd propeller bearing bushing and shaft universal joint lubrication fitting.
 - (3) Using grease gun, apply grease to Fwd propeller drive shaft bearing bushing and shaft universal joint lubrication fitting. Do not over-grease to avoid damaging craft component seals.
 - (4) Remove excess grease from Fwd bearing bushing and universal joint lubrication fitting.

NOTE 5: For craft with Lubrication Access Improvements and Hydraulic Cover Modification proceed to Step 1.l. For craft without Lubrication Access Improvements and Hydraulic Cover Modification omit Step 1.l through Step 1.o.(4) proceed to Step 1.p.

- l. Lubricate hydraulic fill cover hinge and slide-bolts (10 points) (#12 on Figure 2):
 - (1) Inspect Port hydraulic fill cover hinge and slide-bolts for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.l.(2) through Step 1.l.(3) and proceed to Step 1.l.(4). If fresh or excessive grease is not present proceed to Step 1.l.(2).
 - (2) Clean cover hinge and slide-bolts.
 - (3) Using hand, apply grease to cover hinge and slide-bolts. Remove excessive grease.
 - (4) Repeat Step 1.m.(1) through Step 1.l.(4) for Starboard cover hinge and slide-bolts
- m. Lubricate steering cylinder rod pin (1 point) (#13 on Figure 1):
 - (1) Inspect steering cylinder rod pin for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.m.(2) through Step 1.m.(3) and proceed to Step 1.m.(4). If fresh or excessive grease is not present proceed to Step 1.m.(2).
 - (2) Clean steering cylinder rod pin and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote steering cylinder rod pin lubrication fitting. Do not over-grease to avoid craft damage.
 - (4) Remove excessive grease from steering cylinder rod pin and remote lubrication fitting.
- n. Lubricate linear alignment coupling (1 point) (#14 on Figure 1):
 - (1) Inspect linear alignment coupling for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.n.(2) through Step 1.n.(3) and proceed to Step 1.n.(4). If fresh or excessive grease is not present proceed to Step 1.n.(2).
 - (2) Clean linear alignment coupling and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote linear alignment coupling lubrication fitting. Do not over-grease to avoid craft damage.
 - (4) Remove excessive grease from linear alignment coupling and remote lubrication fitting.
- o. Lubricate land steering shaft (1 point) (#15 on Figure 1):
 - (1) Inspect steering shaft for fresh or excessive grease. If fresh or excessive grease is present omit Step 1.o.(2) through Step 1.o.(3) and proceed to Step 1.o.(4). If fresh or excessive grease is not present proceed to Step 1.o.(2).
 - (2) Clean land steering shaft and remote lubrication fitting in Port Fwd compartment.
 - (3) Using a grease gun, apply grease to remote steering shaft lubrication fitting. Do not over-grease to avoid craft damage.
 - (4) Remove excessive grease from steering shaft and remote lubrication fitting.
- p. Remove safety tag.
- q. Take off gloves and then eye protection.
- r. Return LARC to readiness condition.

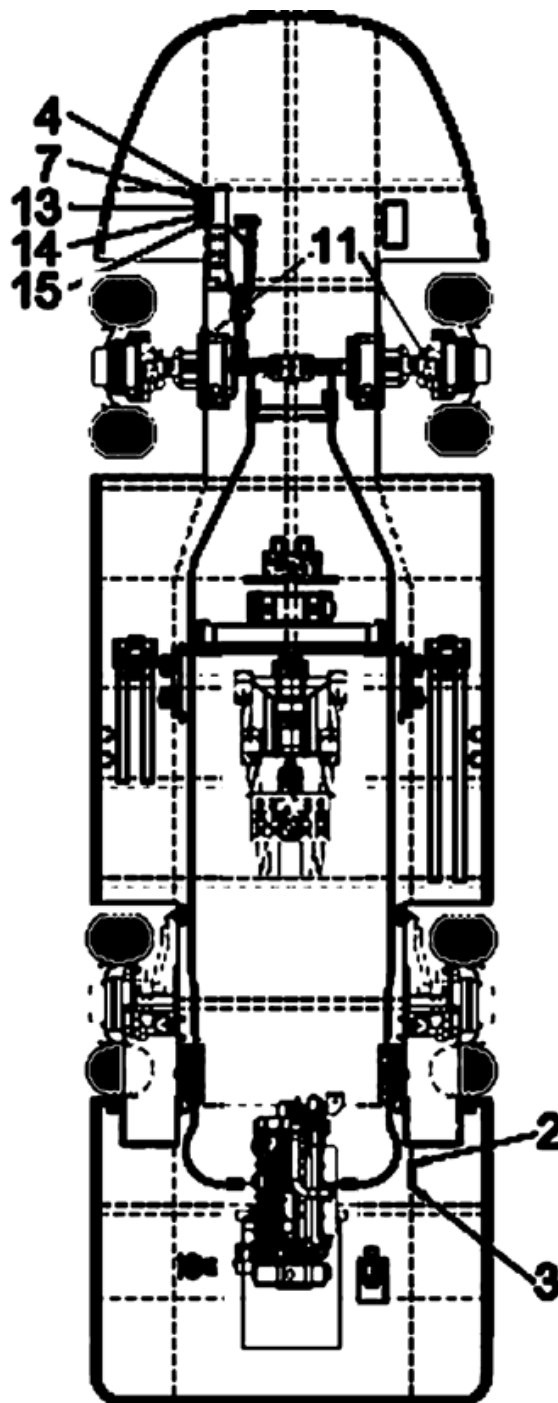
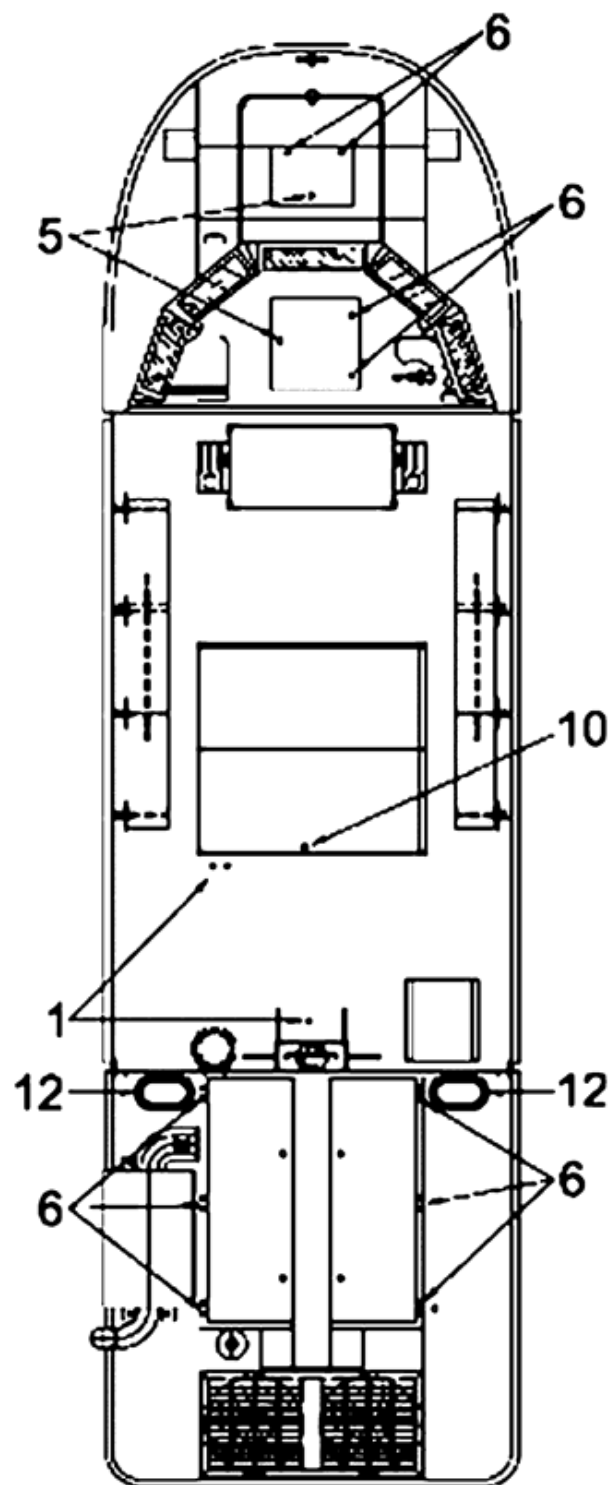


Figure 1 Lube Points

**Figure 2 Lube Points**

SHIP:

NBU-7 (RMC 57078)

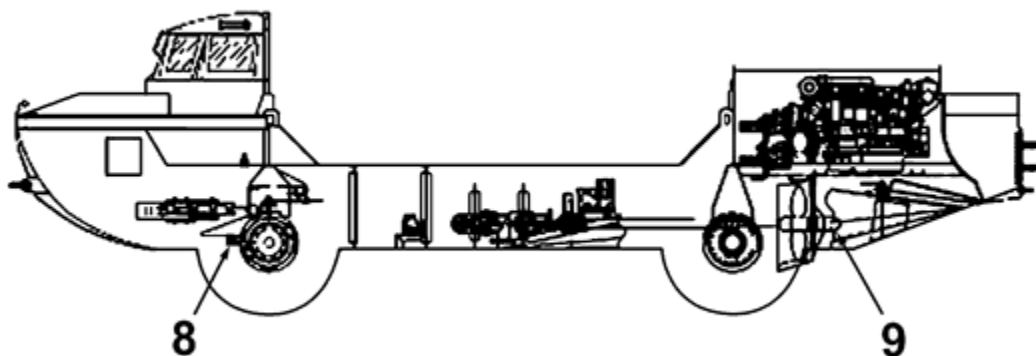


Figure 3 Lube Points

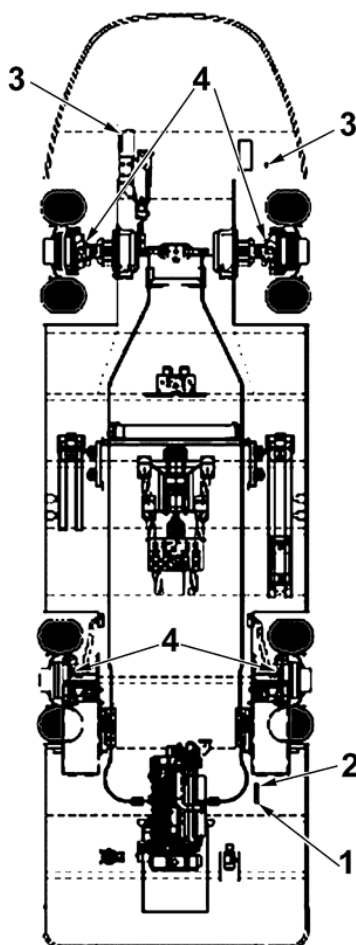


Figure 4 Lube Points

SSP: SASN-003-26

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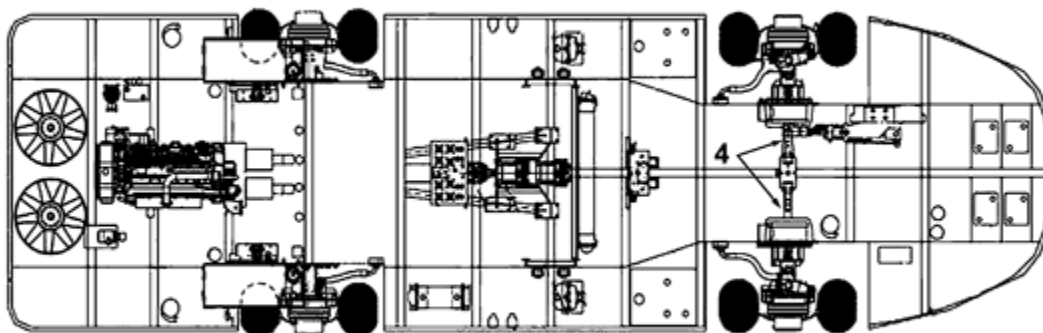


Figure 5 Lube Points

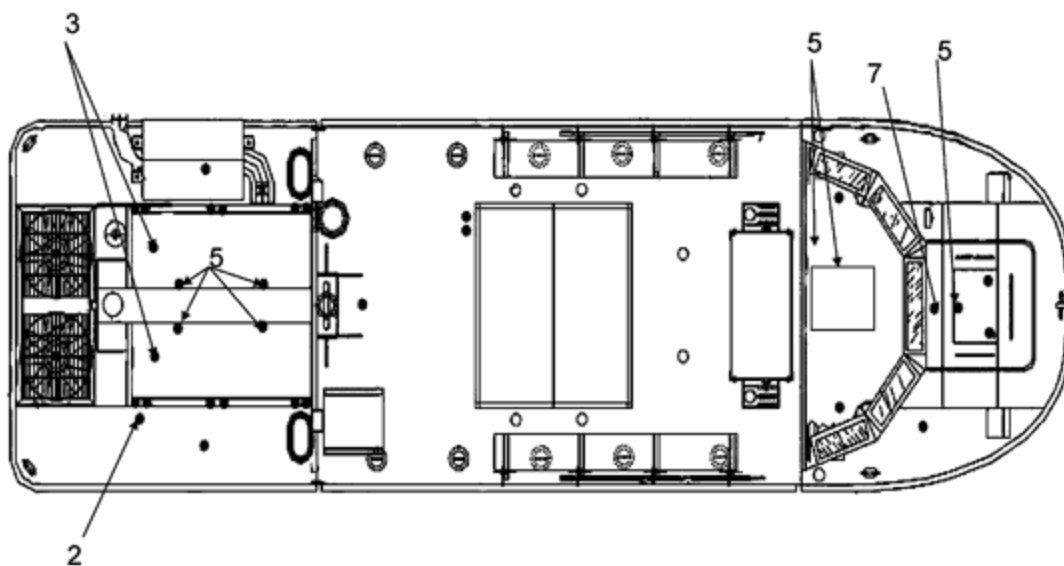


Figure 6 Lube Points

SHIP:

NBU-7 (RMC 57078)

Item Number	Nomenclature
1	Propeller drive shaft lubrication points inside center bay
2	Rudder post (1 Aft remote lubrication fitting)
3	Aft Port/Stbd rudder cable lube points (2 Aft remote lubrication fittings)
4	Forward Port/Stbd steering cable lube points (2 Fwd remote lubrication fittings)
5	Hatch dogging handles
6	Engine/bow/top hatch (16 points)
7	Steering post inside forward bilge area centerline (1 Fwd remote lubrication fitting)
8	Tie rod ends underneath forward hull (between front wheels) (4 points)
9	Aft propeller bearing (2 points)
10	Forward propeller bearing and shaft universal joint (2 points)
11	Wheel motor steering bearing cover plate (4 points)
12	Hydraulic fill cover hinge and slide-bolts (10 points)
13	Land Steering cylinder rod pin (1 Fwd remote lubrication fitting)
14	Land steering linear alignment coupling (1 Fwd remote lubrication fitting)
15	Land steering shaft (1 Fwd remote lubrication fitting)

Table 1 LARC Lubrication Point Identification

CRITICAL COATING

SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>631-00-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L019</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>HAMABE, YUUKI</u>

1. SCOPE:

1.1 Title: Exterior Surface; preserve

1.2 Location of Work:

1.2.1 Exterior Hull (Underwater Hull and Freeboard Area) "CRITICAL COATED AREA" CCNS# LARC-001-26

1.2.2 Exterior Hull (Deck and Equipment Including Inside Cabin)

1.3 Identification:

1.3.1 A Total of (2,500 SF) of Exterior Hull Preservation Area Including Inside Cabin Surface, Located in 1.2.1 and 1.2.2

2. REFERENCES:

2.1 Standard Items, (FY26)

2.2 1600901-4 Rev F, Paint Schedule and Markings

3. REQUIREMENTS:

3.1 Accomplish the requirements of 009-032 of 2.1, including Table One, Line 11, Column A for each area listed in 1.3.1, using 2.2 and manufacture's instruction for guidance. An aluminum oxide abrasive blast media must not be used.

3.1.1 Power tool clean each narrow area to SSPC-SP-11 where blast nozzle in 3.1 cannot reach.

3.1.1.1 Power tool clean each narrow area to SSPC-SP-3 where power tool in 3.1.1 cannot reach.

3.1.1.2 Hand tool clean each narrow area to SSPC-SP-2 where power tool in 3.1.1.1 cannot reach.

3.1.2 Apply one primer coat (Find No. 3 of 2.2, MIL-DTL-24441/20 F-150, Green, DFT 3-6 mils) on each cleaned area.

3.1.3 Apply one top coat (Sher-Cryl High Performance Acrylic, Dark Yellow) on the primed surface area. (See Note 4.2)

3.1.4 Apply new marking on each surface, using 2.2 and manufacturer's instruction for guidance. (See Note 4.3)

3.2 Accomplish the requirements of 009-032 of 2.1 for cleaning and preservation of each new and disturbed surface to match surrounding areas.

4. NOTES:

4.1 Initial Condition - The paint was chipped in some spots and it is needing repainted.

4.2 The use of Sher-Cryl High Performance Acrylic (Dark Yellow color code 23448) by Sherwin-Williams as substitute for Find No.7 of 2.2 has

SHIP:

NBU-7 (RMC 57078)

been approved by S245 under ESR LARC936-0002 response.

4.3 The use of Sher-Cryl High Performance Acrylic (Black and Red) by Sherwin-Williams as substitute for Find No.8 and 13 of 2.2 has been approved by S245 under ESR LARC936-0002 response.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

1. None.

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

SHIP:	<u>NBU-7 (RMC 57078)</u>	ITEM NO:	<u>864-42-001</u>
COAR:	<u>18</u>	PCN:	<u>RS01-L014</u>
BWST FILE NO:	<u>000-001</u>	CMP:	<u>NONE</u>
REVISED:	<u>05 APR 2019</u>	PLANNER:	<u>SAITO, KAZUYUKI</u> <u>NAGAMINE, YUUICHI</u> <u>TERASHITA, SHUUJI</u>

1. SCOPE:

1.1 Title: Bilge Pumping System Component; replace

1.2 Location of Work:

1.2.1 Forward Compartment

1.2.2 Engine Compartment

1.3 Identification:

1.3.1 Quantity (3 EA), Forward, AFT-STBD, and AFT-PORT Bilge Pump located in 1.2.1 and 1.2.2, MFR New York Pyrotechnic Products Company, P/N V11-001-700 (71141-00), Figure 1-52 Index No.8 of 2.3

1.3.2 Quantity (3 EA), Bilge Pump Float Switch located in 1.2.1 and 1.2.2, P/N 43765, Paragraph 3.41.3.2 of 2.2, LS-270 Series - SPST - NO - 20VA, MFR - GEMS, NSN 5930-01-157-4549

1.3.3 Quantity (3 EA), Solenoid Valve for Bilge Pump Supply Pressure installed on STBD and PORT Supply Manifold located in 1.2.2, MFR Parker-Hannifin Co., P/N DS163B/D024WD, Figure 1-10 Index No.15 and Figure 1-11 Index No.9 of 2.3

1.3.4 Quantity (3 EA), Relief Valve for Bilge Pump Supply Pressure installed on STBD and PORT Supply Manifold located in 1.2.2, MFR Parker-Hannifin Co., P/N RAH101T10, Set Press. 1,000 PSI, Figure 1-10 Index No.16 and Figure 1-11 Index No.10 of 2.3

2. REFERENCES:

2.1 NAVSEA Standard Items, (FY26)

2.2 F9YC0-AB-TMM-010 Rev 5, Change U, System Technical Manual for LARC-V A1/A2, Organizational and Intermediate Maintenance

2.3 F9YC0-AL-IPB-010, System Technical Manual for LARC-V A1/A2, Illustrated Parts Breakdown

2.4 1600911-2 Rev W, Hydraulic System Schematic

2.5 1600910-13 Rev E, Bilge System

2.6 1600907-1 Rev AA, Electrical System Installation & Details

2.7 1600907-3 Rev M, LARC V Electronic Transmission Control Schematic Diagram

2.8 Maintenance Requirement Card, MRC Control Number: 95 2UG8 N (Perform LARC V Lubrication) (enclosed)

2.9 Test Procedure, Test No.58331-4-100_26131, Bilge System Component

3. REQUIREMENTS:

(V) (G) "SHOP TEST"

3.1 Accomplish the shop test and set each new relief valve listed in 1.3.4, using ENCLOSURE (B) for guidance. -S130

3.1.1 Ensure the test medium is clean hydraulic oil conforming to MIL-PRF-17672.

3.1.2 Ensure the lifting pressure is 1,000 (+20/-20) PSIG.

3.1.3 Attach a metal tag to each relief valve, stamped with the following information:

3.1.3.1 Ship name and hull number

3.1.3.2 Valve Number or identification

3.1.3.3 Valve lifting pressure

3.1.3.4 Date Valve tested and set

3.1.3.5 Name of repair facility

3.2 Remove existing and install each new component for bilge pumping system listed in 1.3.1 through 1.3.4 installing each new gasket and packing, using Paragraph 6.9 of 2.2, 2.3 through 2.7, ENCLOSURE (A), and ENCLOSURE (B) for guidance.

3.2.1 Matchmark, identify and retain disconnected hardware.

3.2.2 Clean each part exposed by disassembly and removals free of oil and foreign matter leaving no residue or injurious effects.

3.2.3 Inspect each part not specified for renewal for wear and defect.

3.2.4 Remove existing and install each new component in place of those identified to be missing or defective.

3.2.5 Remove defective and install each new electric connector listed in material list of Paragraph 5.

3.2.6 Install retained hardware of 3.2.1.

3.2.7 Apply new sealant to each threaded hardware in accordance with Note One of 2.5.

3.2.8 Tighten each fastener at each torque value stated in Paragraph 6.9 of 2.2.

3.2.9 Apply new grease conforming to A-A-50433A to each bilge pump lubrication point, using 2.8 for guidance.

3.3 Accomplish the requirements of 009-032 of 2.1 for each new and disturbed surface.

(V) (G) "OPERATIONAL TEST"

3.4 Accomplish the requirements of 2.9 (TP) for an operational test of each equipment listed in 1.3.1 through 1.3.4. -S246

4. NOTES:

4.1 Initial Condition - S/F discovered bilge pumping system equipment damage on LARC 936 due to corrosion to primary bilge pumping system components to include: bilge pumps, bilge pump float switches, bilge pump

solenoid valves, bilge pump supply manifold relief valves.

4.2 SRF Tool Room will issue each torque wrench and pressure gage when required.

4.3 Ship's Force will provide fluids for operation such as hydraulic oil when refilling.

4.4 The removal and filling of hydraulic oil from each tank and the hydraulic system flushing will be conducted under the job specification, SSP#SASN-003-26 WN#57078-RS01-L009 WI# 589-53-001.

4.5 GFM listed in 5.1 have been ordered under ICN 38NBA86402-A01.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 LLTM:

	TOTAL QUANTITY PROVIDED	NAME Of PART	PIECE NO.	REF NO.	NATIONAL STOCK NO.	REQ PARA NO.
1.	3 EA	PUMPING UNIT, SEWAGE, PUMPING UNIT, SEWAGE, P/N 5NW	1-52-8	2.3	9B463001 6547282	3.2
2.	3 EA	PUMPING UNIT, SEWAGE, PUMPING UNIT, SEWAGE, P/N 5NW	1-52-8	2.3	9B463001 6547282	3.2
3.	2 PG	SCREW, HEX HD, 3/8"-16UNC X 1- 1/8LG, 18-8 SST, 5 E	13	2.5	9B5305SA 2300318	3.2
4.	9 EA	SCREW, CAP, HEXAGO N H, P/N: 92186A633, 3/8"- 16 UNC X	1-52-7	2.3	9B530501 5261069	3.2
5.	1 HD	WASHER, LOCK, MCMASTER P/N: 91475A031, SIZE; 3/8"	14	2.5	9B531000 9847042	3.2
6.	18 EA	WASHER, FLAT, P/N; 98019A500, SIZE; 3/8"	15	2.5	9B531001 4829259	3.2
7.	9 EA	NUT, PLAIN, HEXAGO N, P/N: 90762A127, SIZE; 3/8-16UNC	16	2.5	9B531000 2607924	3.2
8.	6 EA	CLAMP, HOSE, MCMASTER P/N: 5204K14	1-52-9	2.3	9B4730SA 2300320	3.2
9.	3 EA	VALVE, LINEAR, DIR ECT, P/N: DS163BD024W-A16T	1-10-15	2.3	9B481001 6523187	3.2
10.	3 EA	VALVE, LINEAR, DIR ECT, P/N: DS163B/D024WD	1-10-15	2.3	9B4820SA 2600630	3.2

SHIP:

NBU-7 (RMC 57078)

11.	6 EA	O-RING, P/N: SK16-3	-		9B5331SA 2300176	3.2
12.	3 EA	VALVE,RELIEF,PRE SSU, P/N: RAH101T10	1-10-16	2.3	9B482012 3955626	3.2
13.	3 EA	O-RING, P/N: 3- 916	1-10-30	2.3	9B533101 6420472	3.2
14.	6 EA	PARTS KIT,SEAL REPL, P/N: SK10- 2	-		9B533001 3504474	3.2
15.	5 OZ	CHEMICAL CORROSION, ANTICORROSIVE COMPOUND, TEF-GE	27	2.5	9B685001 4504009	3.2
16.	1 CN	HYDRAULIC FLUID,PET, MIL- PRF-17672, 5 GL/CN	-		9B915000 9857237	3.2
17.	3 EA	VALVE, VALVE, P/N DS163B, FOR SOLENOID VALVE P/N D	For1-10- 15	2.3	9B4820SA 2600671	3.2
18.	3 EA	COIL, DS COIL, P/N 851062- 024VDC, TYP W (DOUBLE WI	For1-10- 15	2.3	9B4820SA 2600672	3.2
19.	3 EA	SWITCH,FLOW, P/N 43765 GEMS LS- 270	-	2.2	3B593001 1574549	3.2
20.	3 EA	FUSE,CARTRIDGE, P/N 5226MDL FUSE 3A ITEM NO.44	44	2.6	9B592000 3778279	3.2
21.	1 HD	SCREW,CAP,HEXAGO N H, P/N 93190A5401/4"-20 UNC X 3/	-		9B530501 3524397	3.2
22.	25 EA	LOCK NUT, P/N 90715A1241/4"-- 20UNC 316 SST	-		9B531001 5045702	3.2
23.	3 OZ	CHEMICAL CORROSION, ANTICORROSIVE COMPOUND, TEF-GE	27	2.5	9B685001 4504009	3.2
24.	1 TU	INSULATING COMPOUND, P/N 3140 RTV	-		9B597001 4871943	3.2
25.	3 PG	CONNECTOR,PLUG,E LEC, P/N DT04-2P	-		9B593501 5065555	3.2
26.	3 FV	RETAINER,ELECTRI CAL, P/N W2P	-		9B593501 5191808	3.2
27.	6 EA	CONTACT,ELECTRIC AL, P/N 0460- 202-16141	-		9B599901 2163648	3.2
28.	3 EA	CONNECTOR	-		9B593501	3.2

SSP: SASN-003-26

4 of 16

ITEM NO: 864-42-001

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SHIP:

NBU-7 (RMC 57078)

		BODY, PLUG, P/N DT06-2S			6657216	
29.	3 PG	RETAINER, ELECTRI CAL, P/N W2S- PO12	-		9B593501 4800358	3.2
30.	6 EA	CONTACT, ELECTRIC AL, P/N 0462- 201-16141	-		9B599901 2036687	3.2
31.	1 LG	INSULATION SLEEVING, HEAT- SHRINK, SLEEVE, ADHESIVE	-		9B597001 2558990	3.2
32.	4 FT	INSULATE, HEAT- SHRINK, SLEEVE, ADHESIVE, BLK, 1"	-		9B597001 1520397	3.2
33.	5 EA	CONDUIT, NONMETAL LIC, 3/8" , P/N 7840K82INSIDE DIAM	-		9B597501 5749289	3.2
34.	100 EA	STRAP, TIE DOWN, ELEC, P/N 7130K42	-		9B597501 5723833	3.2
35.	1 SL	WIRE, P/N 180003, 18 AWG BLACK 35FT/ SPOOL	-		9B6145SA 2300249	3.2
36.	3 EA	ENCLOSURE, P/N J-604W6X4X4 INCHES FIBERGLASS ENCLO	-		9B9999SA 2300250	3.2
37.	3 EA	BACKING PLATE, P/N BP64ALJUNCTION BOX BACKING PLAT	-		9B9999SA 2300251	3.2
38.	10 EA	FUSE LINK, ELECTRICAL, P/N 0217005.HXP5A, GLASS AND	-		9B592001 5635884	3.2
39.	3 EA	SWITCH, LIQUID LEVEL, P/N 43760, SPST - NC - 20VA	-		3B593001 1943442	3.2
40.	2 EA	LENS, LIGHT, P/N 103-1231- 403BILGE ALARM	-		9B621001 3584319	3.2
41.	2 EA	LIGHT, INDICATOR, P/N 103-3502-05- 103LIGHT SOCKET	-		9B621000 7532289	3.2
42.	2 EA	LAMP, INCANDESCEN T, P/N S6S/24V LIGHT BULB	-		9B624000 1437513	3.2

SSP: SASN-003-26

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ITEM NO: 864-42-001

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SHIP:

NBU-7 (RMC 57078)

5.2 PUSH MATERIAL:

1. None.

5.3 KITTED MATERIAL:

1. None.

Catalog 3105/USA
Technical Information

Directional Control Valves
Series DS163

General Description

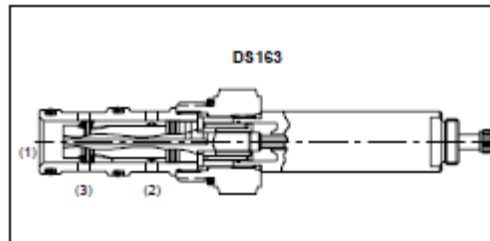
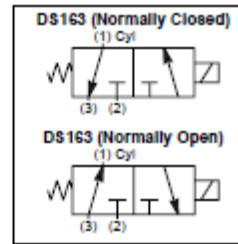
The DS163, 3-way spool valves are used as low flow directional valves, pilot valves, or selector valves. They can be used in normally open or normally closed applications.

Operation

In the de-energized mode, the spool is positioned by spring force. The solenoid, when energized, directly shifts the spool against the spring, thus changing the flow path through the valve.

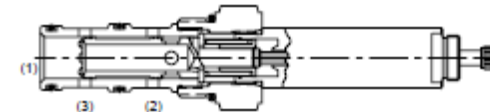
Features

- High flow capacity with reduced space requirements
- Standard valve bodies and common cavities
- One-piece encapsulated coil with minimal amperage draw
- No dynamic seals
- Variety of coil terminations



Construction

DS163B



Specifications

Rated Flow (At 70 PSI ΔP)	DS163
	N.O. 56.3 LPM (15 GPM)
	N.C. 56.3 LPM (15 GPM)
	Selector 48.8 LPM (13 GPM)
	DS163B
	N.O. 56.3 LPM (15 GPM)
	N.C. 56.3 LPM (15 GPM)
	Selector 63.8 LPM (17 GPM)
Maximum Inlet Pressure	210 Bar (3000 PSI)
Leakage	82 cc/min at 210 Bar (3000 PSI)
Operating Temp. Range (Ambient)	-40°C to +93.3°C (Nitrile) (-40°F to +200°F) -31.7°C to +121.1°C (Fluorocarbon) (-25°F to +250°F)
Cartridge Material	All parts steel. All operating parts hardened steel.
Body Material	Steel or Aluminum
Filtration	ISO Code 16/13, SAE Class 4 or better
Mounting	No restrictions
Cavity	Common Cavity No. C16-3

Solenoid Coil Specifications

Voltage	12 VDC	24 VDC
Duty	HD.	HD.
Watts	42	42
Amps	3.5	1.75

Response Time (at nominal flow and maximum pressure)	
Normally Closed	Up to 90ms
Normally Open	Up to 100ms
Wire Class	
"H" for all voltages	
A.C. Rectifiers	
Integral full wave bridge	

ds163.pn6.5, U, I



ENCLOSURE (A) - 1/4
DC107

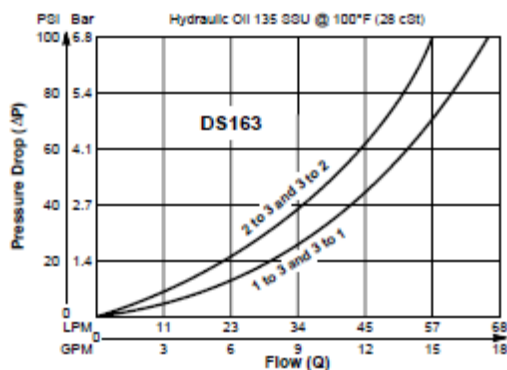
Parker Hannifin Corporation
 Hydraulic Valve Division
 Elyria, Ohio 44035 USA

Catalog 3105/USA
Technical Information

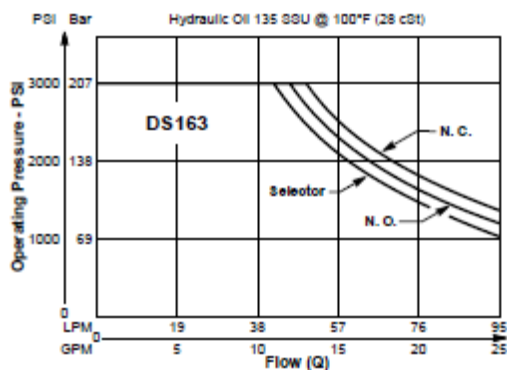
Directional Control Valves
Series DS163

Performance Curves

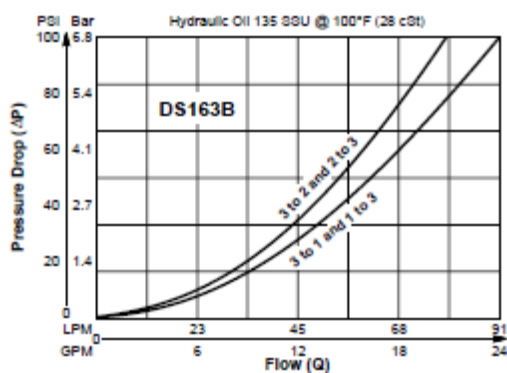
Pressure Drop vs. Flow (Through cartridge only)



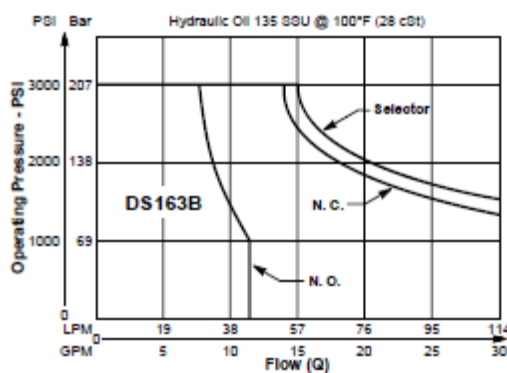
Operating Characteristics (Through cartridge only)



Pressure Drop vs. Flow (Through cartridge only)



Operating Characteristics (Through cartridge only)



ds163.pn6.5, U, I



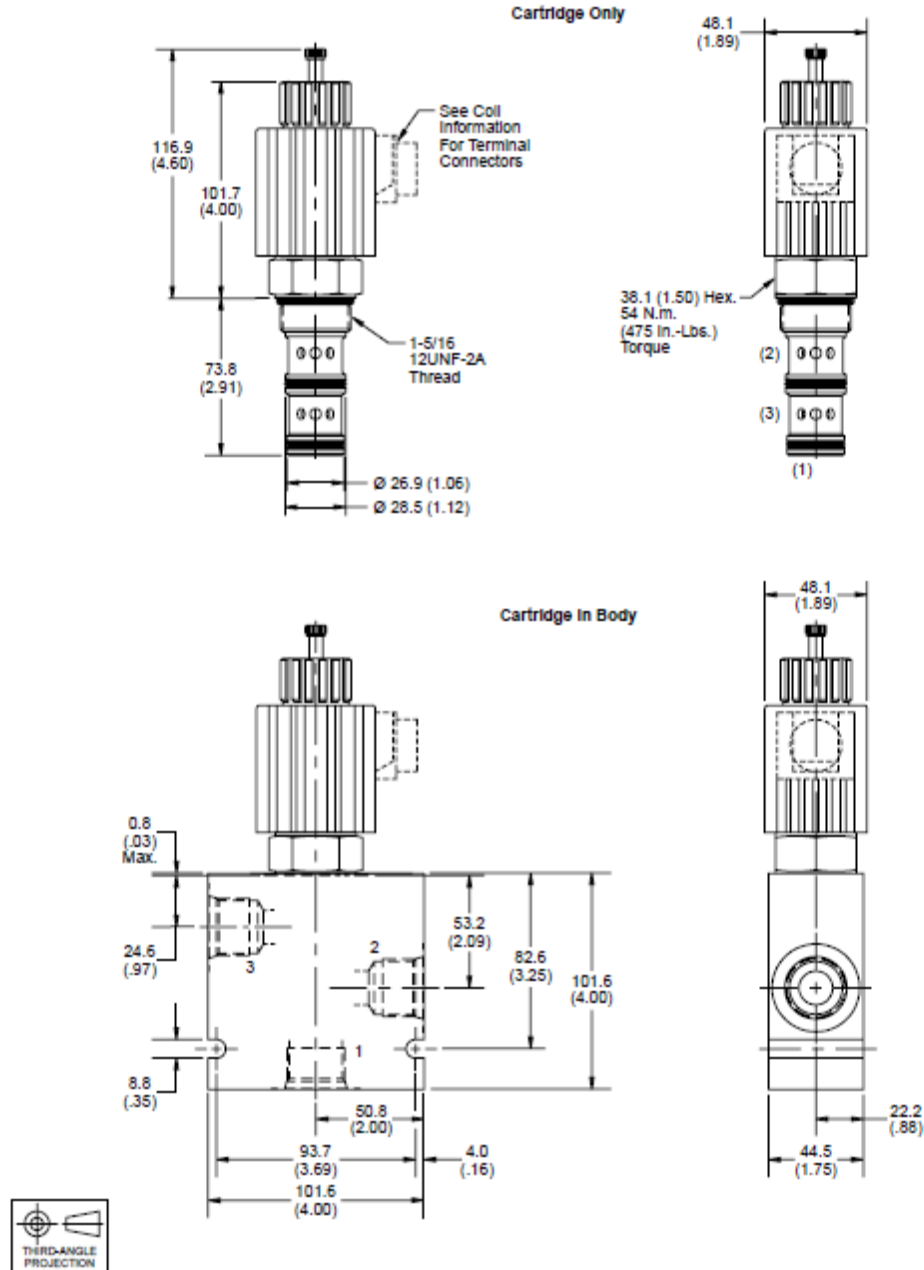
ENCLOSURE (A) - 2/4
DC108

Parker Hannifin Corporation
Hydraulic Valve Division
Elyria, Ohio 44035 USA

Catalog 3105/USA
Dimensions

Directional Control Valves
Series DS163

*Inch equivalents for millimeter dimensions are shown in ("")



ds163.pn6.5, U, I

Parker Hydraulics

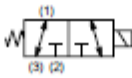
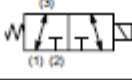
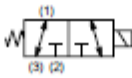
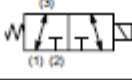
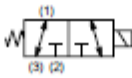
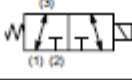
ENCLOSURE (A) - 3/4
DC109

Parker Hannifin Corporation
Hydraulic Valve Division
Elyria, Ohio 44035 USA

SHIP:

NBU-7 (RMC 57078)

Catalog 3105/USA
Ordering InformationDirectional Control Valves
Series DS163

DS	16	3																																		
Solenoid Valve	Size	Spool Type	Style	Seals	Coil Voltage	Coil Termination	Body Option																													
	<table border="1"><thead><tr><th>Code</th><th>Style</th></tr></thead><tbody><tr><td>16</td><td>1-5/16" - 12UNF-2B</td></tr></tbody></table>	Code	Style	16	1-5/16" - 12UNF-2B			<table border="1"><thead><tr><th>Code</th><th>Type</th></tr></thead><tbody><tr><td>Omit</td><td>Nitrile</td></tr><tr><td>V</td><td>Fluorocarbon</td></tr></tbody></table>	Code	Type	Omit	Nitrile	V	Fluorocarbon																						
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SERVICE PARTS

DIN Connector: Grey — 692914 Black — 692915	Coil Assemblies 851057***** Conduit Coil 851058***** DIN (Hirschmann) Coil 851060***** Double Spade Vertically-Oriented Coil 851062***** Double Wire Coil 851064***** Double Screw Vertically-Oriented Coil 851065***** Single Screw Vertically-Oriented Coil
Coil Nut: 118378-00	(*****Calls out voltage. For example: 851060-012VDC is a 12 volt, 42 watt Double Spade Coil.)
Nitrile Seal Kit: SK16-3	
Fluorocarbon Seal Kit: SK16-3V	

Shipping Weight
Cartridge Only .59 kg (1.3 lbs.)
Cartridge In Body 1.4 kg (3.0 lbs.)

ds163.prd 5, 14, 15

ENCLOSURE (A) - 4/4
DC110Parker Hannifin Corporation
Hydraulic Valve Division
Elyria, Ohio 44035 USA

SSP: SASN-003-26

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ITEM NO: 864-42-001

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Catalog HY15-3502/US

Technical Information

Pilot Operated Relief Valve
Series RAH101

CV

Quick
Valves

SH

Shuttle
Valves

LM

Load/Unload
Controls

FC

Flow
Controls

PC

Pressure
Controls

LE

Logic
Elements

DC

Directional
Controls

MV

Manual
Valves

SV

Solenoid
Valves

PV

Proportional
Valves

CE

Cells &
Electronics

BC

Bodies &
Cartridges

TD

Technical
Data

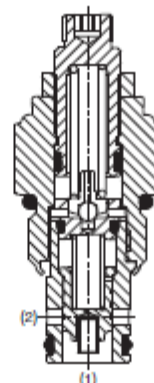
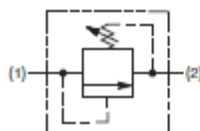
General Description

Pilot Operated Spool-Type Relief Valve
For additional information see Technical
Tips on pages PC1-PC6.



Features

- Hardened, precision ground parts for durability
- Low profile adapter for minimal space requirements
- Fully guided poppet for more consistent reseal
- Steel adapters are coated with yellow zinc dichromate for protection from salt spray
- Polyurethane "D"-Ring eliminates backup rings and prevents hydrolysis
- Internal screening protects pilot spring from debris



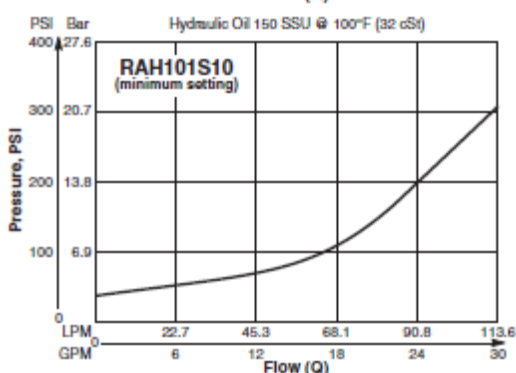
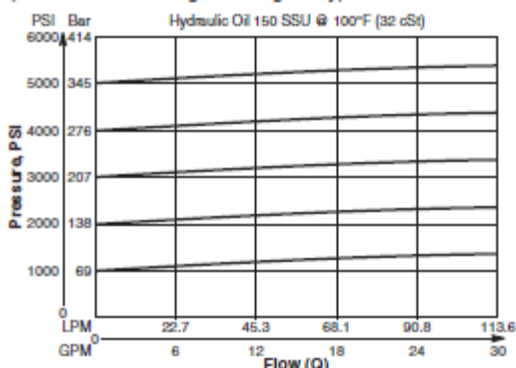
Specifications

Rated Flow	113 LPM (30 GPM)
Maximum Inlet Pressure	380 Bar (5500 PSI)
Maximum Pressure Setting	350 Bar (5000 PSI)
Maximum Tank Pressure	350 Bar (5000 PSI)
Reseat Pressure	90% of crack pressure
Leakage at 150 SSU (32 cSt)	5 cc per 100 PSI (6.8 Bar) setting
Cartridge Material	All parts steel. All operating parts hardened steel.
Operating Temp. Range/Seals	-45°C to +93.3°C ("D"-Ring) (-50°F to +200°F) -31.7°C to +121.1°C (Fluorocarbon) (-25°F to +250°F)
Fluid Compatibility/Viscosity	Mineral-based or synthetic with lubricating properties at viscosities of 45 to 2000 SSU (6 to 420 cSt)
Filtration	ISO Code 16/13, SAE Class 4 or better
Approx. Weight	.23 kg (.50 lbs.)
Cavity	C10-2 (See BC Section for more details)
Form Tool	Rougher None Finisher NFT10-2F

Performance Curves

Flow vs. Inlet Pressure

(Pressure rise through cartridge only)



PC37

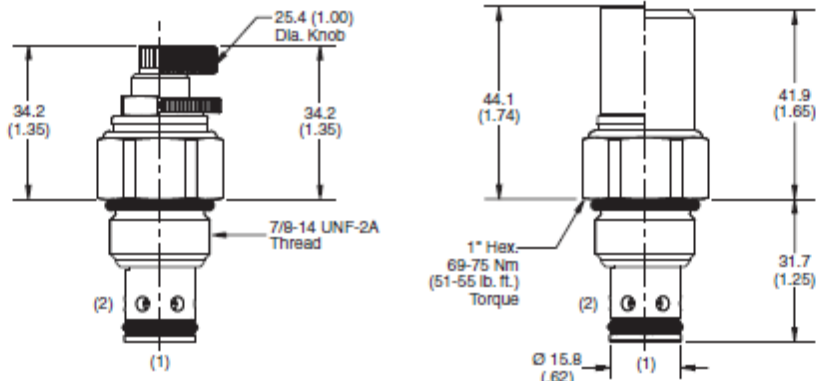
ENCLOSURE (B) - 1/2

Parker Hannifin Corporation
Hydraulic Cartridge Systems

Catalog HY15-3502/US
Technical Information

Pilot Operated Relief Valve
Series RAH101

Dimensions Millimeters (Inches)



Screw/Knob Version

Fixed Cap/Tamper Resistant Version

Ordering Information

RAH101

10 Size Pilot Operated Relief Valve

Adjustment Style

Pressure Range

Seals

Optional Pressure Setting

Body Material

Port Size

Code	Adjustment Style / Kit No.
F	Fixed style, preset at factory.
K	Knob Adjust (717784-10)
S	Screw Adjust
T	Tamper Resistant Cap (718083)

Code	Seals / Kit No.
Omit	"O"-Ring / (SK10-2)
N	Nitrile / (SK10-2N)
V	Fluorocarbon / (SK10-2V)

Code	Body Material
Omit	Steel
A	Aluminum

Code	Pressure Range
10	6.9 - 69 Bar (100 - 1000 PSI) Standard Setting: 34.5 Bar (500 PSI) @ crack pressure, approximately .95 LPM (.25 GPM)
20	6.9 - 138 Bar (100 - 2000 PSI) Standard Setting: 69 Bar (1000 PSI) @ crack pressure, approximately .95 LPM (.25 GPM)
30	13.8 - 207 Bar (200 - 3000 PSI) Standard Setting: 103.5 Bar (1500 PSI) @ crack pressure, approximately .95 LPM (.25 GPM)
50	13.8 - 345 Bar (200 - 5000 PSI) Standard Setting: 172.4 Bar (2500 PSI) @ crack pressure, approximately .95 LPM (.25 GPM)

Optional Pressure Setting
Pressure + 10 i.e. 235 - 2350 PSI (Omit if standard setting is used)
Setting Range: 100 to 5000 PSI
All settings at crack pressure, approximately .95 LPM (.25 GPM)

Code	Port Size	Body Part No.
Omit	Cartridge Only	
4P	1/4" NPTF	(B10-2-*4P)
6P	3/8" NPTF	(B10-2-*6P)
8P	1/2" NPTF	(B10-2-*8P)
6T	SAE-6	(B10-2-*6T)
8T	SAE-8	(B10-2-*8T)
T8T	SAE-8	(B10-2-T8T)†
6B	3/8" BSPG	(B10-2-6B)†

* Add "A" for aluminum, omit for steel.
† Steel body only.



PC38
ENCLOSURE (B) - 2/2

Parker Hannifin Corporation
Hydraulic Cartridge Systems

CV
Check Valves
SH
Shuttle Valves
LM
Load/Unload Controls
FC
Flow Controls
PC
Pressure Controls
LE
Logic Elements
DC
Directional Controls
MV
Manual Valves
SV
Solenoid Valves
PV
Proportional Valves
CE
Calls & Electronics
BC
Bodies & Cartridges
TD
Technical Data

SHIP: NBU-7 (RMC 57078)

DISTRIBUTION STATEMENT A:

Approved for public release: distribution unlimited.

Date: September 2025 **MIP** 59A3 **MRC:** 95 2UG8 N **Periodicity:** M-2
Series:

Location:

Ship System: Special Purpose Systems 59000
System: Logistics Over The Shore (LOTS) Support 59A00
SubSystem: Logistics Over the Shore (LOTS), Personnel Transport 59A30
Equipment: Lighter, Amphibious, Resupply, Cargo V (LARC V) 59A32

Rates	Man-Hours	Rates	Man-Hours	Rates	Man-Hours
CMCN	2.0				
Total Man-Hours:	2.0	Elapsed Time:	2.0		

MAINTENANCE REQUIREMENT DESCRIPTION

1. [Perform LARC V Lubrication.](#)

SAFETY PRECAUTIONS

1. Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.
2. To prevent personnel injury, Tag-Out procedures shall be in accordance with the Tag-out Users Manual (TUM), and/or local tag-out instruction.
3. Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.

TOOLS, PARTS, MATERIALS, TEST EQUIPMENT

MATERIALS

1. [\[000073\]](#)Grease, Termalene 2, water-wash resistant, CID A-A-50433
Hazardous Material User's Guide (HMUG) Group 13
2. [\[001144\]](#)Tag, safety
3. [\[035110\]](#)Grease, General Purpose
Hazardous Material User's Guide (HMUG) Group 13

TOOLS

1. [\[000771\]](#)Lubricating gun, hand, Lever-operated, 16 OZ 12" rub hose ext & adpt kit incl
2. [\[020301\]](#)Tool kit, CESE maintenance, basic

MISCELLANEOUS

1. [\[010949\]](#)Spectacles, industrial with side cover
2. [\[017934\]](#)Gloves, 4-mil Nitrile, Light use only
3. [\[001365\]](#)Technical manuals/drawings, W/C provide (F9YC0-AM-TMM-030 LARC V; Technical Manual Supplement for Lighter Amphibious Resupply Cargo, Five Ton (LARC-V A1/A2))
4. [\[011922\]](#)S9086-WK-STM-020/NSTM 670 Vol II, Hazardous Material Users Guide (HMUG)
5. [\[015244\]](#)S0400-AD-URM-010/TUM; TAG OUT USER'S MANUAL

NOTE: Numbers in brackets can be referenced to Standard PMS Materials Identification Guide (SPMIG) for identification.

HAZARDOUS MATERIALS CONTROL STATEMENT(U)

For all Navy commands, units, and activities regardless of warfare community, OPNAV M-5100.23 Navy Safety and

SSP: SASN-003-26

13 of 16

ITEM NO: 864-42-001

EXPORT CONTROLLED DATA: This document contains technical information whose export is governed by U.S. International Traffic in Arms Regulation (ITAR). This information must not be transferred to a foreign person/entity without proper authorization of the U.S. Government. Violations may result in administrative, civil, or criminal penalties.

Occupational Health Manual provides guidance for Hazardous Material Control and Management (HMC&M) and directs Safety Data Sheets (SDS) be maintained and accessible to review, for all HM at the command. For forces afloat, the Hazardous Material Users Guide (HMUG), S9086-WK-STM-020, NSTM Chapter 670 Volume II supplements the technical data found in SDSs by providing additional control measures, precautions, personal protective equipment (PPE), and spill controls for the hazardous material(s) identified in the Tools, Parts, Materials, Test, Equipment block. Maintenance personnel shall determine if additional

PPE is necessary to accomplish the MRC and take appropriate action to obtain and wear such PPE to ensure the safety of maintenance personnel. Final disposition of all HAZMAT, both shipboard and ashore, shall be IAW local HAZMINCEN/CHRIMP guidance as required by OPNAV-M 5090.1 series Table 35-3.

PROCEDURE

WARNING: Comply with Navy Safety and Occupational Health (SOH) Manual, OPNAV M-5100.23 as well as any additional guidance from other applicable policy.

Preliminary

- a. Ensure that the LARC is in a maintenance-ready condition.

NOTE 1: The steering ball valve handle is accessible in the operator's cabin under the port passenger seat. During normal operation the ball valve handle should be in the OPEN position.

- b. Ensure the emergency steering ball valve is in the OPEN position, if applicable.

WARNING: To prevent personnel injury, Tag-Out procedures shall be in accordance with the Tag-out Users Manual (TUM), and/or local tag-out instruction.

- c. De-energize/isolate LARC and attach safety tag(s) in accordance with the Tag-out User's Manual (TUM) and/or local tag-out instruction.

WARNING: Avoid repeated/prolonged skin contact with hazardous materials. Wash affected areas with soap and water upon completion of task or prior to eating, drinking, smoking, or applying cosmetics.

- d. Put on eye protection and gloves.

1. Perform LARC V Lubrication.

NOTE 2: Craft with the Bilge Pump Replacement Modification omit Step [1.a.](#) through Step [1.c.](#) and proceed to Step [1.d.](#)

CAUTION: Do not over-grease to avoid damaging craft components.

- a. Lubricate Aft Port Bilge Pump (#1 on Figure [1](#)):
- (1) Craft without the lubrication access improvement modification, clean Aft Port Bilge Pump grease fitting. Craft with lubrication access improvement modification, clean Aft Port Bilge Pump remote grease fitting.
 - (2) Apply grease [000073] to bilge pump grease fitting.
 - (3) Remove excess grease from grease fitting.
- b. Lubricate Aft Starboard Bilge Pump (#2 on Figure [1](#)):
- (1) Craft without the lubrication access improvement modification, clean Aft Starboard Bilge Pump grease fitting. Craft with lubrication access improvement modification, clean Aft Starboard Bilge Pump remote grease fitting.
 - (2) Apply grease [000073] to bilge pump grease fitting.
 - (3) Remove excess grease from grease fitting.
- c. Lubricate Fwd Bilge Pump (#4 on Figure [1](#)):
- (1) Craft without the lubrication access improvement modification, clean Fwd Bilge Pump grease

fitting. Craft with lubrication access improvement modification, clean Fwd Bilge Pump remote grease fitting.

- (2) Apply grease [000073] to bilge pump grease fitting.
 - (3) Remove excess grease from grease fitting.
- d. Lubricate Wheel Motors (Four locations) (#3 on Figure 1):

NOTE 3:

There are two grease fittings on each wheel motor.

- (1) Clean the grease fitting to be used for lubrication and remove the opposite grease fitting. Inspect fittings for damage.
- (2) Apply grease [000073] to wheel motor lubrication fitting until grease exits the open hole.
- (3) Remove excess grease from lubrication fittings.
- (4) Reinstall lubrication fitting.
- (5) Repeat Step 1.d.(1) through Step 1.d.(3) for remaining wheel motors.

NOTE 4:

Craft with the Linemaster Double Pole Double Throw (DPDT) Dead Man Foot Pedal Modification proceed to Step 1.e. Craft without the Linemaster DPDT Dead Man Foot Pedal Modification omit Step 1.e. through Step 1.e.(2) and proceed to Step 1.f.

- e. Lubricate Dead Man Foot Pedal inside operator's cabin (One location) (Figure 2).
- (1) Clean and lubricate treadle hinge rod, actuator button and rod with grease [000073] (this passes through the housing).
 - (2) Remove excess grease from treadle hinge rod, actuator button and rod.

NOTE 5: Hydraulic fan motors are located below the radiators on the AFT side of the LARC. Removal of the AFT bulkhead pipe guard and grill will be required to access the fan motor housing lubrication fittings.

CAUTION: Do not pump more than two pumps of synthetic grease into hydraulic fan motor housing or damage to the seals may occur.

- f. Lubricate both hydraulic fan motors.
 - (1) Clean and lubricate hydraulic fan motors with one to two pumps of grease [030654] per fitting.
 - (2) Remove excess grease from hydraulic fan motor lubrication fittings.
- g. Take off gloves and then eye protection.
- h. Remove safety tag(s) and energize/align LARC in accordance with the Tag-out User's Manual (TUM) and/or local tag-out instruction.
- i. Return LARC to readiness condition.

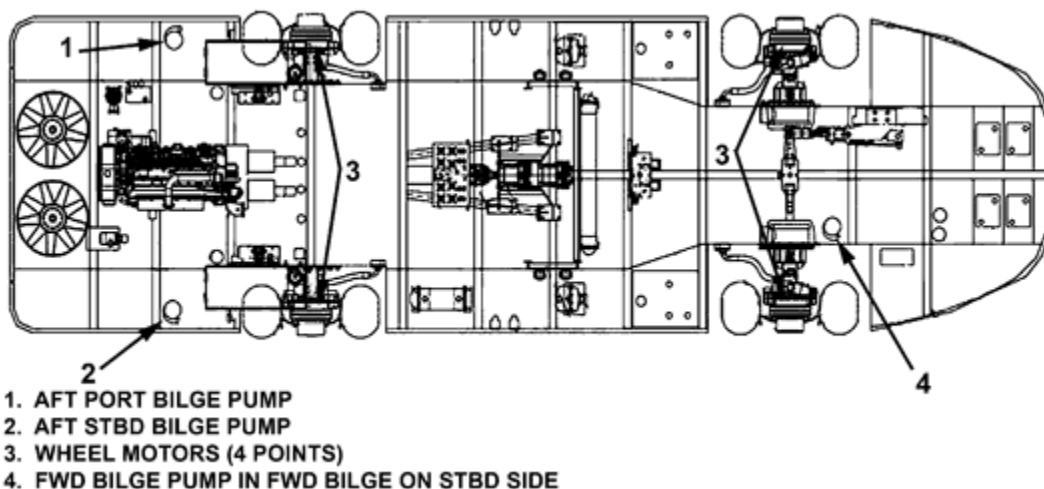
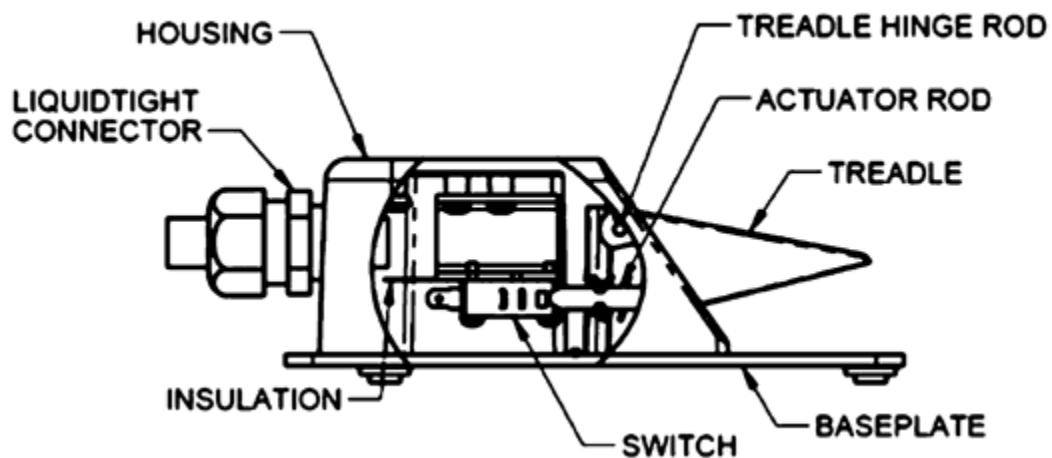


Figure 1 Lube Points**Figure 2 Dead Man Foot Pedal Lube Points.**